Linear and Generalised Linear Models (MATH30013)

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TB2 2025/26

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1 Introduction

Regression analysis is a statistical procedure for inferring the relationship between the variables, say Y (response variable) and X (covariate, explanatory variable, regressor, predictor), so that we can predict Y for given X. Linear regression is ubiquitous in statistical analysis, adopted extensively to evaluate interventions in experimental data as well as analysing observational data collected in many fields including natural and applied sciences. As such, Y and X can be of variety of different forms:

X \ Y	$\mathbb{R}$	$\{0,1\}$	K groups	$\mathbb{N}\cup\{0\}$	$\cdots$
$\mathbb{R}$	*				
$\{0,1\}$	*				
K groups	*				
$\mathbb{N}\cup\{0\}$	*				
$\vdots$					

Part 1 of this unit focuses on modelling the relationship between Y and X of the types corresponding to the *first column* of the table above by means of the **linear models**. We cover the theory and the practice of linear modelling, including estimation and computation, model building, data transformation, model diagnosis and inference, as well as the use of statistical computing environment `R` (especially the function `lm` and its methods). We will also briefly discuss methods for high-dimensional linear regression, which has received considerable attention in the last couple of decades with the emergence of 'big data'.

Example 1.1. The `Prestige` data has 102 observations on the various aspects of occupations, as reported in the Census of Canada conducted in 1971. The dataset contains the following variables among many: `income` (average income of incumbents in $), `education` (average education of occupational incumbents in years), `women` (percentage of incumbents who are women) and `prestige` (Pineo-Porter prestige score for occupation, from a social survey). We would like to see if there is any relationship between `income` and other observed variables. The scatter plots in Figure 1.1 indicate that indeed, there may be association among the variables.

Linear modelling assumes that the response variable is continuous and has constant variance. Its applicability can be extended through transformations and weighting, but this can

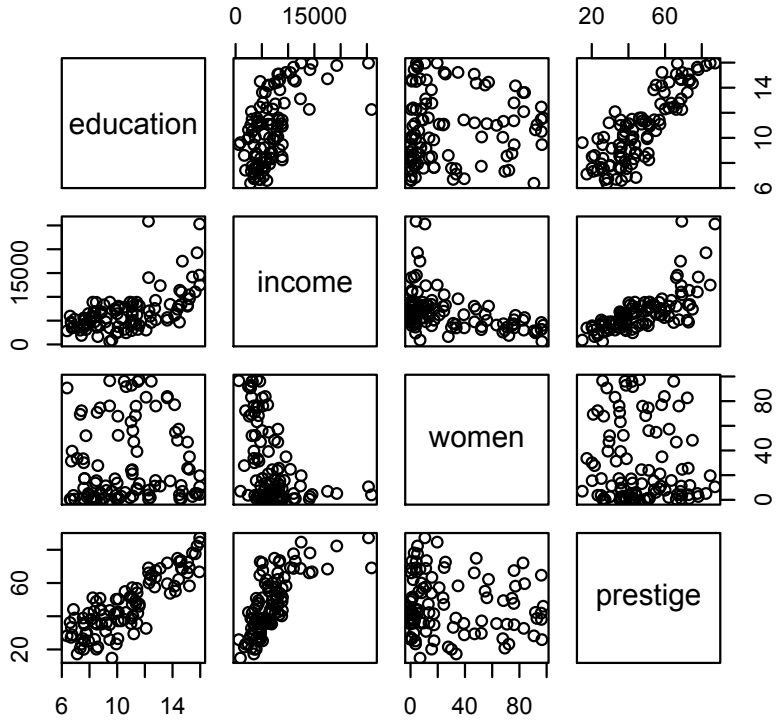


Figure 1.1: A matrix of scatterplots between `income`, `education`, `women` and `prestige`.

make the interpretation of the outcomes difficult. Furthermore, the applicability of linear modelling is still limited as it does not allow for discrete responses, e.g. $Y \in \{1, \ldots, K\}$. In Part 2 of this unit, we will consider how the main components of linear models can be extended to situations where the distribution of the response is a member of a much broader class of distributions named the *exponential family*. The resultant *generalised linear modelling* is applicable to learn the relationship between Y and X that spans the entire table above, and provides a unified framework for statistical inference. As special cases, they embed linear models and analysis-of-variance models ($Y \in \mathbb{R}$), and popular models for binary ($Y \in \{0, 1\}$, logit and probit models), and count ($Y \in \mathbb{N} \cup \{0\} = \{0, 1, 2, \ldots\}$), log-linear models) responses, which will be discussed in due course.

2 Preliminaries

2.1 Matrix algebra

2.1.1 Eigendecomposition

Let $\mathbf{A}$ be a square $p \times p$ matrix with p linearly independent eigenvectors $\mathbf{e}_j$, $1 \leq j \leq p$, and the associated eigenvalues λ_j, $1 \leq j \leq p$, i.e.

$$\mathbf{A}\mathbf{e}_j = \lambda_j \mathbf{e}_j, \quad j = 1, \ldots, p.$$

Then, $\mathbf{A}$ admits the decomposition

$$\mathbf{A}\begin{bmatrix}\mathbf{e}_1 & \mathbf{e}_2 & \ldots & \mathbf{e}_p\end{bmatrix} = \begin{bmatrix}\lambda_1\mathbf{e}_1 & \lambda_2\mathbf{e}_2 & \ldots & \lambda_p\mathbf{e}_p\end{bmatrix} = \begin{bmatrix}\mathbf{e}_1 & \mathbf{e}_2 & \ldots & \mathbf{e}_p\end{bmatrix}\begin{bmatrix}\lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ & & \ddots & \\ 0 & 0 & \ldots & \lambda_p\end{bmatrix},$$

$$\text{i.e.} \quad \mathbf{A} = \mathbf{E}\mathbf{\Lambda}\mathbf{E}^{-1},$$

where $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_p]$ denotes the matrix of eigenvectors and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_p)$ denotes the diagonal matrix containing the eigenvalues on its diagonal. As a special case, for any real symmetric matrices, the eigenvalues are real and the eigenvectors can be chosen such that they are orthogonal to each other. In other words, a real symmetric matrix $\mathbf{A}$ admits the decomposition

$$\mathbf{A} = \mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top, \quad \text{where} \quad \mathbf{E}\mathbf{E}^\top = \mathbf{E}^\top\mathbf{E} = \mathbf{I}_p.$$

2.1.2 QR decomposition

Any real $n \times p$ (let $p \leq n$ for convenience) matrix $\mathbf{X}$ can be written as

$$\mathbf{X} = \mathbf{Q}\begin{bmatrix}\mathbf{R} \\ \mathbf{O}\end{bmatrix} \tag{2.1}$$

where $\mathbf{Q}$ is an $n \times n$ orthogonal matrix ($\mathbf{Q}\mathbf{Q}^\top = \mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_n$) and $\mathbf{R}$ is a $p \times p$ upper triangular matrix ($\mathbf{R} = [r_{ij}]_{i,j=1}^p$ where $r_{ij} = 0$ for $i > j$). The Gram-Schmidt process that orthogonalises a set of vectors in an inner product space, can be adopted to compute the

QR decomposition of any given matrix. It is possible to write the QR decomposition as $\mathbf{X} = \mathbf{Q}_f\mathbf{R}$, where $\mathbf{Q}_f$ consists of the first p columns of $\mathbf{Q}$ (hence $\mathbf{Q}_f^\top\mathbf{Q}_f = \mathbf{I}_p$).

2.1.3 Blockwise matrix inversion

Any invertible square matrix can be inverted in a blockwise manner, by using the following analytic inversion formula:

$$\begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{C} & \mathbf{D}\end{bmatrix}^{-1} = \begin{bmatrix}\mathbf{A}^{-1}+\mathbf{A}^{-1}\mathbf{B}(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\\ -(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{C}\mathbf{A}^{-1} & (\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\end{bmatrix}, \quad (2.2)$$

where $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are sub-matrices of compatible dimensions. Notice that $\mathbf{A}$ and $\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B}$ must be square and invertible.
It is further seen that

$$(\mathbf{A}-\mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} = \mathbf{A}^{-1}+\mathbf{A}^{-1}\mathbf{B}(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{C}\mathbf{A}^{-1},$$

which is termed the Sherman-Morrison-Woodbury formula. Substituting $\mathbf{D}^{-1} = -\mathbf{W}$, $\mathbf{B} = \mathbf{U}$ and $\mathbf{C} = \mathbf{V}$, we yield:

$$(\mathbf{A}+\mathbf{U}\mathbf{W}\mathbf{V})^{-1} = \mathbf{A}^{-1}-\mathbf{A}^{-1}\mathbf{U}\left(\mathbf{W}^{-1}+\mathbf{V}\mathbf{A}^{-1}\mathbf{U}\right)^{-1}\mathbf{V}\mathbf{A}^{-1}. \quad (2.3)$$

2.2 Random vectors

Let $\mathbf{X} = (X_1,\ldots,X_p)^\top$ denote a *random vector* which contains random variables X_i, $1 \leq i \leq p$, as its elements. The mean and covariance matrix of $\mathbf{X}$ are given by

$$\boldsymbol{\mu}_X = \mathsf{E}(\mathbf{X}) = \begin{bmatrix}\mathsf{E}(X_1)\\ \mathsf{E}(X_2)\\ \vdots\\ \mathsf{E}(X_p)\end{bmatrix}, \quad \text{and}$$

$$\boldsymbol{\Sigma}_X = \mathsf{Cov}(\mathbf{X}) = \mathsf{E}((\mathbf{X}-\boldsymbol{\mu}_X)(\mathbf{X}-\boldsymbol{\mu}_X)^\top)$$

$$= \begin{bmatrix} \mathsf{Cov}(X_1, X_1) \equiv \mathsf{Var}(X_1) & \mathsf{Cov}(X_1, X_2) & \dots & \mathsf{Cov}(X_1, X_p) \\ \mathsf{Cov}(X_2, X_1) & \mathsf{Var}(X_2) & \dots & \mathsf{Cov}(X_2, X_p) \\ & & \ddots & \\ \mathsf{Cov}(X_p, X_1) & \mathsf{Cov}(X_p, X_2) & \dots & \mathsf{Var}(X_p) \end{bmatrix},$$

respectively. The matrix $\boldsymbol{\Sigma}_X$ is symmetric ($\boldsymbol{\Sigma}_X = \boldsymbol{\Sigma}_X^\top$) and positive semi-definite:

$$\text{all its eigenvalues } \geq 0 \quad \Longleftrightarrow \quad \mathbf{a}^\top \boldsymbol{\Sigma}_X \mathbf{a} \geq 0 \text{ for any } \mathbf{a} \in \mathbb{R}^p.$$

If $\mathbf{A}$ is a matrix of fixed real coefficients and $\mathbf{Y} = \mathbf{A}\mathbf{X}$, we have $\boldsymbol{\mu}_Y = \mathsf{E}(\mathbf{Y}) = \mathbf{A}\boldsymbol{\mu}_X$, and

$$\begin{aligned} \boldsymbol{\Sigma}_Y &= \mathsf{E}((\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top) = \mathsf{E}((\mathbf{A}\mathbf{X} - \mathbf{A}\boldsymbol{\mu}_X)(\mathbf{A}\mathbf{X} - \mathbf{A}\boldsymbol{\mu}_X)^\top) \\ &= \mathbf{A}\mathsf{E}((\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^\top)\mathbf{A}^\top = \mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}^\top. \end{aligned} \tag{2.4}$$

2.3 Multivariate normal distribution

Let the random vector $\mathbf{X} = (X_1, \dots, X_p)^\top$ have its elements satisfy $X_i \sim_{\text{i.i.d}} \mathcal{N}(0, 1)$, $1 \leq i \leq p$. Then, $\mathbf{X}$ follows a multivariate normal distribution, and we often write

$$\mathbf{X} \sim \mathcal{N}_p(\mathbf{0}, \mathbf{I}_p).$$

The density function of $\mathbf{X}$ at $\mathbf{x} = (X_1, \dots, X_p)^\top$, is given by

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^{p} f_{X_i}(x_i) = \prod_{i=1}^{p} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x_i^2}{2}\right) = \frac{1}{\sqrt{(2\pi)^p}} \exp\left(-\frac{1}{2}\mathbf{x}^\top\mathbf{x}\right).$$

2.3.1 Linear transform

Let $\mathbf{A}$ denote an $m \times p$ matrix of real coefficients and $\mathbf{b}$ a m-vector of real coefficients. Then, $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ follows a multivariate normal distribution $\mathcal{N}_m(\boldsymbol{\mu}_Y, \boldsymbol{\Sigma}_Y)$ where

$$\boldsymbol{\mu}_Y = \mathsf{E}(\mathbf{Y}) = \mathbf{b} \quad \text{and} \quad \boldsymbol{\Sigma}_Y = \mathsf{Cov}(\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}^\top = \mathbf{A}\mathbf{A}^\top,$$

with the second inequality following from (2.4). When $\mathbf{\Sigma}_Y$ is invertible, $\mathbf{Y}$ has the following density function

$$f_{\mathbf{Y}}(\mathbf{y}; \boldsymbol{\mu}_Y, \mathbf{\Sigma}_Y) = \frac{1}{\sqrt{(2\pi)^m |\mathbf{\Sigma}_Y|}} \exp\left[-\frac{1}{2}(\mathbf{y}-\boldsymbol{\mu}_Y)^\top \mathbf{\Sigma}_Y^{-1}(\mathbf{y}-\boldsymbol{\mu}_Y)\right],$$

where $|\mathbf{\Sigma}_Y|$ denotes the determinant of $\mathbf{\Sigma}_Y$. Suppose that $\mathbf{Y}$ is partitioned into two parts, $\mathbf{Y} = \begin{bmatrix}\mathbf{Y}_1 \\ \mathbf{Y}_2\end{bmatrix}$. Then, we write

$$\begin{bmatrix}\mathbf{Y}_1 \\ \mathbf{Y}_2\end{bmatrix} \sim \mathcal{N}_m\left(\begin{bmatrix}\boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2\end{bmatrix}, \begin{bmatrix}\mathbf{\Sigma}_{11} & \mathbf{\Sigma}_{12} \\ \mathbf{\Sigma}_{21} & \mathbf{\Sigma}_{22}\end{bmatrix}\right)$$

where $\mathbf{\Sigma}_{ii} = \mathsf{Cov}(\mathbf{Y}_i)$, $i = 1, 2$ and $\mathbf{\Sigma}_{12} = \mathbf{\Sigma}_{21}^\top = \mathsf{Cov}(\mathbf{Y}_1, \mathbf{Y}_2)$. The random vectors $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are *independent* if and only if

$$\mathsf{Cov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{\Sigma}_{12} = \underline{\quad\mathbf{O}.\quad}$$

The linear transform of $\mathbf{Y}$ also follows a multivariate normal distribution, i.e.

$$\mathbf{C}\mathbf{Y} + \mathbf{d} \sim \mathcal{N}_q(\mathbf{C}\boldsymbol{\mu}_Y + \mathbf{d}, \mathbf{C}\mathbf{\Sigma}_Y\mathbf{C}^\top) \tag{2.5}$$

for any $q \times m$ matrix $\mathbf{C}$ and $\mathbf{d} \in \mathbb{R}^q$.

2.3.2 Quadratic form

Let $\mathbf{Y} \sim \mathcal{N}_p(\boldsymbol{\mu}_Y, \mathbf{\Sigma}_Y)$ and suppose that $\mathbf{\Sigma}_Y$ is invertible. Then

$$(\mathbf{Y}-\boldsymbol{\mu}_Y)^\top \mathbf{\Sigma}_Y^{-1}(\mathbf{Y}-\boldsymbol{\mu}_Y) \sim \chi^2_p.$$

Proof. Denote the eigendecomposition of $\mathbf{\Sigma}_Y$ as $\mathbf{\Sigma}_Y = \mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top$, with the notations as in page 3. Then,

$$\mathbf{\Sigma}_Y^{-1} = (\mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top)^{-1} = (\mathbf{E}^\top)^{-1}\mathbf{\Lambda}^{-1}\mathbf{E}^{-1} = \underline{\quad\mathbf{E}\mathbf{\Lambda}^{-1}\mathbf{E}^\top\quad} =: \mathbf{\Gamma}\mathbf{\Gamma}^\top$$

since $\mathbf{E}\mathbf{E}^\top = \mathbf{I}_p$ and thus $\underline{\quad\mathbf{E}^{-1} = \mathbf{E}^\top\quad}$, where with $\mathbf{\Lambda}^{-1} = \mathrm{diag}(\lambda_1^{-1}, \ldots, \lambda_p^{-1})$ and

$\boldsymbol{\Gamma} = \mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top$. Therefore,

$$(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top \boldsymbol{\Sigma}_Y^{-1}(\mathbf{Y} - \boldsymbol{\mu}_Y) = (\boldsymbol{\Gamma}^\top(\mathbf{Y} - \boldsymbol{\mu}_Y))^\top \boldsymbol{\Gamma}^\top(\mathbf{Y} - \boldsymbol{\mu}_Y) = \mathbf{X}^\top\mathbf{X},$$

where $\mathbf{X} = \boldsymbol{\Gamma}^\top(\mathbf{Y} - \boldsymbol{\mu}_Y)$ satisfies

$$\begin{aligned}
\mathsf{E}(\mathbf{X}) &= \underline{\qquad \boldsymbol{\Gamma}^\top(\mathsf{E}(\mathbf{Y}) - \boldsymbol{\mu}_Y) = \mathbf{0} \qquad} \quad \text{and} \\
\mathsf{Cov}(\mathbf{X}) &= \boldsymbol{\Gamma}^\top \mathsf{Cov}(\mathbf{Y})\, \boldsymbol{\Gamma} = \boldsymbol{\Gamma}^\top \boldsymbol{\Sigma}_Y \boldsymbol{\Gamma} \\
&= \underline{\qquad (\mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top)^\top \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^\top(\mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top) \qquad} = \mathbf{I}_p, \quad \text{thus} \\
\mathbf{X} &\sim \underline{\qquad \mathcal{N}_p(\mathbf{0}, \mathbf{I}_p) \qquad} \quad \text{by (2.5).}
\end{aligned}$$

Therefore, we may write

$$(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top \boldsymbol{\Sigma}_Y^{-1}(\mathbf{Y} - \boldsymbol{\mu}_Y) = \mathbf{X}^\top\mathbf{X} = \sum_{i=1}^{p} X_i^2 \sim \chi_p^2$$

from that $\mathbf{X} = (X_1, \ldots, X_p)^\top$ with $\underline{\qquad X_i \sim_{\text{i.i.d}} \mathcal{N}(0,1), \qquad}$ by the definition of χ_p^2, the Chi-squared distribution with p degrees of freedom. □

3 Linear models

3.1 What is a linear model?

Let us suppose that we have collected n observed pairs $(Y_i, \mathbf{x}_i)$, $1 \leq i \leq n$, of the scalar response Y_i and the vector of covariances $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top$ for some $p \geq 1$. For the i-th observation, Y_i typically represents the outcome of an experiment or an observational study and x_{ij}, $1 \leq j \leq p$, the attributes of the i-th observation that are possibly related to Y_i. In a linear model, Y_i is related to $\mathbf{x}_i$ linearly in the vector of parameters $\boldsymbol{\beta}_0 = (\beta_1, \beta_2, \ldots, \beta_p)^\top$ and the random error ε_i, so that

$$Y_i = \sum_{j=1}^{p} \beta_j x_{ij} + \varepsilon_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i.$$

It is conventional to allow for the intercept by letting $x_{i1} = 1$ for all i, and often we assume that ε_i's are i.i.d. random variables with $\mathsf{E}(\varepsilon_i) = 0$ and $\mathsf{Var}(\varepsilon_i) = \sigma^2$, which we will denote by $\varepsilon_i \sim_{\text{i.i.d}} (0, \sigma^2)$. We can re-write the model as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}, \tag{3.1}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

The *design matrix* or *model matrix* $\mathbf{X}$ is an $n \times p$ matrix, whose rows represent the vectors of covariates $\mathbf{x}_i$. The j-th column of $\mathbf{X}$ contains all the observations made on the j-th variable included in the model, for $j = 1, \ldots, p$.

Throughout this course, unless stated otherwise (as in Section 8), we assume that $n > p$ and $\mathbf{X}$ has full column rank ($\text{rank}(\mathbf{X}) = p$) such that $\mathbf{X}^\top\mathbf{X}$ is invertible. Also, $\mathbf{X}$ (and thus $\mathbf{x}_i$) is treated as fixed rather than random. Then, the response variable Y_i satisfies $\mathsf{E}(Y_i) = \mu_i =$ $\mathbf{x}_i^\top\boldsymbol{\beta}_0$ and $\mathsf{Var}(Y_i) =$ σ^2; sometimes we write

$$Y_i \sim (\mu_i, \sigma^2) \quad \text{or, equivalently,} \quad \mathbf{Y} \sim (\boldsymbol{\mu}, \sigma^2\mathbf{I}_n) \quad \text{with} \quad \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}_0.$$

For the purpose of statistical inference, such as testing hypotheses about β_j and deriving

their confidence intervals, we may further assume that the random vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^\top$ follows a multivariate normal distribution,

$$\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

Under this additional assumption of the normality on ε_i, we have

$$Y_i \sim \mathcal{N}(\mu_i, \sigma^2) \quad \text{or, equivalently,} \quad \underline{\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n).}$$

While the observations $(Y_i, \mathbf{x}_i)$, $1 \leq i \leq n$, are typically assumed to be *independent* of one another, the covariates are often correlated so that $\mathbf{X}^\top \mathbf{X}$ is not diagonal.
Covariates, a.k.a. explanatory variables or regressors, can be either *numeric* or *categorical*.

(i) Numeric variables take on meaningful numeric quantities. They are further classified into *continuous* and *discrete* variables. Continuous variables can take values anywhere in $\mathbb{R}$, whilst discrete variables can only take values from a countable set, e.g. $\mathbb{N}^+ = \{0\} \cup \mathbb{N} = \{0, 1, 2, \ldots\}$.

(ii) Categorical variables, sometimes called *factors*, are not numeric and take on specified levels, e.g. house type $\in$ {bungalow, detached, semi-detached}. If the levels are ordered, we refer to the variable as an *ordered categorical variable*, e.g. house size $\in$ {small, medium, large}. While it is clear that small $<$ medium $<$ large, we do not know e.g. how much a medium-sized house is larger than a small-sized house.

Example 3.1 (Construction of $\mathbf{X}$)**.** Suppose that we observe the response Y_i, $i = 1, \ldots, n$, with $n = 9$, along with numeric variables w_i and z_i, and a factor variable g_i that contains three levels partitioning Y_i into three groups. For convenience, let $g_1 = g_2 = g_3 = 1$, $g_4 = g_5 = g_6 = 2$ and $g_7 = g_8 = g_9 = 3$ (we can always arrange the index so that this holds). We further assume that the following model is considered to be appropriate for the data, with some (unknown) parameters $(\delta, \mu_1, \mu_2, \mu_3, \beta_1, \beta_2, \beta_3, \beta_4)$:

$$Y_i = \delta + \mu_{g_i} + \beta_1 w_i + \beta_2 z_i + \beta_3 z_i^2 + \beta_4 w_i z_i + \varepsilon_i, \quad i = 1, \ldots, n.$$

Then, we may re-write the model as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon},$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \\ . \\ . \\ . \\ Y_9 \end{bmatrix} = \underline{\begin{bmatrix} 1 & 1 & 0 & 0 & w_1 & z_1 & z_1^2 & w_1 z_1 \\ 1 & 1 & 0 & 0 & w_2 & z_2 & z_2^2 & w_2 z_2 \\ 1 & 1 & 0 & 0 & w_3 & z_3 & z_3^2 & w_3 z_3 \\ 1 & 0 & 1 & 0 & w_4 & z_4 & z_4^2 & w_4 z_4 \\ . & . & . & . & . & . & . & . \\ . & . & . & . & . & . & . & . \\ . & . & . & . & . & . & . & . \\ 1 & 0 & 0 & 1 & w_9 & z_9 & z_9^2 & w_9 z_9 \end{bmatrix}} \underline{\begin{bmatrix} \delta \\ \mu_1 \\ \mu_2 \\ \mu_3 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix}} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ . \\ . \\ . \\ \varepsilon_9 \end{bmatrix}.$$

Notice how each of the factor levels/groups is assigned an indicator variable that corresponds to a column in $\mathbf{X}$, where $x_{ij} = 1$ for $j = 2, 3, 4$, indicates the membership of the i-th observation to the $(j-1)$-th group. In other words, $x_{ij} \in \{0, 1\}$ and $\sum_{j=2}^{4} x_{ij} =$ __1__ for $j = 2, 3, 4$. Also, w_i and z_i have entered the model non-linearly. This shows that the model is linear in the parameters and error, but not necessarily in the covariates. Notice that $\mathbf{X}$ is __not__ full rank. While *apparently* the model contains $p = 8$ parameters to estimate, an important question is whether such a model is *identifiable*. If we set e.g. $\mu_1 = 0$ and remove the column of $\mathbf{X}$ corresponding to μ_1, we have a full rank matrix with $p = 7$ parameters to estimate. We re-visit the handling of factor variables in Section 3.3 where we will learn that this approach does not lose any modelling power.

3.2 Least squares estimation

3.2.1 Least squares estimation and its geometry

An estimator of the vector of model parameters $\boldsymbol{\beta}_0$ can be obtained by the method of least squares, which is one of the most widely used method of estimation besides the maximum likelihood estimation. The aim is to find some $\boldsymbol{\beta} \in \mathbb{R}^p$ which minimise the 'distance' between $\mathbf{Y}$ and $\mathbf{X}\boldsymbol{\beta}$, by minimising the residual sum of squares (RSS)

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{n} (Y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \underline{\quad \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 \quad}$$

with respect to $\boldsymbol{\beta}$. Here, $\|\cdot\|$ denotes the Euclidean norm. By identifying the minimiser of $S(\boldsymbol{\beta})$, say $\widehat{\boldsymbol{\beta}}$, we find the linear combination of the column vectors of $\mathbf{X}$, say $\mathbf{X}\widehat{\boldsymbol{\beta}}$, which __minimises__ the Euclidean distance between $\mathbf{Y}$ and $\mathbf{X}\boldsymbol{\beta}$ among all possible $\boldsymbol{\beta} \in \mathbb{R}^p$.

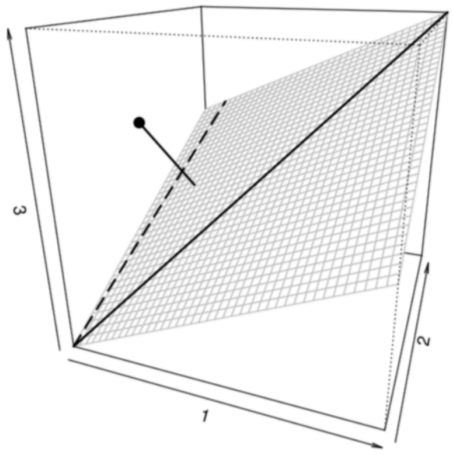


Figure 3.1: Taken from Wood (2015). With $n = 3$ and $p = 2$, the point represents $\mathbf{Y}$ while the solid and broken lines represent the vectors in the directions of the two columns of $\mathbf{X}$, and the subspace spanned by the two vectors is shaded in grey. The projection of $\mathbf{Y}$ on this subspace represents $\mathbf{X}\widehat{\boldsymbol{\beta}}$. See also the main text.

Figure 3.1 illustrates this with a toy example, where

$$\mathbf{Y} = \begin{bmatrix} 0.05 \\ 0.40 \\ 0.65 \end{bmatrix} \quad \text{and} \quad \mathbf{X} = \begin{bmatrix} 1 & 0.1 \\ 1 & 1 \\ 1 & 0.6 \end{bmatrix}.$$

The vector $\mathbf{Y}$ is represented by a single point, and the columns of the model matrix span the subspace shown in grey. The least squares estimator of $\boldsymbol{\mu} = \mathsf{E}(\mathbf{Y})$, say $\widehat{\boldsymbol{\mu}} = \mathbf{X}\widehat{\boldsymbol{\beta}}$, is the orthogonal projection of the data point $\mathbf{Y}$ onto this subspace, i.e. $\widehat{\boldsymbol{\mu}}$ is the point in the subspace spanned by the columns of $\mathbf{X}$ which is the closest to $\mathbf{Y}$ according to the Euclidean distance.

Besides having the nice geometrical interpretation, the least squares estimator of $\boldsymbol{\beta}_0$ has an analytical expression. Let us re-write the residual sum of squares as

$$\begin{aligned} S(\boldsymbol{\beta}) &= \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}. \end{aligned}$$

Since the transpose of a scalar is also a scalar, we have $\mathbf{Y}^\top\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{Y}$ and therefore

$$S(\boldsymbol{\beta}) = \underline{\quad \mathbf{Y}^\top\mathbf{Y} - 2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{Y} + \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}. \quad}$$

$S(\boldsymbol{\beta})$ is a *convex* function in $\boldsymbol{\beta}$, and therefore the least squares estimator minimising the residual sum of squares, say $\widehat{\boldsymbol{\beta}}$, is obtained as a solution to the following equation:

$$\frac{\partial}{\partial\boldsymbol{\beta}}S(\widehat{\boldsymbol{\beta}}) = \underline{\quad -2\mathbf{X}^\top\mathbf{Y} + 2\mathbf{X}^\top\mathbf{X}\widehat{\boldsymbol{\beta}} = 0, \quad}$$

which follows from

$$\frac{\partial}{\partial\mathbf{x}}\mathbf{x}^\top\mathbf{a} = \mathbf{a} \quad \text{and} \quad \frac{\partial}{\partial\mathbf{x}}\mathbf{x}^\top\mathbf{A}\mathbf{x} = 2\mathbf{A}\mathbf{x}.$$

Solving this equation gives

$$\widehat{\boldsymbol{\beta}} = \underline{\quad (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}. \quad} \tag{3.2}$$

From the assumption $\boldsymbol{\varepsilon} \sim (\mathbf{0}, \sigma^2\mathbf{I}_n)$, from which it follows that $\mathbf{Y} \sim (\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$, so that

$$\begin{aligned}
\mathsf{E}(\widehat{\boldsymbol{\beta}}) &= \underline{\quad (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathsf{E}(\mathbf{Y}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}_0 \quad} = \boldsymbol{\beta}_0, \\
\mathsf{Cov}(\widehat{\boldsymbol{\beta}}) &= \underline{\quad (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathsf{Cov}(\mathbf{Y})\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} \quad} \\
&= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.
\end{aligned} \tag{3.3}$$

Note that $\widehat{\boldsymbol{\beta}}$ is a *linear* transformation of $\mathbf{Y}$. Hence, under the additional assumption of the normality, i.e. $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2\mathbf{I}_n)$ (and thus $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$), we have $\widehat{\boldsymbol{\beta}}$ also follow a normal distribution (see page 5),

$$\widehat{\boldsymbol{\beta}} \sim \mathcal{N}_p(\boldsymbol{\beta}_0, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}).$$

The point estimator of $\mathsf{E}(\mathbf{Y})$ is obtained as

$$\begin{aligned}
\widehat{\boldsymbol{\mu}} = \mathbf{X}\widehat{\boldsymbol{\beta}} &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \\
&= \text{projection of } \mathbf{Y} \text{ onto } \underline{\quad \text{the column space of } \mathbf{X} \quad}.
\end{aligned}$$

The so-called 'hat' matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ (a.k.a. influence matrix), with which we

can write $\widehat{\boldsymbol{\mu}} = \mathbf{HY}$, plays an important role as discussed later.

3.2.2 Estimation using QR decomposition

The analytical representation of the least squares estimator $\widehat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ involves inverting the $p \times p$-matrix $\mathbf{X}^\top\mathbf{X}$ which, due to the potentially high computational cost, should be avoided if possible particularly when p is large. In this section, we represent $\widehat{\boldsymbol{\beta}}$ using the QR decomposition of $\mathbf{X}$, which provides a more efficient way to compute the estimator. Additionally, it allows us to investigate the distribution of $\widehat{\boldsymbol{\beta}}$ with ease, with which we perform statistical inference.
Recall the method of least squares for estimating $\boldsymbol{\beta}_0$, which minimises

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$$

with respect to $\boldsymbol{\beta} \in \mathbb{R}^p$. Since $S(\boldsymbol{\beta})$ is simply the squared (Euclidean) length of the vector $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$, its value will be unchanged if $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$ is rotated or reflected. Next, recall that $\mathbf{X}$ (with $n \geq p$) admits the following QR decomposition (see (2.1) on page 3):

$$\mathbf{X} = \mathbf{Q}\begin{bmatrix}\mathbf{R}\\ \mathbf{O}\end{bmatrix} = \mathbf{Q}_f\mathbf{R}$$

where $\mathbf{R}$ is a $p \times p$ upper triangular matrix and $\mathbf{Q}$ is an $n \times n$ orthogonal matrix ($\mathbf{QQ}^\top = \mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_n$), the first p columns of which form its $n \times p$ sub-matrix $\mathbf{Q}_f$. Orthogonal matrices rotate/reflect vectors, but do not change their length, i.e.

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\| = \|\mathbf{Q}^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\|.$$

Multiplying $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$ by $\mathbf{Q}^\top$,

$$\mathbf{Q}^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{Q}^\top\mathbf{Y} - \mathbf{Q}^\top\mathbf{Q}\begin{bmatrix}\mathbf{R}\\ \mathbf{O}\end{bmatrix}\boldsymbol{\beta} = \underline{\begin{bmatrix}\mathbf{f}\}_p\\ \mathbf{r}\}_{n-p}\end{bmatrix} - \begin{bmatrix}\mathbf{R}\boldsymbol{\beta}\\ \mathbf{0}\end{bmatrix}},$$

$$\therefore \quad S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \underline{\|\mathbf{f} - \mathbf{R}\boldsymbol{\beta}\|^2 + \|\mathbf{r}\|^2.}$$

Since $\|\mathbf{r}\|^2$ does not involve $\boldsymbol{\beta}$, we have $S(\widehat{\boldsymbol{\beta}})$ attain the minimum value by solving $\|\mathbf{f} -$

$\mathbf{R}\boldsymbol{\beta}\|^2 = 0$, which leads to the following representation of the least squares estimator

$$\widehat{\boldsymbol{\beta}} = \underline{\quad \mathbf{R}^{-1}\mathbf{f}. \quad}$$

This gives $\|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2 = \underline{\quad \|\mathbf{r}\|^2, \quad}$ the RSS for the model fit.
It is easily shown that, from $\mathbf{X} = \mathbf{Q}_f\mathbf{R}$,

$$(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} = (\mathbf{R}^\top\mathbf{Q}_f^\top\mathbf{Q}_f\mathbf{R})^{-1}\mathbf{R}^\top\mathbf{Q}_f^\top\mathbf{Y} = \mathbf{R}^{-1}(\mathbf{R}^\top)^{-1}\mathbf{R}^\top\mathbf{f} = \mathbf{R}^{-1}\mathbf{f},$$

i.e. the estimator via QR decomposition agrees with that obtained in page 12.

3.2.3 Estimation of σ^2

Under the assumption that $\boldsymbol{\varepsilon} \sim (\mathbf{0}, \sigma^2\mathbf{I}_n)$ or, equivalently $\mathbf{Y} \sim (\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$, it follows that

$$\mathsf{E}(\mathbf{Q}^\top\mathbf{Y}) = \mathbf{Q}^\top\mathsf{E}(\mathbf{Y}) = \underline{\quad \mathbf{Q}^\top\mathbf{X}\boldsymbol{\beta}_0 = \mathbf{Q}^\top\mathbf{Q}\begin{bmatrix}\mathbf{R}\\ \mathbf{O}\end{bmatrix}\boldsymbol{\beta}_0 = \begin{bmatrix}\mathbf{R}\boldsymbol{\beta}_0\\ \mathbf{0}\end{bmatrix}, \quad}$$

$$\mathsf{Cov}(\mathbf{Q}^\top\mathbf{Y}) = \underline{\quad \sigma^2\mathbf{Q}^\top\mathbf{I}_n\mathbf{Q} = \sigma^2\mathbf{I}_n \quad} \text{ (see (2.4))},$$

$$\therefore \quad \mathbf{Q}^\top\mathbf{Y} = \begin{bmatrix}\mathbf{f}\\ \mathbf{r}\end{bmatrix} \sim \left(\begin{bmatrix}\mathbf{R}\boldsymbol{\beta}_0\\ \mathbf{0}\end{bmatrix}, \begin{bmatrix}\sigma^2\mathbf{I}_p & \mathbf{O}\\ \mathbf{O} & \sigma^2\mathbf{I}_{n-p}\end{bmatrix}\right),$$

which leads to

$$\mathbf{f} \sim (\mathbf{R}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_p), \quad \mathbf{r} \sim (\mathbf{0}, \sigma^2\mathbf{I}_{n-p}) \quad \text{and} \quad \mathsf{Cov}(\mathbf{f}, \mathbf{r}) = \mathbf{O}_{p\times(n-p)}.$$

Therefore, the RSS $S(\widehat{\boldsymbol{\beta}}) = \|\mathbf{r}\|^2$ satisfies

$$\mathsf{E}(\|\mathbf{r}\|^2) = (n-p)\sigma^2,$$

which suggests

$$\widehat{\sigma}^2 = \underline{\quad \frac{\|\mathbf{r}\|^2}{n-p} \quad} = \frac{S(\widehat{\boldsymbol{\beta}})}{n-p}, \tag{3.4}$$

as an unbiased estimator of σ^2.

3.2.4 The influence matrix

The influence matrix (sometimes called the *hat* matrix) is a useful tool in model diagnosis, which will be discussed in Section 5. It is defined as the matrix that yields the vector of *fitted values* $\widehat{\boldsymbol{\mu}}$ (a point estimator of $\boldsymbol{\mu} = \mathsf{E}(\mathbf{Y})$) when right-multiplied by $\mathbf{Y}$, i.e. $\widehat{\boldsymbol{\mu}} = \mathbf{H}\mathbf{Y} = \mathbf{X}\widehat{\boldsymbol{\beta}}$. From that $\widehat{\boldsymbol{\beta}} = \mathbf{R}^{-1}\mathbf{f} = \mathbf{R}^{-1}\mathbf{Q}_f^\top\mathbf{Y}$, we have

$$\widehat{\boldsymbol{\mu}} = \mathbf{X}\widehat{\boldsymbol{\beta}} = \underline{\qquad \mathbf{Q}_f\mathbf{R}\mathbf{R}^{-1}\mathbf{Q}_f^\top\mathbf{Y} = \mathbf{Q}_f\mathbf{Q}_f^\top\mathbf{Y}, \qquad}$$

such that the influence matrix can be written as $\mathbf{H} = \underline{\qquad \mathbf{Q}_f\mathbf{Q}_f^\top \qquad}$. Also, as seen in page 12, we can write $\mathbf{H}$ in terms of $\mathbf{X}$, namely $\mathbf{H} = \underline{\qquad \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \qquad}$.
The influence matrix $\mathbf{H}$ has several interesting properties:

(i) The trace of $\mathbf{H}$ is the number of identifiable parameters in the model,

$$\mathsf{tr}(\mathbf{H}) = \mathsf{tr}(\mathbf{Q}_f\mathbf{Q}_f^\top) = \underline{\qquad \mathsf{tr}(\mathbf{Q}_f^\top\mathbf{Q}_f) = \mathsf{tr}(\mathbf{I}_p) = p \qquad}$$

using the orthogonality of $\mathbf{Q}_f$. This coincides with the *rank* of $\mathbf{H}$.

(ii) $\mathbf{H}$ is *idempotent*,

$$\mathbf{H}\mathbf{H} = \underline{\qquad \mathbf{Q}_f\mathbf{Q}_f^\top\mathbf{Q}_f\mathbf{Q}_f^\top = \mathbf{Q}_f\mathbf{Q}_f^\top = \mathbf{H}. \qquad}$$

(iii) $\mathbf{H}$ is a *projection* matrix onto the space spanned by the columns of $\mathbf{X}$, and hence

$$\mathbf{H}\mathbf{X} = \underline{\qquad \mathbf{Q}_f\mathbf{Q}_f^\top\mathbf{Q}_f\mathbf{R} = \mathbf{Q}_f\mathbf{R} = \mathbf{X}. \qquad}$$

Additionally, the matrix $\mathbf{I}_n - \mathbf{H}$ is also a projection matrix onto a space *orthogonal* to the space spanned by the columns of $\mathbf{X}$, and satisfies

(i') $\mathsf{tr}(\mathbf{I}_n - \mathbf{H}) = n - p$,

(ii') $(\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H}) = (\mathbf{I}_n - \mathbf{H})$,

(iii') $(\mathbf{I}_n - \mathbf{H})\mathbf{X} = \mathbf{O}$.

3.3 Factors

Suppose that we observe (Y_i, g_i), $i = 1, \ldots, n = 6$, where g_i denotes the factor variable with three levels. For convenience, let $g_1 = g_2 = 1$, $g_3 = g_4 = 2$ and $g_5 = g_6 = 3$. We consider the following model

$$\mathsf{E}(Y_i) = \delta + \beta_{g_i}, \tag{3.5}$$

where the mean of the i-th response variable is determined by the factor level g_i. If we write the design matrix and the vector of linear coefficients as

$$\widetilde{\mathbf{X}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \widetilde{\boldsymbol{\beta}}_0 = \begin{bmatrix} \delta \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix},$$

we encounter the problem of rank deficiency, i.e. the matrix $\widetilde{\mathbf{X}}$ does not have the full column rank, such that $\widetilde{\mathbf{X}}^\top \widetilde{\mathbf{X}}$ is not invertible. This problem originates from that the model in (3.5) is not *identifiable*, i.e. we cannot determine $\widetilde{\boldsymbol{\beta}}_0$ uniquely. To see this, for any constant α, we have

$$\delta + \beta_{g_i} = \delta' + \beta'_{g_i}, \quad \text{where} \quad \delta' = \delta + \alpha \quad \text{and} \quad \beta'_{g_i} = \beta_{g_i} - \alpha,$$

regardless of the true values of β_j. In other words, there are *infinitely many* $\widetilde{\boldsymbol{\beta}}_0 = (\delta, \beta_1, \beta_2, \beta_3)^\top$ that lead to identical $\mathsf{E}(\mathbf{Y})$ under the model (3.5), unless an extra constraint is imposed.

One remedy for this issue is to remove one of the columns of $\widetilde{\mathbf{X}}$, treating the corresponding

parameter as zero. For example, setting $\beta_1 = 0$ gives

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{\beta}_0 = \begin{bmatrix} \delta \\ \beta_2 \\ \beta_3 \end{bmatrix},$$

where $\mathbf{X}$ is of full column rank. By default, `R` handles factor variables by dropping the column of $\mathbf{X}$ that correspond to the ___first___ level of each factor, in this example, by setting $\beta_1 = 0$. This is by no means the only way of handling factor variables; for example, we may assume that $\sum_{j=1}^{3} \beta_j = 0$, which also leads to an identifiable model. In the single factor example in (3.5), δ represents the mean of Y_i's corresponding to the level 1, while β_j quantifies the differences between the effects of levels j and 1 for each $j = 2, 3$. More generally, adding a single factor with K levels to the model adds ___$K-1$___ parameters to estimate.

3.4 Multiple R^2 and adjusted R^2

The multiple R^2 is an estimate of the proportion of the variance in the response that is explained by the regression: Given the observations $(Y_i, \mathbf{x}_i)$, $1 \leq i \leq n$,

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(Y_i - \widehat{\mu}_i)^2/n}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2/n} \quad \text{where} \quad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i \quad \text{and} \quad \widehat{\mu}_i = \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}.$$

This fraction accounts for the *proportion* of the sample variance of Y_i, $1 \leq i \leq n$, that is not explained by the linear model, and it always holds that $0 \leq R^2 \leq 1$. The closer R^2 is to one, the more of variability in Y_i, $1 \leq i \leq n$, is explained by the fitted model.

An issue with R^2 is that it always ___*increases*___ when a new variable is added to the model, no matter how relevant that variable is for predicting Y_i (mathematical explanation is easily given by the arguments in Section 3.6.3). Note that as an estimator of $\sigma^2 = \mathsf{Var}(\varepsilon_i)$, the term $n^{-1}\sum_{i=1}^{n}(Y_i - \widehat{\mu}_i)^2$ is biased (see (3.4)) and tends to inflate R^2. Hence the *adjusted*

R^2 is defined as

$$R^2_{\text{adj}} = 1 - \frac{\sum_{i=1}^n (Y_i - \widehat{\mu}_i)^2 / (\quad n-p \quad)}{\sum_{i=1}^n (Y_i - \bar{Y})^2 / (\quad n-1 \quad)}.$$

A large value of R^2_{adj} indicates that the model well-explains the the variability in the response variable. Note that while R^2 always lies between 0 and 1 (proportion), it is not the case with R^2_{adj}.

3.5 Gauss-Markov theorem

Minimising the residual sum of squares, $S(\boldsymbol{\beta})$, is closely related to maximising the log-likelihood under the normality assumption $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ [1] which leads to $\widehat{\boldsymbol{\beta}}$ being the maximum likelihood estimator. However, even *without* assuming the normality, the Gauss-Markov theorem states that $\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} S(\boldsymbol{\beta})$ is the best (minimum variance) linear unbiased estimator (BLUE) of $\boldsymbol{\beta}_0$.

Theorem 3.1 (Gauss-Markov theorem)**.** Suppose $\mathbf{Y} \sim (\mathbf{X}\boldsymbol{\beta}_0, \sigma^2 \mathbf{I}_n)$. Let $\mathbf{a}^\top \mathbf{Y}$ be any unbiased estimator of $\mathbf{c}^\top \boldsymbol{\beta}_0$ for some $\mathbf{c} \in \mathbb{R}^p$, then

$$\mathsf{Var}(\mathbf{a}^\top \mathbf{Y}) \geq \mathsf{Var}(\mathbf{c}^\top \widehat{\boldsymbol{\beta}}).$$

Proof. By assumption,

$$\begin{aligned} \mathsf{E}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \mathbf{X} \boldsymbol{\beta}_0 &= \mathbf{c}^\top \boldsymbol{\beta}_0, \\ \therefore \ \mathbf{a}^\top \mathbf{X} &= \mathbf{c}^\top, \end{aligned}$$

and therefore, recalling that $\mathsf{Cov}(\widehat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ from (3.3),

$$\mathsf{Var}(\mathbf{c}^\top \widehat{\boldsymbol{\beta}}) = \mathbf{c}^\top \mathsf{Cov}(\widehat{\boldsymbol{\beta}}) \mathbf{c} = \sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c} = \sigma^2 \mathbf{a}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{a} = \underline{\quad \sigma^2 \mathbf{a}^\top \mathbf{H} \mathbf{a}. \quad}$$

Also note that

$$\mathsf{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \mathsf{Cov}(\mathbf{Y}) \mathbf{a} = \sigma^2 \mathbf{a}^\top \mathbf{a}.$$

[1] Verify this in problem sheet Q3.

The variances of the two estimators can directly be compared, as

$$\begin{aligned}\mathsf{Var}(\mathbf{a}^\top\mathbf{Y}) - \mathsf{Var}(\mathbf{c}^\top\widehat{\boldsymbol{\beta}}) &= \sigma^2\mathbf{a}^\top(\mathbf{I}_n - \mathbf{H})\mathbf{a} \\ &= \sigma^2((\mathbf{I}_n - \mathbf{H})\mathbf{a})^\top(\mathbf{I}_n - \mathbf{H})\mathbf{a} = \sigma^2\|(\mathbf{I}_n - \mathbf{H})\mathbf{a}\|^2 \geq 0,\end{aligned}$$

using the fact that $\mathbf{I}_n - \mathbf{H}$ is idempotent and symmetric, see (ii') in Section 3.2.4. □

Notice that $\mathbf{c}^\top\widehat{\boldsymbol{\beta}} = \mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ is a linear combination of $\mathbf{Y}$. The theorem thus states that if you can find another unbiased and linear estimator of $\mathbf{c}^\top\boldsymbol{\beta}_0$ in the form of $\mathbf{a}^\top\mathbf{Y}$, its variance must be at least as large as that of $\mathbf{c}^\top\widehat{\boldsymbol{\beta}}$. Also important is what is *not* mentioned in the theorem: The theorem is valid as long as $\mathsf{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}_n$ but it does not require normality.

3.6 Hypothesis testing

3.6.1 The distribution of the estimators

The **unbiasedness** of $\widehat{\boldsymbol{\beta}} = \mathbf{R}^{-1}\mathbf{f}$ follows as

$$\mathsf{E}(\widehat{\boldsymbol{\beta}}) = \mathsf{E}(\mathbf{R}^{-1}\mathbf{f}) = \mathbf{R}^{-1}\mathsf{E}(\mathbf{f}) = \mathbf{R}^{-1}\mathbf{R}\boldsymbol{\beta}_0 = \boldsymbol{\beta}_0.$$

Furthermore,

$$\mathsf{Cov}(\widehat{\boldsymbol{\beta}}) = \mathbf{R}^{-1}\mathsf{Cov}(\mathbf{f})(\mathbf{R}^{-1})^\top = \mathbf{R}^{-1}(\sigma^2\mathbf{I}_p)(\mathbf{R}^{-1})^\top = \sigma^2\mathbf{R}^{-1}(\mathbf{R}^{-1})^\top.$$

Under the additional normality assumption, $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2\mathbf{I}_n)$, we have $\mathbf{Q}^\top\mathbf{Y} = \begin{bmatrix}\mathbf{f}\\ \mathbf{r}\end{bmatrix}$, $\mathbf{f}$ and $\mathbf{r}$ follow normal distributions (of dimension n, p and $n-p$, respectively). In particular, noting that

$$\mathbf{f} \sim \mathcal{N}_p(\mathbf{R}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_p) \quad \text{and} \quad \mathbf{r} \sim \mathcal{N}_{n-p}(\mathbf{0}, \sigma^2\mathbf{I}_{n-p}), \tag{3.6}$$

its linear transform $\widehat{\boldsymbol{\beta}} = \mathbf{R}^{-1}\mathbf{f}$ also follows a multivariate normal distribution:

$$\widehat{\boldsymbol{\beta}} \sim \mathcal{N}_p(\boldsymbol{\beta}_0, \sigma^2\mathbf{R}^{-1}(\mathbf{R}^{-1})^\top). \tag{3.7}$$

This result is not directly applicable to statistical inference about $\boldsymbol{\beta}_0$ since σ^2 is typically

unknown and needs to be estimated, which introduces an extra component of uncertainty that should be accounted for. From (3.6), we have

$$\frac{\|\mathbf{r}\|^2}{\sigma^2} = \sum_{i=1}^{n-p} \frac{r_i^2}{\sigma^2} \sim \underline{\quad \chi^2_{n-p} \quad}$$

(see Section 2.3.2), which leads to

$$\frac{(n-p)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p}. \tag{3.8}$$

Finally, $\mathbf{f}$ and $\mathbf{r}$ are independent under the normality assumption since $\mathsf{Cov}(\mathbf{f}, \mathbf{r}) = \underline{\quad \mathbf{O} \quad}$. This implies that $\widehat{\boldsymbol{\beta}}$ and $\widehat{\sigma}^2$ are also $\underline{\quad \text{independent.} \quad}$

We utilise these properties of $\widehat{\boldsymbol{\beta}}$ and $\widehat{\sigma}^2$ to infer about $\boldsymbol{\beta}_0$ by answering questions such as follows:

(i) For some fixed $j = 1, \ldots, p$, does the data support that the j-th variable (associated with the j-th column of $\mathbf{X}$) contributes to explaining $\mathsf{E}(Y_i)$?
$\Longrightarrow$ Test $\mathcal{H}_0 : \beta_j = 0$ vs. $\mathcal{H}_1 : \beta_j \neq 0$ (Wald test, Section 3.6.2).

(ii) Partitioning the variables as $\mathbf{X} = \begin{bmatrix} \mathbf{X}_0 & \mathbf{X}_1 \end{bmatrix}$, does the data support that the variables corresponding to $\mathbf{X}_1$ *collectively* contribute to explaining $\mathsf{E}(\mathbf{Y})$?
$\Longrightarrow$ Partitioning $\boldsymbol{\beta}_0$ accordingly as $\boldsymbol{\beta}_0 = (\boldsymbol{\beta}_{0,0}^\top, \boldsymbol{\beta}_{0,1}^\top)^\top$, test $\mathcal{H}_0 : \boldsymbol{\beta}_{0,1} = \mathbf{0}$ vs. $\mathcal{H}_1 : \boldsymbol{\beta}_{0,1} \neq \mathbf{0}$ (F-test, Section 3.6.3).

3.6.2 Wald test

Recall that under the assumption that $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$, for some fixed $j = 1, \ldots, p$,

$$\widehat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2_{\widehat{\beta}_j}) \quad \text{with} \quad \sigma^2_{\widehat{\beta}_j} = \underline{\quad \sigma^2[\mathbf{R}^{-1}(\mathbf{R}^{-1})^\top]_{j,j}. \quad}$$

By replacing σ^2 with its unbiased estimator $\widehat{\sigma}^2 = \|\mathbf{r}\|^2/(n-p)$ in (3.4), we obtain an unbiased estimator of $\sigma_{\widehat{\beta}_j}$ as

$$\widehat{\sigma}_{\widehat{\beta}_j} = \widehat{\sigma}\sqrt{[\mathbf{R}^{-1}(\mathbf{R}^{-1})^\top]_{j,j}}$$

which, from the independence between $\widehat{\boldsymbol{\beta}}$ and $\widehat{\sigma}^2$, is independent from $\widehat{\beta}_j$.

These observations enable us to construct a confidence interval around β_j, and is the basis for testing any hypothesis about individual β_j:

$$\mathcal{H}_0 : \beta_j = 0 \qquad \text{vs.} \qquad \mathcal{H}_1 : \beta_j \neq 0.$$

We would reject $\mathcal{H}_0 : \beta_j = 0$ if $|\widehat{\beta}_j|$ is 'large' relative to the noise level. In order to construct a statistical hypothesis test, note that from (3.7) and (3.8), and the independence between $\widehat{\boldsymbol{\beta}}$ and $\widehat{\sigma}^2$, we have

$$\begin{aligned} t = \frac{\widehat{\beta}_j - \beta_j}{\widehat{\sigma}_{\widehat{\beta}_j}} = \frac{(\widehat{\beta}_j - \beta_j)/\sigma_{\widehat{\beta}_j}}{\sqrt{\widehat{\sigma}^2/\sigma^2}} &= \frac{(\widehat{\beta}_j - \beta_j)/\sigma_{\widehat{\beta}_j}}{\sqrt{\frac{1}{\sigma^2}\|\mathbf{r}\|^2/(n-p)}} \\ &\sim \frac{\mathcal{N}(0,1)}{\sqrt{\chi^2_{n-p}/(n-p)}} =_d t_{n-p}. \end{aligned} \tag{3.9}$$

Therefore, writing

$$t_{\text{obs}} = \frac{\widehat{\beta}_j}{\widehat{\sigma}_{\widehat{\beta}_j}},$$

we reject $\mathcal{H}_0 : \beta_j = 0$ at the significance level $\alpha \in (0,1)$ if the associated p-value

$$\Pr\left(|T| > |t_{\text{obs}}| \right) = \texttt{2*(1 - pt(abs(t), n-p))}$$

is less than α, where T denotes a random variable following the t_{n-p} distribution. Equivalently, we reject $\mathcal{H}_0$ if $|t_{\text{obs}}| > t_{\alpha/2,n-p}$, where $t_{\alpha/2,n-p}$ denotes the $100(1-\alpha/2)$-quantile of the t_{n-p}-distribution (i.e. $\mathsf{P}(|T| \geq t_{\alpha/2,n-p}) = \alpha$). From (3.9), it is straightforward to see that $100(1-\alpha)\%$ confidence interval for β_j can be constructed as

$$(\widehat{\beta}_j - t_{\alpha/2,n-p}\widehat{\sigma}_{\widehat{\beta}_j}, \qquad \widehat{\beta}_j + t_{\alpha/2,n-p}\widehat{\sigma}_{\widehat{\beta}_j}).$$

As we see later, the function `lm` in `R` routinely performs the Wald test for individual variables and report the outcome.

3.6.3 F-test

It is also of interest to test, for example, whether *a set of variables* makes a meaningful contribution in explaining $\mathsf{E}(Y_i)$. Such a test is particularly useful when making inference about a factor variable and interactions of multiple variables, because the contribution from each factor (or interaction) is typically represented by several parameters (recall Example 3.1). Suppose that we want to test

$$\mathcal{H}_0 : \mathsf{E}(\mathbf{Y}) = \mathbf{X}_0\boldsymbol{\beta}_{0,0} \quad \text{vs.} \quad \mathcal{H}_1 : \mathsf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}_0,$$

where $\mathbf{X}_0$ is *nested* within $\mathbf{X}$ (meaning that $\mathbf{X}_0$ is a sub-matrix of $\mathbf{X}$ containing a subset of the columns of $\mathbf{X}$). Without loss of generality, we assume that

$$\mathbf{X} = \begin{bmatrix} \underbrace{\mathbf{X}_0}_{q} & \underbrace{\mathbf{X}_1}_{p-q} \end{bmatrix} \quad \text{and} \quad \boldsymbol{\beta}_0 = \begin{bmatrix} \boldsymbol{\beta}_{0,0}\}_q \\ \boldsymbol{\beta}_{0,1}\}_{p-q} \end{bmatrix},$$

as we can always re-parameterise the model for this to hold. Then, the null hypothesis can be re-written as $\mathcal{H}_0 : \boldsymbol{\beta}_{0,1} = \mathbf{0}$.
We can write the QR decomposition of $\mathbf{X}$ in a partitioned form:

$$\begin{aligned} \mathbf{X} &= \mathbf{Q}\begin{bmatrix} \mathbf{R} \\ \mathbf{O} \end{bmatrix} \\ \implies \mathbf{Q}^\top\mathbf{X} &= \mathbf{Q}^\top\begin{bmatrix} \mathbf{X}_0 & \mathbf{X}_1 \end{bmatrix} = \begin{bmatrix} \widetilde{\mathbf{R}}_0 & \widetilde{\mathbf{R}}_1 \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \\ \implies \mathbf{Q}^\top\mathbf{X}_0 &= \begin{bmatrix} \widetilde{\mathbf{R}}_0 \\ \mathbf{O} \end{bmatrix}, \end{aligned}$$

where $\widetilde{\mathbf{R}}_0$ denotes the first q columns of $\mathbf{R}$, and $\widetilde{\mathbf{R}}_1$ the remaining $p-q$ columns. Since $\mathbf{R}$ is upper triangular, letting $\mathbf{R}_0$ denote the first q rows of $\widetilde{\mathbf{R}}_0$ (i.e. the sub-matrix consisting of the first q rows and columns of $\mathbf{R}$),

- $\mathbf{R}_0$ is also upper triangular, and
- the last $p-q$ rows of $\widetilde{\mathbf{R}}_0$ are zero.

Recalling the partitioning of $\mathbf{Q}^\top\mathbf{Y}$ into $\mathbf{f}$ and $\mathbf{r}$, let us further partition $\mathbf{f}$ into the vector of its first q elements, $\mathbf{f}_0$, and the remainder $\mathbf{f}_1$ so that $\mathbf{f}^\top = (\mathbf{f}_0^\top, \mathbf{f}_1^\top)$. Then, together

with (3.6),

$$\mathbf{f} = \begin{bmatrix} \mathbf{f}_0\}_q \\ \mathbf{f}_1\}_{p-q} \end{bmatrix} \sim \mathcal{N}_p\left(\mathbf{R}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_p\right)$$
$$= \mathcal{N}_p\left(\begin{bmatrix} \mathbf{R}_0\boldsymbol{\beta}_{0,0} \\ \mathbf{0} \end{bmatrix}, \sigma^2 \begin{bmatrix} \mathbf{I}_q & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_{p-q} \end{bmatrix}\right) \quad (\text{under } \mathcal{H}_0 : \boldsymbol{\beta}_{0,1} = \mathbf{0}). \tag{3.10}$$

Under $\mathcal{H}_0$, the least squares estimator of $\boldsymbol{\beta}_{0,0}$ is obtained via minimising $S_0(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}_0\boldsymbol{\beta}\|^2$ over $\boldsymbol{\beta} \in \mathbb{R}^q$. Rotating $\mathbf{Y} - \mathbf{X}_0\boldsymbol{\beta}$ using $\mathbf{Q}^\top$ yields

$$S_0(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}_0\boldsymbol{\beta}\|^2 = \|\mathbf{Q}^\top\mathbf{Y} - \mathbf{Q}^\top\mathbf{X}_0\boldsymbol{\beta}\|^2 = \left\| \begin{bmatrix} \mathbf{f}_0\}_q \\ \mathbf{f}_1\}_{p-q} \\ \mathbf{r}\}_{n-p} \end{bmatrix} - \begin{bmatrix} \widetilde{\mathbf{R}}_0\boldsymbol{\beta}\}_p \\ \mathbf{0}\}_{n-p} \end{bmatrix} \right\|^2$$
$$= \left\| \begin{bmatrix} \mathbf{f}_0\}_q \\ \mathbf{f}_1\}_{p-q} \\ \mathbf{r}\}_{n-p} \end{bmatrix} - \begin{bmatrix} \mathbf{R}_0\boldsymbol{\beta}\}_q \\ \mathbf{0}\}_{p-q} \\ \mathbf{0}\}_{n-p} \end{bmatrix} \right\|^2 = \underline{\qquad \|\mathbf{f}_0 - \mathbf{R}_0\boldsymbol{\beta}\|^2 + \|\mathbf{f}_1\|^2 + \|\mathbf{r}\|^2. \qquad}$$

From this, we derive that

$$\widehat{\boldsymbol{\beta}}_{0,0} = \underset{\boldsymbol{\beta}\in\mathbb{R}^q}{\arg\min}\ S_0(\boldsymbol{\beta}) = \mathbf{R}_0^{-1}\mathbf{f}_0.$$

The RSS for this null model is $\|\mathbf{f}_1\|^2 + \|\mathbf{r}\|^2$: The increase in the RSS by the amount of $\|\mathbf{f}_1\|^2$ under $\mathcal{H}_0 : \boldsymbol{\beta}_{0,1} = \mathbf{0}$, results from dropping $\mathbf{X}_1$ from the model. That is, $\|\mathbf{f}_1\|^2$ is the difference in RSS between the *full* model (under $\mathcal{H}_1$, namely $S(\widehat{\boldsymbol{\beta}})$) and the *reduced* model (under $\mathcal{H}_0$, namely $S_0(\widehat{\boldsymbol{\beta}}_{0,0})$). If $\|\mathbf{f}_1\|^2$ is 'large' relative to the level of noise, we would prefer the full model since the contribution from $\mathbf{X}_1$ in explaining $\mathsf{E}(\mathbf{Y})$ is non-negligible, and hence reject $\mathcal{H}_0$. Using that $\mathbf{f}_1 \sim \mathcal{N}_{p-q}(\mathbf{0}, \sigma^2\mathbf{I}_{p-q})$ under $\mathcal{H}_0$ from (3.10), we have

$$\sigma^{-2}\|\mathbf{f}_1\|^2 \sim_{\mathcal{H}_0} \underline{\qquad \chi^2_{p-q}. \qquad}$$

Also, we know that $\mathbf{f}_1$ and $\mathbf{r}$ are independent. Thus we construct an F-ratio statistic for testing $\mathcal{H}_0 : \boldsymbol{\beta}_{0,1} = \mathbf{0}$, as

$$F = \frac{\|\mathbf{f}_1\|^2/(p-q)}{\widehat{\sigma}^2} = \underline{\qquad \frac{\sigma^{-2}\|\mathbf{f}_1\|^2/(p-q)}{\sigma^{-2}\|\mathbf{r}\|^2/(n-p)} \qquad} \tag{3.11}$$

$$\sim_{\mathcal{H}_0} \frac{\chi^2_{p-q}/(p-q)}{\chi^2_{n-p}/(n-p)} =_d \underline{\quad F_{p-q,n-p.} \quad}$$

The associated p-value for the test is derived from the $F_{p-q,n-p}$-distribution,

$$\Pr\left(F_{p-q,n-p} > F\right) = \texttt{1 - pf(F, p-q, n-p)}.$$

When the p-value is smaller than a given significance level $\alpha \in (0,1)$ or, equivalently, when F is greater than a critical value derived from $F_{p-q,n-p}$, there is an evidence supporting that *some* elements of $\boldsymbol{\beta}_{0,1}$ are non-zero and we reject $\mathcal{H}_0$.
Recalling that

$$\|\mathbf{f}_1\|^2 = S_0(\widehat{\boldsymbol{\beta}}_{0,0}) - S(\widehat{\boldsymbol{\beta}}) = \|\mathbf{Y} - \mathbf{X}_0\widehat{\boldsymbol{\beta}}_{0,0}\|^2 - \|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2,$$

represents the difference in RSS under $\mathcal{H}_0$ and $\mathcal{H}_1$, we can write F as

$$F = \frac{(\|\mathbf{Y} - \mathbf{X}_0\widehat{\boldsymbol{\beta}}_{0,0}\|^2 - \|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2)/(\dim(\boldsymbol{\beta}_0) - \dim(\boldsymbol{\beta}_{0,0}))}{\|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2/(n - \dim(\boldsymbol{\beta}_0))}.$$

The `lm` function in `R` routinely tests

$$\mathcal{H}_0 : \mathsf{E}(Y_i) = \beta_1 \quad \text{vs.} \quad \mathcal{H}_1 : \mathsf{E}(Y_i) = \mathbf{x}_i^\top \boldsymbol{\beta}_0,$$

i.e. whether *any* of the $p-1$ covariates (other than the intercept) is associated with Y_i.

3.7 Prediction

After fitting a model to the observed $(Y_i, \mathbf{x}_i)$, $1 \leq i \leq n$, one common task is to predict the expected response from the model at a new covariate value $\mathbf{x}_*$. Let us denote by Y_* the response associated with $\mathbf{x}_*$, and $\mu_* = \mathbf{x}_*^\top \boldsymbol{\beta}_0$ the mean of Y_*. A natural estimator of μ_* is

$$\widehat{\mu}_* = \mathbf{x}_*^\top \widehat{\boldsymbol{\beta}}, \quad \text{which satisfies} \quad \widehat{\mu}_* \sim \mathcal{N}(\mathbf{x}_*^\top \boldsymbol{\beta}_0,\, \sigma^2 \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*). \tag{3.12}$$

Rather than reporting a point estimator, it is more informative to report a range of values that μ_* can take by giving a confidence interval about μ_*. From (3.12), we derive a $100(1-$

$\alpha)\%$ *confidence* interval for μ_* as

$$(\widehat{\mu}_* - t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{\mathbf{x}_*^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_*}, \quad \widehat{\mu}_* + t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{\mathbf{x}_*^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_*}).$$

A *prediction* interval is similar in spirit to a confidence interval, except that the prediction interval is designed to cover a 'moving' target, $Y_* = \mu_* + \varepsilon_*$, with the random error $\varepsilon_* \sim (0, \sigma^2)$. The prediction interval is constructed as

$$\widehat{\mu}_* \pm t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{1 + \mathbf{x}_*^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_*}.$$

Although both intervals are centerd around $\widehat{\mu}_*$, the prediction interval is wider than the confidence interval for given $\mathbf{x}_*$ and confidence level. This makes sense, since the prediction interval must take account of the tendency of Y_* to fluctuate from its mean value (due to the noise ε_*), while the confidence interval simply needs to account for the uncertainty in estimating the mean value.

3.8 `R` example

3.8.1 Basic functions for linear modelling on R

Table 3.1 contains some standard linear modelling functions on R. Typing `?function name` or `help(function name)` on `R` console loads a help page for the function which contains a more complete description with examples.

Table 3.1: Some standard linear modelling functions. Strictly, all of these functions except `lm` itself end with `.lm`, but when calling them with an object of class `lm`, this may be omitted (taken from Chapter 1 of Wood (2015)).

`lm`	Estimates a linear model by least squares. Returns a fitted object of class `lm` containing parameter estimates plus other auxiliary results for use by other functions.
`plot`	Produces model checking plots from a fitted model object.
`summary`	Produces summary information about a fitted model, including parameter estimates, associated standard errors and p-values, R^2, etc.
`anova`	Used for model comparison based on F ratio testing.
`AIC`	Extract Akaike information criterion for a model fit.
`residuals`	Extract an array of model residuals form a fitted model.
`fitted`	Extract an array of fitted values from a fitted model object.
`predict`	Obtain predicted values from a fitted model, either for new values of the variables, or for the original values. Standard errors of the predictions can also be returned.

3.8.2 Example: Prestige data

The `Prestige` data frame from the `car` package has 102 observations on the six aspects of occupations, as reported in the Census of Canada conducted in 1971. The 6 columns of the data frame are:

`education`	Average education of occupational incumbents in years.
`income`	Average income of incumbents in $.
`women`	Percentage of incumbents who are women.
`prestige`	Pineo-Porter prestige score for occupation, from a social survey conducted in the mid-1960s.
`census`	Canadian Census occupational code.
`type`	Type of occupation. A factor with levels: `bc`, Blue Collar; `prof`, Professional, Managerial, and Technical; `wc`, White Collar.

Let's install and load the package `car`, which contains the dataset in the data frame `Prestige` and have a look at the data.

```
library(car) # load the package 'car'
  # if not installed, install.packages('car')
help(Prestige) # access information about the data frame
Prestige[1:5,] # print the first 5 observations in the data frame
                    education income women prestige census type
gov.administrators      13.11  12351 11.16     68.8   1113 prof
general.managers        12.26  25879  4.02     69.1   1130 prof
accountants             12.77   9271 15.70     63.4   1171 prof
purchasing.officers     11.42   8865  9.11     56.8   1175 prof
chemists                14.62   8403 11.68     73.5   2111 prof
```

Setting `income` as the response and `education`, `women` and `prestige` as regressors, we first perform explanatory analysis to observe the relationship between the response and regressors.

```
newdata = Prestige[,c(1:4)]
pairs(newdata)
```

produces a matrix of scatterplots between pairs of the response and regressor variables, see Figure 1.1. For example, we can see how income and education are related (see first column, second row): With increasing years of education, income also increases.

R function lm

We first fit a linear model

$$\mathsf{E}(\texttt{income}_i) = \beta_1 + \beta_2 \texttt{education}_i + \beta_3 \texttt{prestige}_i + \beta_4 \texttt{women}_i$$

on this dataset and see how well it models the observed data.

```
pres.fit0 <- lm(income ~ education + prestige + women,
                                data = Prestige)
print(pres.fit0) # print the fitted regression coefficients
```

```
Coefficients:
(Intercept)    education     prestige        women
     -253.8         177.2        141.4        -50.9
```

It is not easy to interpret the intercept $= -253.8$ when all explanatory variables are recorded in different units. To have a meaningful intercept estimate, we center each variable and re-fit the linear model.

```
education.c <- scale(Prestige$education, center = TRUE, scale = FALSE)
prestige.c <- scale(Prestige$prestige, center = TRUE, scale = FALSE)
women.c <- scale(Prestige$women, center = TRUE, scale = FALSE)
newdata <- cbind(newdata, cbind(education.c, prestige.c, women.c))
names(newdata)[5:7] <- c('education.c', 'prestige.c', 'women.c')
pres.fit1 <- lm(income ~ education.c + prestige.c + women.c, data = newdata)
print(pres.fit1)
```

```
Coefficients:
(Intercept)  education.c   prestige.c      women.c
     6797.9        177.2        141.4        -50.9
```

We can observe that while centring does not alter the coefficient estimates other than the intercept, it allows us to state that the estimated income is $\$6,798$ for an occupation of the average prestige score, whose incumbents have the average number of years of education and the percentage of women at the overall average.
From the model output and the scatter plot, we can make some interesting observations:

- For any given level of education and prestige in a profession, increasing female incumbents by one percentage point in a given profession will see the average income decline by \$50.9.
- Similarly, for two occupations with identical level of education and percentage of women, the profession with the larger prestige score by one point has an extra \$141.4 in average income.

`lm` computes the test statistics for testing

$$\mathcal{H}_0 : \beta_j = 0 \quad \text{vs.} \quad \mathcal{H}_1 : \beta_j \neq 0 \quad \text{for individual } j$$

(Wald test) and

$$\mathcal{H}_0 : \beta_j = 0 \, \forall \, j \neq 1 \quad \text{vs.} \quad \mathcal{H}_1 : \text{ some } \beta_j \neq 0 \text{ for } j = 2, 3, 4$$

(F-test), as well as R^2 and R^2_{adj}. We can access the results by using `summary`.

```
summary(pres.fit1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 6797.902    254.934  26.665  < 2e-16 ***
education.c  177.199    187.632   0.944    0.347
prestige.c   141.435     29.910   4.729 7.58e-06 ***
women.c      -50.896      8.556  -5.948 4.19e-08 ***

Residual standard error: 2575 on 98 degrees of freedom
Multiple R-squared:  0.6432,Adjusted R-squared:  0.6323
F-statistic: 58.89 on 3 and 98 DF,  p-value: < 2.2e-16
```

Note that $R^2_{\text{adj}} = 0.6323$, implying that the model does reasonably well in explaining the variability of the income. From the F-statistic 58.59 (on 3 and 98 degrees of freedom, p-value $< 2.2\times10^{-16}$), there is a strong evidence that at least one of the variables is associated with `income`.

The last two columns of the table show the t-statistics and p-values from conducting the Wald test for the hypotheses associated with individual variables, namely $\mathcal{H}_0 : \beta_j = 0$ for each $j = 1, \ldots, 4$. Notice `education` does not display a significantly small p-value, whereas when a simple linear regression model is fitted with `education` only, it is considered to be highly significant:

```
edu.fit <- lm(income ~ education.c, data=newdata)
summary(edu.fit)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   6797.9      344.9  19.709  < 2e-16 ***
education.c    898.8      127.0   7.075 2.08e-10 ***
```

`education.c` in `pres.fit1` represents the average effect of the duration of education on `income`, while holding the other variables `women` and `prestige` fixed. From Figure 1.1, we can see that `education` exhibits correlations with other variables, in particular with `prestige`. See also

```
cor(newdata[, 5:7])

            education.c prestige.c     women.c
education.c  1.00000000  0.8501769  0.06185286
prestige.c   0.85017689  1.0000000 -0.11833419
women.c      0.06185286 -0.1183342  1.00000000
```

where off-diagonal elements close to one (in modulus) indicate strong correlations among the variables. Therefore, when all the variables are included in the model, `education` is no longer significant after adjusting for `prestige`. When two or more variables are strongly correlated, we face the problem of collinearity (some variables are linear combinations of

the others). Hence we fit a new model without `education`, and perform the F-test for the hypotheses

$$\mathcal{H}_0 : \mathsf{E}(\texttt{income}_i) = \beta_1 + \beta_2 \texttt{prestige}_i + \beta_3 \texttt{women}_i \quad \text{vs.}$$
$$\mathcal{H}_1 : \mathsf{E}(\texttt{income}_i) = \beta_1 + \beta_2 \texttt{prestige}_i + \beta_3 \texttt{women}_i + \beta_4 \texttt{education}_i$$

using the function `anova`.

```
pres.fit2 <- lm(income ~ prestige.c + women.c, data=newdata)
summary(pres.fit2)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 6797.902    254.795  26.680  < 2e-16 ***
prestige.c   165.875     14.988  11.067  < 2e-16 ***
women.c      -48.385      8.128  -5.953 4.02e-08 ***

Residual standard error: 2573 on 99 degrees of freedom
Multiple R-squared:   0.64,Adjusted R-squared:  0.6327
F-statistic: 87.98 on 2 and 99 DF,  p-value: < 2.2e-16

anova(pres.fit2, pres.fit1)

Model 1: income ~ prestige.c + women.c
Model 2: income ~ education.c + prestige.c + women.c
  Res.Df       RSS Df Sum of Sq      F Pr(>F)
1     99 655566665
2     98 649654265  1   5912400 0.8919 0.3473
```

Section 7 discusses more systematic ways of performing model selection. For this example, however, it is reasonable to conclude that `pres.fit2` provides a better fit to the data with fewer variables when compared to `pres.fit1`, since:

- Excluding `education` has in fact improved our F-statistic from 58.89 to 87.98, as well as marginally increasing R^2_{adj} from 0.6323 to 0.6327.

- ANOVA table confirms that adding `education` to `pres.fit2` does not improve the fit greatly (p-value $= 0.3473$). [2]

We can perform prediction based on `pres.fit2` using the function `predict`. Suppose that we observe a new occupation which, *after centering*, has `prestige.c = 20` and `women.c = 5`. Then, we can predict the average income of incumbents of the occupation as well as computing the confidence and prediction intervals at 95% level, as

```
xstar <- data.frame(prestige.c = 20, women.c = 5)
predict(pres.fit2, newdata = xstar, interval = 'confidence')

       fit     lwr      upr
1 9873.474 9081.49 10665.46

predict(pres.fit2, newdata = xstar, interval = 'prediction')

       fit      lwr      upr
1 9873.474 4706.427 15040.52
```

Computation via QR decomposition

Let's verify the outcome from `pres.fit2` by directly implementing the least squares estimation procedure based on the QR decomposition of $\mathbf{X}$. [3]

```
X <- cbind(1, prestige.c, women.c); y <- Prestige$income
n <- nrow(X); p <- ncol(X)
qrx <- qr(X)
R <- qr.R(qrx)
qy <- qr.qty(qrx, y)
f <- qy[1:p]; r <- qy[-(1:p)]
invR <- backsolve(R, diag(ncol(R)))
bhat <- invR%*%f
bhat
```

[2] Notice the p-values associated with `education.c` in `pres.fit1` and the F-test above! Can you derive a relationship between the Wald test statistic and the F-test statistic?

[3] Look up the functions such as `qr`, `qr.R` and `qr.qty` and learn their roles.

```
           [,1]
[1,] 6797.90196
[2,]  165.87480
[3,]  -48.38473
```

We can reproduce the columns of `Coefficients` table from `summary` output:

```
sigma.hat <- sum(r^2)/(n-p)
sqrt(sigma.hat)
```

```
[1] 2573.302
```

```
var.bhat <- sigma.hat*invR%*%t(invR)
sqrt(diag(var.bhat))
```

```
[1] 254.794910  14.988241   8.128149
```

```
t <- bhat/sqrt(diag(var.bhat))
t
```

```
          [,1]
[1,] 26.679897
[2,] 11.066996
[3,] -5.952737
```

```
2*(1-pt(abs(t), n-p))
```

```
             [,1]
[1,] 0.000000e+00
[2,] 0.000000e+00
[3,] 4.016443e-08
```

Can you similarly reproduce the F-test results? [4]

[4] See problem sheet Q4.

Finally, we can manually compute the predicted value, the confidence and prediction intervals.

```
xnew <- c(1, 20, 5)
# confidence interval
sum(bhat*xnew) + qt(1-.05/2, n-p)*
               sqrt(t(xnew)%*%var.bhat%*%xnew)

[1,] 10665.46

sum(bhat*xnew) - qt(1-.05/2, n-p)*
               sqrt(t(xnew)%*%var.bhat%*%xnew)

[1,] 9081.49

# prediction interval
sum(bhat*xnew) + qt(1-.05/2, n-p)*
               sqrt(sigma.hat+t(xnew)%*%var.bhat%*%xnew)

[1,] 15040.52

sum(bhat*xnew) - qt(1-.05/2, n-p)*
               sqrt(sigma.hat+t(xnew)%*%var.bhat%*%xnew)

[1,] 4706.427
```

4 ANOVA models for categorical covariates

An ANOVA model is a linear model with categorical covariates (factors) rather than with continuous ones. In this chapter, we discuss ANOVA models with one or more factor variables.

4.1 One-way ANOVA

In one-way ANOVA, we have a single factor with J levels at which repeated measurements are made.

factor level	responses
1	$y_{11}, y_{21}, \ldots, y_{m1}$
2	$y_{12}, y_{22}, \ldots, y_{m2}$
$\vdots$	$\vdots$
J	$y_{1J}, y_{2J}, \ldots, y_{mJ}$

The ANOVA model is

$$Y_{ij} = \mu + \alpha_j + \varepsilon_{ij}, \quad 1 \leq i \leq m,\, 1 \leq j \leq J,$$

where $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ for all i and j. We impose a constraint on α_j so that e.g. $\alpha_j = 0$ for a single index j or $\sum_{j=1}^{J} \alpha_j = 0$. In other words, for a factor with J levels, we introduce $J - 1$ parameters for model identifiability. For further discussion, see Section 3.3. The function `lm` in `R` sets $\alpha_1 = 0$, the level 1 of factor A. Then, the intercept μ corresponds to the average effect of level 1, and α_j, $2 \leq j \leq J$, capture the departure from the mean at level 1 in the mean at level j.

Example 4.1 (Fertiliser and crop yields)**.** In this example, we study the effects of different fertilisers on the crop yields. There are three yields for each treatment. We are interested in how the yield varies with respect to the treatment.

Fertiliser	Crop yields
A	33.63, 37.80, 36.58
B	35.83, 38.18, 37.89
C	42.95, 40.43, 41.46
D	38.02, 39.62, 35.99

In R, the steps for fitting an ANOVA model are as follows.

Step 1: Read in/type the response as a vector:

```
yield <- c(33.36, 37.80, 36.58, 35.83, 38.18, 37.89,
42.95, 40.43, 41.46, 38.02, 39.62, 35.99)
```

Step 2: Define a factor by `factor.x <- gl(n1, n2, n3)`, where `n1` is the number of levels, `n2` is the number of responses in each level, and `n3` is the length of the vector. So in our example, `factor.x <- gl(` 4,3,12 `)`.

Step 3: Fit the linear model using the command `lm`.

```
fit1 <- lm(yield ~ factor.x)
summary(fit1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  35.9133     0.9924  36.188 3.73e-10
factor.x2     1.3867     1.4035   0.988  0.35208
factor.x3     5.7000     1.4035   4.061  0.00363
factor.x4     1.9633     1.4035   1.399  0.19939

Residual standard error: 1.719 on 8 degrees of freedom
Multiple R-squared:  0.6931,Adjusted R-squared:  0.578
F-statistic: 6.022 on 3 and 8 DF,  p-value: 0.01896
```

We can check the constraint imposed on the factor variable by looking up the design matrix used for `fit1` using `model.matrix(fit1)` (cropped).

```
   (Intercept) factor.x2 factor.x3 factor.x4
1            1         0         0         0
2            1         0         0         0
3            1         0         0         0
4            1         1         0         0
5            1         1         0         0
```

Step 4: The p-values associated with α_2, α_3 and α_4 are obtained as the probabilities of observing a t_8-distributed random variable more extreme than the Wald statistic that is obtained under the following null hypothesis

$$\mathcal{H}_0 : \alpha_j = 0 \quad \text{vs.} \quad \mathcal{H}_1 : \alpha_j \neq 0 \qquad \text{for each } j = 2, 3, 4.$$

The p-values suggest that we should reject the null hypothesis that $\alpha_3 = 0$, but we cannot reject the other two null hypotheses. The conclusion is that there is no significant difference between the fertilisers ___A, B and D,___ but that fertiliser ___C___ produces higher yields than the others.

Step 5: We can perform an analysis of variance on this model. This allows us to test the hypotheses

$$\mathcal{H}_0 : \alpha_j = 0 \text{ for all } j = 2, 3, 4 \qquad \text{vs.}$$
$$\mathcal{H}_1 : \text{___At least___ one of } j = 2, 3, 4, \text{ leads to } \alpha_j \neq 0$$

using the F-test. The two nested models

$$\mathcal{H}_0 : \mathsf{E}(Y_{ij}) = \mu \quad \text{vs.} \quad \mathcal{H}_1 : \mathsf{E}(Y_{ij}) = \mu + \alpha_j$$

are compared using the function `anova`:

```
anova(fit1, test = 'F')
```

```
          Df Sum Sq Mean Sq F value  Pr(>F)
factor.x   3 53.376 17.7919  6.0218 0.01896 *
Residuals  8 23.637  2.9546
```

The F-statistic is calculated as $6.0218 =$ ___17.7919/2.9546___, which is large compared with the 5% critical values of the $F_{3,8}$ distribution, and thus we reject the null hypothesis.

4.2 Two-way ANOVA

Example 4.2 (Carbon monoxide emission)**.** The following table shows measurements of carbon monoxide emission (in centigrams per miles) obtained in an experiment, to eval-

uate the effect of two fuel types and three driving conditions on exhaust emissions. Two independent measurements were recorded for each fuel-location combination.

	City	Motorway	Mountain
Fuel 1	140, 156	128, 134	152, 158
Fuel 2	126, 138	124, 130	146, 156

We are interested in answering the following questions.

(i) Do exhaust emissions vary depending on the driving conditions?

(ii) Is there a difference in exhaust emissions when using Fuel 1 and Fuel 2?

(iii) If there is a difference for Fuel 1 and Fuel 2, is it dependent on the driving conditions?

Denoting by Y_{ijk} the i-th measurement in location j using fuel k, the linear model including interaction is

$$\mathsf{E}(Y_{ijk}) = \mu + \alpha_j + \beta_k + \gamma_{jk}, \quad i = 1, 2;\ j = 1, 2, 3;\ k = 1, 2,$$

where

- μ: overall mean
- α_1, α_2 and α_3: location effects
- β_1 and β_2: fuel effects
- γ_{11}, γ_{12}, γ_{13}, γ_{21}, γ_{22} and γ_{23}: location-fuel interactions.

The function `lm` in `R` imposes conditions on the parameters such that

$$\alpha_1 = 0, \quad \beta_1 = 0 \quad \text{and} \quad \gamma_{jk} = 0 \text{ if either } j = 1 \text{ or } k = 1,$$

i.e. all the coefficients involving level 1 of any factor are set to zero.
The data is entered into `R` as follows.

```
carbon <- c(140, 156, 128, 134, 152, 158, 126, 138, 124, 130, 146, 156)
loc <- gl(  ,    ,   )
fuel <- gl(   ,    ,    )
```

```
fit2 <- lm(carbon ~ loc + fuel + loc * fuel)
anova(fit2, test = 'F')
```

```
          Df Sum Sq Mean Sq F value   Pr(>F)
loc        2 1154.7  577.33 11.3947 0.009052
fuel       1  192.0  192.00  3.7895 0.099519
loc:fuel   2   96.0   48.00  0.9474 0.438976
Residuals  6  304.0   50.67
```

In the first line of the ANOVA table, we test the following.

$$\mathcal{H}_0 : \mathsf{E}(Y_{ijk}) = \mu \quad \text{vs.} \quad \mathcal{H}_1 : \mathsf{E}(Y_{ijk}) = \mu + \alpha_j.$$

The test statistic is computed using $\widehat{\sigma}^2$ estimating $\mathsf{Var}(Y_{ijk})$ associated with the last line to reduce the possibility of early termination, as

$$11.3947 = \frac{1154.7/2}{304.0/6} \sim_{\mathcal{H}_0} F_{2,6}.$$

The p-value associated with the null hypothesis is very small ($0.009 < 0.01$), so we reject the null model and conclude that `loc` is an important factor on carbon monoxide emissions. Next, we perform the hypothesis test where

$$\mathcal{H}_0 : \mathsf{E}(Y_{ijk}) = \mu + \alpha_j \quad \text{vs.} \quad \mathcal{H}_1 : \mathsf{E}(Y_{ijk}) = \mu + \alpha_j + \beta_k.$$

The p-value associated with this null hypothesis is not sufficiently small (> 0.05) to reject the null hypothesis. Hence the finally chosen model based on the analysis of variance is Model 1. There is strong evidence of differences in exhaust emissions from different locations, but little evidence of difference between fuels or meaningful interactions.

4.3 Linear model with mixed types of covariates

In this section, we discuss the normal linear model with mixed types of covariates.

Example 4.3 (Vitamin B in turnip greens)**.** The data below are from a study of the amount of vitamin B in turnip greens.

- `sun`: Sunlight during the preceding half-day.
- `soil`: Soil moisture tension.
- `temp`: Air temperature.
- `vit`: Milligrams of vitamin B2 per gram of turnip green; this is the response variable.

sun	soil	temp	vit	sun	soil	temp	vit	sun	soil	temp	vit
1.76	0.07	7.8	110.4	1.98	0.02	7.6	96.6	2.03	0.474	7.6	74.0
1.55	0.07	8.9	102.8	0.59	0.02	6.5	99.4	1.91	0.474	8.3	65.7
2.73	0.07	8.9	101.0	0.80	0.02	6.7	96.2	1.91	0.474	8.2	56.8
2.73	0.07	7.2	108.4	0.80	0.02	6.2	99.0	1.91	0.474	6.9	62.1
2.56	0.07	8.4	100.7	1.05	0.02	7.0	88.4	0.76	0.474	7.4	61.0
2.80	0.07	8.7	100.3	1.80	0.02	7.3	75.3	2.13	0.474	7.6	53.2
2.80	0.07	7.4	102.0	1.80	0.02	6.5	92.0	2.13	0.474	6.9	59.4
1.84	0.07	8.7	93.7	1.77	0.02	7.6	82.4	1.51	0.474	7.5	58.7
2.16	0.07	8.8	98.9	2.30	0.02	8.2	77.1	2.05	0.474	7.6	58.0

First reading the data into R, as

```
turnip <- read.table('[file location]/turnip.data', header = FALSE)
names(turnip) <- c('sun', 'soil', 'temp', 'vit')
```

we visualise the relationship between different variables by generating the scatter plots of `vit` against all three covariates, see Figure 4.1.
From Figure 4.1, we notice that

- `soil` may be the most important covariate.
- The dependence of `vit` on `soil` is not described by a linear relationship.
- There are only ___three___ distinct values for `soil`.

To account for the non-linearity, we may include a quadratic effect of `soil` in the model yet there are too few distinct values of soil moisture tension to properly locate the curve. Instead, we opt to treat `soil` as a factor with three levels. One of the following two commands can be used to declare `soil` a qualitative variable:

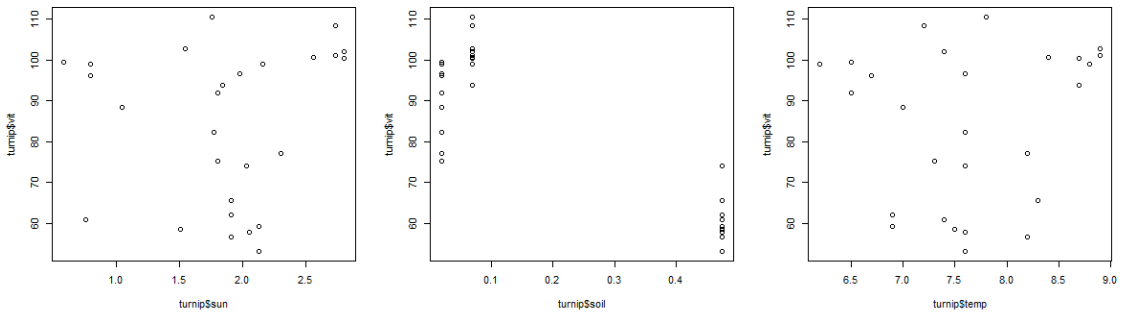


Figure 4.1: `vit` against `sun` (left), `soil` (middle) and `temp` (right).

`turnip$factor.soil <- gl(3, 9, 27)` or `turnip$factor.soil <- factor(soil)`

Let Y_{ij} denote the vitamin B2 level for the i-th observation with `factor.soil`$= j$, for $1 \leq i \leq 9$ and $1 \leq j \leq 3$. First, we fit the following one-way ANOVA model with the soil factor as the covariate

$$\mathsf{E}(Y_{ij}) = \mu + \alpha_j.$$

```
soil.fit <- lm(vit ~ factor.soil, data = turnip)
anova(soil.fit)
```

```
            Df Sum Sq Mean Sq F value    Pr(>F)
factor.soil  2 7969.9  3985.0  81.009 2.128e-11 ***
Residuals   24 1180.6    49.2
```

In the table above, the line `factor.soil` performs the hypothesis testing with

$$\mathcal{H}_0 : \mathsf{E}(Y_{ij}) = \underline{\quad \mu \quad} \text{ vs. } \mathcal{H}_1 : \mathsf{E}(Y_{ij}) = \underline{\quad \mu + \alpha_j. \quad}$$

Since the p-value is very small, we reject the null hypothesis and include `factor.soil` in the model.
Plotting the residuals from `soil.fit` against the other variables `sun` and `temp`, Figure 4.2 indicates that there may be a relationship between the residuals and the other covariates, especially `temp`. Therefore, we consider whether to add the two covariates to the model:

```
all.fit <- lm(vit ~ factor.soil + temp + sun, data = turnip)
```

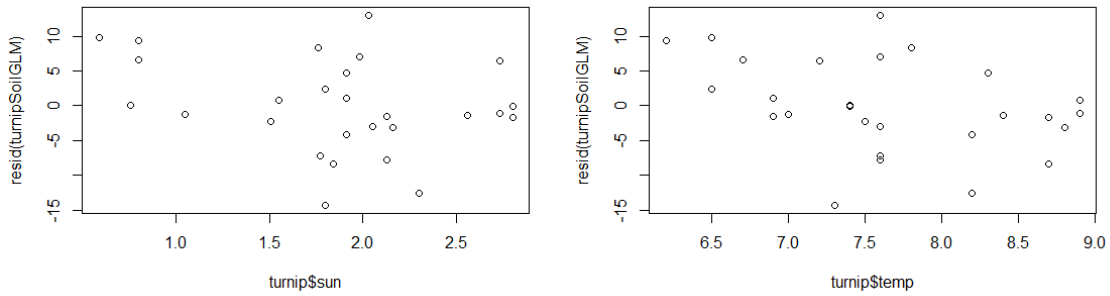


Figure 4.2: Residuals from `soil.fit` against `sun` (left) and `temp` (right).

```
anova(all.fit)
```

```
             Df Sum Sq Mean Sq  F value    Pr(>F)
factor.soil   2 7969.9  3985.0 106.9548 4.639e-12
temp          1  289.8   289.8   7.7791    0.0107
sun           1   71.1    71.1   1.9079    0.1811
Residuals    22  819.7    37.3
```

From the above analysis of deviance table, we can sequentially test whether the covariate in question should be included in the model. In the line `temp`, we test

$$\mathcal{H}_0 : \mathsf{E}(Y_{ij}) = \underline{\quad \mu + \alpha_j, \quad}$$
$$\mathcal{H}_1 : \mathsf{E}(Y_{ij}) = \underline{\quad \mu + \alpha_j + \beta_1 \texttt{temp}_{ij}. \quad}$$

The p-value 0.0107 indicates a significant departure from the $\underline{\quad F_{1,22} \quad}$ distribution than that we would expect under $\mathcal{H}_0$, hence we reject the null hypothesis and keep `temp` in our model. In the last line with `sun`, we test

$$\mathcal{H}_0 : \mathsf{E}(Y_{ij}) = \underline{\quad \mu + \alpha_j + \beta_1 \texttt{temp}_{ij}, \quad}$$
$$\mathcal{H}_1 : \mathsf{E}(Y_{ij}) = \underline{\quad \mu + \alpha_j + \beta_1 \texttt{temp}_{ij} + \beta_2 \texttt{sun}_{ij}. \quad}$$

With the p-value 0.1811, there is insufficient evidence for us to reject $\mathcal{H}_0$.
Finally, let us explore whether the interaction between `factor.soil` and `temp` has any effect on `vit`, i.e. whether the effect of temperature on the amount of vitamin in turnip

greens differs with respect to soil moisture tension level.

```
interaction.fit <- lm(vit ~ factor.soil * temp, data = turnip)
anova(interaction.fit)
```

```
                Df Sum Sq Mean Sq  F value    Pr(>F)
factor.soil      2 7969.9  3985.0 112.7905 5.858e-12
temp             1  289.8   289.8   8.2035  0.009287
factor.soil:temp 2  148.8    74.4   2.1062  0.146678
Residuals       21  741.9    35.3
```

We conclude that the interaction is not significant enough to be included in the model. Re-indexing the response variable so that Y_i denotes the i-th observation in the data, the final fitted model becomes

$$\mathsf{E}(Y_i) = 129.993 + 19.535 \cdot \mathbb{I}_{\{\texttt{soil}_i = 0.07\}} - 25.817 \cdot \mathbb{I}_{\{\texttt{soil}_i = 0.474\}} - 5.716 \cdot \texttt{temp}_i.$$

```
final.fit <- lm(vit ~ factor.soil + temp,  data = turnip)
summary(final.fit)
```

```
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      129.993     14.910   8.718 9.51e-09
factor.soil0.07   19.535      3.920   4.983 4.86e-05
factor.soil0.474 -25.817      3.106  -8.311 2.22e-08
temp              -5.716      2.089  -2.736   0.0118

Residual standard error: 6.223 on 23 degrees of freedom
Multiple R-squared:  0.9027,Adjusted R-squared:   0.89
F-statistic: 71.09 on 3 and 23 DF,  p-value: 8.747e-12
```

We can make the following conclusions:

- 19.535 and -25.817 are the treatment effects of the soil moisture tension at level 2 (`soil` $= 0.07$) and level 3 (`soil` $= 0.474$) compared to that at level 1 (`soil` $= 0.02$). The amount of `vit` of the turnip greens planted in the soil with `soil` $= 0.07$ is __more__ than the ones planted in the soil with `soil` $= 0.02$ or `soil` $= 0.474$, with the average difference at 19.535 and 45.352, respectively.

- $\beta = -5.716$ implies that the higher temperature is, the ___lower___ the amount of `vit` is. On average, increasing `temp` by one degree, will ___decrease___ `vit` by ___5.716___ units.

5 Model diagnosis

In fitting linear models, several assumptions are made, such as

(i) the regression function is linear, i.e. for given $(Y_i, \mathbf{x}_i)$, we have $\mathsf{E}(Y_i) =$ $\mathbf{x}_i^\top \boldsymbol{\beta}_0$

(ii) the errors ε_i, $i = 1, \ldots, n$, have zero mean and constant variance

(iii) the errors ε_i, $i = 1, \ldots, n$ are independent

(iv) the errors are normally distributed, i.e. $\varepsilon_i \sim_{\text{i.i.d}} \mathcal{N}(0, \sigma^2)$

It is important to check the plausibility of these assumptions on the data, prior to conduct any statistical inference. Often, the first step in model diagnosis is to plot the residuals against various quantities to verify these assumptions.

5.1 Residuals for model diagnosis

5.1.1 Raw residuals

Recall the influence ('hat') matrix $\mathbf{H}$ and that the fitted values for $\mathbf{Y}$ are given by $\widehat{\boldsymbol{\mu}} = \mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{Y}$ (see Section 3.2.4). The vector of raw residuals is defined as

$$\mathbf{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} Y_1 - \widehat{\mu}_1 \\ \vdots \\ Y_n - \widehat{\mu}_n \end{bmatrix} = \mathbf{Y} - \widehat{\boldsymbol{\mu}} = \mathbf{Y} - \mathbf{H}\mathbf{Y} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}$$

and, given that $\mathbf{Y} \sim (\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$ (assumptions in (i), (ii) and (iii)), it satisfies

$$\mathsf{E}(\mathbf{e}) = (\mathbf{I}_n - \mathbf{H})\mathsf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}_0 - \mathbf{H}\mathbf{X}\boldsymbol{\beta}_0 = \mathbf{X}\boldsymbol{\beta}_0 - \mathbf{X}\boldsymbol{\beta}_0 = \mathbf{0}, \tag{5.1}$$

$$\mathsf{Cov}(\mathbf{e}) = \sigma^2(\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H})^\top = \sigma^2(\mathbf{I}_n - \mathbf{H}) \tag{5.2}$$

using that $\mathbf{H}$ is a projection matrix onto the column space of $\mathbf{X}$, as well as its idempotence and symmetry. With the additional normality (assumption in (iv)), i.e. $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$, the distribution of $\mathbf{e}$ is also normal:

$$\mathbf{e} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2(\mathbf{I}_n - \mathbf{H})).$$

The normal distribution is a degenerate one, since some eigenvalues of $\mathbf{I}_n - \mathbf{H}$ are zero and hence $\mathbf{I}_n - \mathbf{H}$ is not positive definite: If $\mathbf{a} = \mathbf{H}\mathbf{c}$ for some $\mathbf{c} \in \mathbb{R}^n$, we have $\mathbf{a}^\top(\mathbf{I}_n - \mathbf{H})\mathbf{a} = 0$, which implies that some eigenvalues of $\mathbf{I}_n - \mathbf{H}$ are zero.

5.1.2 Standardised residuals

Results from Section 5.1.1 imply that for the purpose of model checking, we need to *standardise* the raw residuals to have constant variance so that the residuals from different observations are comparable. From (5.1)–(5.2), we have $\mathsf{E}(e_i) = 0$ and $\mathsf{Var}(e_i) = \sigma^2(1 - h_{ii})$, where h_{ii} denotes the i-th diagonal element of $\mathbf{H}$. Then, we *standardise* the raw residual $e_i = Y_i - \widehat{\mu}_i$ to derive the standardised residual

$$s_i = \frac{e_i}{\sqrt{\widehat{\sigma}^2(1 - h_{ii})}}, \quad 1 \leq i \leq n.$$

Standardised residuals are useful in detecting anomalous observations or *outliers* (see Figure 5.2 for an example of outliers). In general, any observation with $|s_i| > 2$ is considered worthy of further scrutiny, although such observations are not necessarily outliers.

5.1.3 Jackknife residuals

The standardised residuals s_i depend on an estimate of the standard deviation $\widehat{\sigma}$. In the presence of outliers, $\widehat{\sigma}$ may over-estimate σ, which in turn shrinks $|s_i|$, and thus some anomalous observations may escape detection. A remedy to this problem is to standardise the i-th residual e_i using an estimate of the error variance obtained after omitting the i-th observation: In doing so, if the i-th observation is an outlier, we obtain an error variance estimate that is not influenced by its presence. This produces the *jackknife* (a.k.a externally standardised) residual

$$t_i = \frac{e_i}{\sqrt{\widehat{\sigma}^2_{(i)}(1 - h_{ii})}},$$

where $\widehat{\sigma}^2_{(i)}$ denotes the estimate of the error variance obtained by fitting the model using $n - 1$ observations excluding the i-th observation $(Y_i, \mathbf{x}_i)$, and is based on an RSS with $n - 1 - p$ degrees of freedom. To be precise, let $\widehat{\boldsymbol{\beta}}_{(i)}$ denote the least squares

estimator obtained with the $(n-1)$ observations $(Y_\ell, \mathbf{x}_\ell)$, $\ell \neq i$, only, i.e.

$$\widehat{\boldsymbol{\beta}}_{(i)} = \underset{\boldsymbol{\beta}\in\mathbb{R}^p}{\arg\min} \sum_{\ell\neq i, \ell=1}^{n} (Y_\ell - \mathbf{x}_\ell^\top \boldsymbol{\beta})^2.$$

Then we generate $\widehat{\sigma}^2_{(i)}$ as

$$\widehat{\sigma}^2_{(i)} = \frac{1}{n-1-p} \sum_{\ell\neq i,\, \ell=1}^{n} (Y_\ell - \mathbf{x}_\ell^\top \widehat{\boldsymbol{\beta}}_{(i)})^2. \tag{5.3}$$

Note that the rest of the quantities such as e_i and $\mathbf{H}$ are still calculated from the model fitted with all n observations.

A natural question to ask is: How about using the residual calculated from the model fitted without the i-th observation

$$e_i^{(i)} = Y_i - \widehat{\mu}_i^{(i)} = Y_i - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_{(i)},$$

itself for model diagnosis? We how show that in fact

$$t_i = \frac{e_i^{(i)}}{\sqrt{\widehat{\mathsf{Var}}(e_i^{(i)})}}. \tag{5.4}$$

Proof of (5.4). Let $\mathbf{X}_{(i)}$ denote the $(n-1) \times p$ matrix which is a sub-matrix of $\mathbf{X}$ without the i-th row, and $\mathbf{Y}_{(i)}$ denote the $(n-1)$-vector which is a sub-vector of $\mathbf{Y}$ without the i-th element. In order to understand the behaviour of $e_i^{(i)} = Y_i - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_{(i)}$, we first investigate the least squares estimator from the $(n-1)$ observations $\widehat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} \mathbf{X}_{(i)}^\top \mathbf{Y}_{(i)}$, where

$$\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)} = \mathbf{X}^\top \mathbf{X} - \mathbf{x}_i \mathbf{x}_i^\top \quad \text{and} \quad \mathbf{X}_{(i)}^\top \mathbf{Y}_{(i)} = \mathbf{X}^\top \mathbf{Y} - \mathbf{x}_i Y_i.$$

The influence matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = [h_{\ell\ell'}]_{\ell,\ell'=1}^n$, has its elements satisfy $h_{\ell\ell'} = \mathbf{x}_\ell^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{\ell'}$.

Recall the Sherman-Morrison-Woodbury formula given in (2.3):

$$(\mathbf{A} + \mathbf{UWV})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\left(\mathbf{W}^{-1} + \mathbf{VA}^{-1}\mathbf{U}\right)^{-1} \mathbf{VA}^{-1}. \tag{5.5}$$

Let us replace $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$, $\mathbf{U} = \mathbf{x}_i$, $\mathbf{W} = -1$ and $\mathbf{V} = \mathbf{x}_i^\top$ into the formula (5.5), so that

$$\mathbf{A} + \mathbf{UWV} = \underline{\quad \mathbf{X}^\top\mathbf{X} - \mathbf{x}_i\mathbf{x}_i^\top \quad} = \mathbf{X}_{(i)}^\top\mathbf{X}_{(i)}.$$

This gives

$$\begin{aligned}(\mathbf{X}_{(i)}^\top\mathbf{X}_{(i)})^{-1} &= (\mathbf{X}^\top\mathbf{X})^{-1} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i(1 - \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i)^{-1}\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1} \\ &= (\mathbf{X}^\top\mathbf{X})^{-1} + \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}}{1 - h_{ii}}\end{aligned}$$

by using that $h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i$. Therefore,

$$\begin{aligned}&\widehat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}_{(i)}^\top\mathbf{X}_{(i)})^{-1}\mathbf{X}_{(i)}^\top\mathbf{Y}_{(i)} \\ &= \left((\mathbf{X}^\top\mathbf{X})^{-1} + \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}}{1 - h_{ii}}\right)(\mathbf{X}^\top\mathbf{Y} - \mathbf{x}_iY_i) \\ &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} - (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_iY_i + \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}}{1 - h_{ii}} \\ &\qquad - \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_iY_i}{1 - h_{ii}} \\ &= \widehat{\boldsymbol{\beta}} - (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_iY_i + \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i^\top\widehat{\boldsymbol{\beta}}}{1 - h_{ii}} - \frac{h_{ii}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_iY_i}{1 - h_{ii}} \\ &= \widehat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_iY_i}{1 - h_{ii}} + \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\widehat{\mu}_i}{1 - h_{ii}} \\ &= \widehat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_ie_i}{1 - h_{ii}}. \qquad (5.6)\end{aligned}$$

Hence

$$\begin{aligned}\mathbf{x}_i^\top\widehat{\boldsymbol{\beta}}_{(i)} = \mathbf{x}_i^\top\widehat{\boldsymbol{\beta}} - \frac{\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_ie_i}{1 - h_{ii}} &= \mathbf{x}_i^\top\widehat{\boldsymbol{\beta}} - \frac{h_{ii}e_i}{1 - h_{ii}}, \quad \text{and} \\ e_i^{(i)} = Y_i - \mathbf{x}_i^\top\widehat{\boldsymbol{\beta}}_{(i)} = \underline{\quad e_i + \frac{h_{ii}e_i}{1 - h_{ii}} = \frac{e_i}{1 - h_{ii}}. \quad} &\qquad (5.7)\end{aligned}$$

Then, from the independence between Y_i and $\widehat{\boldsymbol{\beta}}_{(i)}$ (as the latter depends only on Y_ℓ, $\ell \neq i$), we have

$$\mathsf{Var}(e_i^{(i)}) = \mathsf{Var}(Y_i - \mathbf{x}_i^\top\widehat{\boldsymbol{\beta}}_{(i)}) = \mathsf{Var}(Y_i) + \mathsf{Var}(\mathbf{x}_i^\top\widehat{\boldsymbol{\beta}}_{(i)}) = \mathsf{Var}(Y_i) + \mathbf{x}_i^\top\mathsf{Cov}(\widehat{\boldsymbol{\beta}}_{(i)})\mathbf{x}_i$$

$$= \sigma^2 \left(1 + \mathbf{x}_i^\top (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} \mathbf{x}_i\right),$$

which can be estimated by replacing σ^2 with its estimator $\widehat{\sigma}^2_{(i)}$ in (5.3), as

$$\begin{aligned}
\widehat{\mathsf{Var}}(e_i^{(i)}) &= \widehat{\sigma}^2_{(i)}(1 + \mathbf{x}_i^\top (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} \mathbf{x}_i), \quad \text{where} \\
\mathbf{x}_i^\top (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} \mathbf{x}_i &= \mathbf{x}_i^\top \left((\mathbf{X}^\top \mathbf{X})^{-1} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1}}{1 - h_{ii}} \right) \mathbf{x}_i \\
&= h_{ii} + \frac{h_{ii}^2}{1 - h_{ii}} = \frac{h_{ii}}{1 - h_{ii}}, \quad \text{and therefore} \\
\widehat{\mathsf{Var}}(e_i^{(i)}) &= \frac{\widehat{\sigma}^2_{(i)}}{1 - h_{ii}}.
\end{aligned}$$

Thus it follows that

$$\frac{e_i^{(i)}}{\sqrt{\widehat{\mathsf{Var}}(e_i^{(i)})}} = \frac{e_i/(1 - h_{ii})}{\widehat{\sigma}_{(i)}/\sqrt{1 - h_{ii}}} = \frac{e_i}{\widehat{\sigma}_{(i)}\sqrt{1 - h_{ii}}} = t_i.$$

□

It can further be shown that

$$t_i = s_i \sqrt{\frac{n - 1 - p}{n - p - s_i^2}},$$

which indicates that the jackknife residuals can be computed without having to fit the model n times, each time omitting a single observation. Further, jackknife residuals can be used in testing for outliers, see Section 5.3.3.

Hint: Start from the observation that

$$(n - 1 - p)\widehat{\sigma}^2_{(i)} = \mathbf{Y}_{(i)}^\top (\mathbf{I}_{n-1} - \mathbf{X}_{(i)} (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} \mathbf{X}_{(i)}^\top) \mathbf{Y}_{(i)}.$$

Since t_i and s_i differ only in the estimate of σ^2, the proof follows similarly from the application of (5.5). □

5.2 Plotting residuals for model checking

By plotting the standardised or jackknife residuals, we can evaluate the overall fit of the model to the data and check whether various assumptions given at the beginning of this chapter are valid.

Linearity in (i): Plot the residuals (e_i, s_i or t_i) vs. $\widehat{\mu}_i = \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}$ and check if the residuals are uniformly distributed around 0 without any special patterns/trends.

- **Remedies when the assumption is violated:** Transform the response (see Section 6) or include polynomial terms of the variables.

Independence in (iii): Plot the residuals vs. $i = 1, \ldots, n,$ and check if the residuals are uniformly distributed around 0 without any special patterns/trends. Also, the Durbin-Watson test can be adopted to detect the presence of correlations in the residuals.

Equal variance in (ii): Plot the residuals vs. $\widehat{\mu}_i$ and check if the residuals are uniformly distributed around 0 without any special patterns/trends. Violation of the equal variance assumption results in a cloud shaped like a megaphone, e.g. more clustered residuals for small $\widehat{\mu}_i$ and more dispersed residuals for large $\widehat{\mu}_i$.

- **Remedies:** Transform the response.

Normality in (iv): Plot the normal Q-Q plot of the residuals (quantiles of the residuals vs. quantiles of a standard normal distribution) and check if a straight line fits it well.

- In general, Q-Q plots are used to study the *skewness* and *tail-behaviour* (S-shaped Q-Q plots indicate heavy tails, or an excess of extreme values, relative to the normal distribution) of the observations.
- Normality of the residuals should also be checked, if the other assumptions are appropriate for the data, but it is of only secondary importance to the other assumptions when n is large, thanks to the central limit theorem.

Example 5.1 (Prestige data (cont'd))**.** We continue with the analysis of `Prestige` data frame example introduced in Section 3.8.2. Plotting an `lm` object provides default model-checking residual plots.

```
plot(pres.fit2)
```

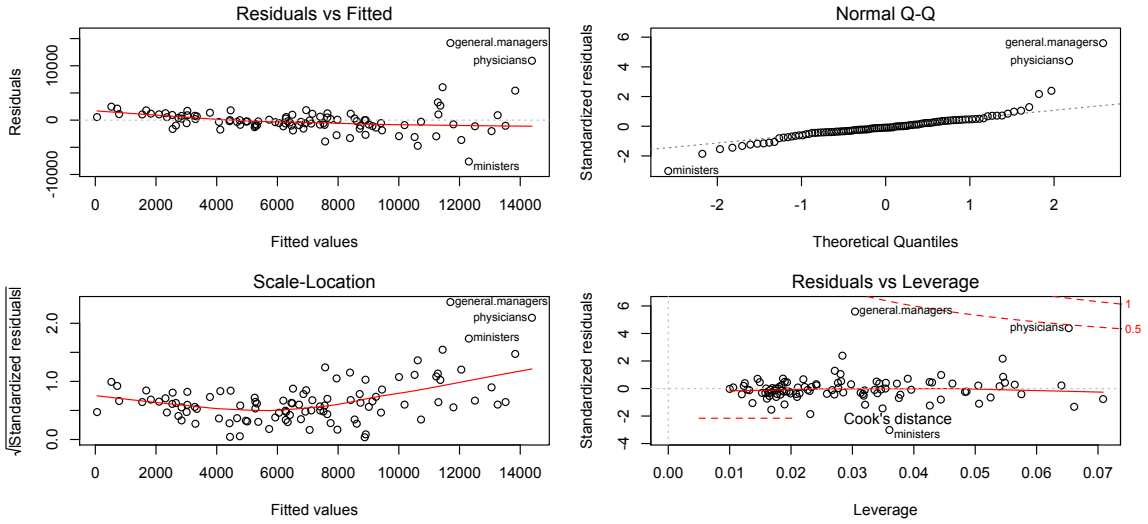


Figure 5.1: Residual plots from `pres.fit2`. Top left: e_i vs. $\widehat{\mu}_i$; bottom left: $\sqrt{|s_i|}$ vs. $\widehat{\mu}_i$; top right: normal Q-Q plot of s_i; bottom right: s_i vs. h_{ii} along with contours of Cook's distance.

For now, we consider the first three panels excluding the bottom right panel. Both of the left-hand plots suggest possible patterns in the variance with increasing mean, suggesting that the linearity/homoscedasticity assumptions may not be met by the data. The top right plot indicates some departure from normality in the residuals (heavy-tail). Note that apart from the top left plot, the standardised residuals s_i are used in producing these plots, which should have identical variance when the fitted model is appropriate for the data.

5.3 Influential observations and outliers

Another point to check is whether some observations exert unduly large influence on the modelling results. Some observations have large influence due to an unusual combination of explanatory variables (unusual x-values), which makes the whole fit sensitive to the value of the corresponding observation. Observations with very large residuals (unusual y-values) can sometimes be problematic and large outliers should be investigated. This point is illustrated in Figure 5.2.

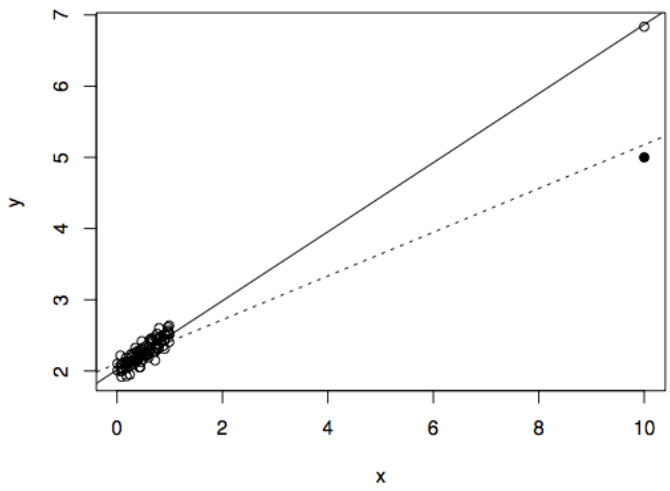


Figure 5.2: Taken from Figure 7.2 of Wood (2006). The solid black line shows the least squares straight line fit to the 100 data shown as open circles. In this case the fit is reasonable for all the data. The dashed line shows the least squares straight line fit when the point at $x = 10$ is moved to the position shown as a solid black disc. To accommodate the observation far from the other points, the line has been forced to fit the remaining 99 data rather poorly.

5.3.1 Leverage

How do we tell if $\mathbf{x}_i$ is 'unusual' compared to $\mathbf{x}_\ell$, $\ell \neq i$? Recall that

$$\mathsf{Var}(e_i) = \mathsf{Var}(Y_i - \widehat{\mu}_i) = \mathsf{Var}([(\mathbf{I}_n - \mathbf{H})\mathbf{Y}]_i) = [\sigma^2(\mathbf{I}_n - \mathbf{H})]_{i,i} = \sigma^2(1 - h_{ii}).$$

As $h_{ii} \to 1$, the variance of e_i approaches zero, indicating that $\widehat{\mu}_i$ is forced to come close to Y_i. In view of this result, the quantity h_{ii} is closely related to the *leverage* or the potential influence of the i-th observation. The i-th observation is usually considered to have high leverage if h_{ii} exceeds $2p/n$, which comes from that

$$\mathsf{tr}(\mathbf{H}) = \mathsf{tr}(\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top) = \mathsf{tr}((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}) = \mathsf{tr}(\mathbf{I}_p) = p.$$

Therefore, $n^{-1}\sum_{i=1}^n h_{ii} = p/n$, and $2p/n$ is the twice of the average h_{ii}, $1 \leq i \leq n$.

5.3.2 Cook's distance

A combination of large residual and leverage indicates a point with substantial influence on the fit. A standard way of measuring this is via *Cook's distance*, a measure of influence based on the extent to which parameter estimates would change if one omitted the i-th observation. Recalling the notations from Section 5.1.3, Cook's distance is the standardised distance between $\widehat{\boldsymbol{\beta}}_{(i)}$ and $\widehat{\boldsymbol{\beta}}$, namely

$$D_i = \frac{(\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})^\top (\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}}))^{-1} (\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})}{p}. \tag{5.8}$$

D_i also admits a representation as a function of the Euclidean distance between $\widehat{\boldsymbol{\mu}}_{(i)} = (\widehat{\mu}_1^{(i)}, \ldots, \widehat{\mu}_n^{(i)})^\top = \mathbf{X}\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{\mu}} = \mathbf{X}\widehat{\boldsymbol{\beta}}$, i.e.

$$D_i = \frac{\sum_{\ell=1}^n (\widehat{\mu}_\ell^{(i)} - \widehat{\mu}_\ell)^2}{p\widehat{\sigma}^2}, \tag{5.9}$$

as well as a function of h_{ii} and s_i,

$$D_i = s_i^2 \frac{h_{ii}}{(1 - h_{ii})p}. \tag{5.10}$$

Proof of the equivalence between (5.8)–(5.10). First, we show that (5.8) = (5.9). From the fact that $\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}}) = \widehat{\sigma}^2(\mathbf{X}^\top\mathbf{X})^{-1}$, we have

$$D_i = \frac{(\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})^\top (\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}}))^{-1} (\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})}{p} = \frac{(\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})^\top \mathbf{X}^\top\mathbf{X} (\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})}{p\widehat{\sigma}^2}$$
$$= \frac{(\mathbf{X}\widehat{\boldsymbol{\beta}}_{(i)} - \mathbf{X}\widehat{\boldsymbol{\beta}})^\top (\mathbf{X}\widehat{\boldsymbol{\beta}}_{(i)} - \mathbf{X}\widehat{\boldsymbol{\beta}})}{p\widehat{\sigma}^2} = \frac{\sum_{\ell=1}^n (\widehat{\mu}_\ell^{(i)} - \widehat{\mu}_\ell)^2}{p\widehat{\sigma}^2}.$$

Next, we show that (5.8) = (5.10). From (5.6) and the definition of s_i, we have

$$\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}} = -\frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i e_i}{1 - h_{ii}} = -\frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i s_i \widehat{\sigma}}{\sqrt{1 - h_{ii}}},$$

and thus

$$D_i = \frac{(\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})^\top (\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}}))^{-1} (\widehat{\boldsymbol{\beta}}_{(i)} - \widehat{\boldsymbol{\beta}})}{p}$$

$$= \frac{\widehat{\sigma}^2 s_i^2}{1-h_{ii}} \cdot \frac{\mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i}{p\widehat{\sigma}^2} = s_i^2 \frac{h_{ii}}{(1-h_{ii})p}.$$

□

Equation (5.10) demonstrates that Cook's distance combines the residual (s_i) and leverage (h_{ii}) in a single measure of influence, increasing as either $|s_i|$ or h_{ii} increases. Also, this expression shows that we can compute D_i, $1 \leq i \leq n$, by fitting a *single* linear model with all n observations, without having to fit n linear models omitting a single observation each time for all i. Any observation with D_i over 0.5 is considered borderline problematic, and observations with $D_i \geq 1$ are usually considered highly influential. In order to justify this rule of thumb, notice that D_i has the form of an F-statistic for testing $\mathcal{H}_0$: $\boldsymbol{\beta}_0 = \widehat{\boldsymbol{\beta}}_{(i)}$, since

$$D_i = \frac{(\widehat{\boldsymbol{\beta}} - \widehat{\boldsymbol{\beta}}_{(i)})^\top (\mathsf{Cov}(\widehat{\boldsymbol{\beta}}))^{-1} (\widehat{\boldsymbol{\beta}} - \widehat{\boldsymbol{\beta}}_{(i)})/p}{\widehat{\sigma}^2/\sigma^2} \sim_{\mathcal{H}_0} \frac{\chi^2_p/p}{\chi^2_{n-p}/(n-p)} =^d F_{p,n-p}.$$

The value one is close to the median of F distributions with a large range of values for degrees of freedom. Since $\mathcal{H}_0$ indicates that the i-th observation does *not* have excessive influence, we consider an observation to have large influence if deleting it would move this F-statistic from zero to the median, which is equivalent to moving the point estimate to the edge of a 50% confidence region.

Example 5.2 (Prestige data (cont'd))**.** The bottom right panel of Figure 5.1 plots the contours of Cook's distance values 0.5 and 1, as a function of s_i and h_{ii}. Points outside these contours warrant further investigation. In such cases it may be wise to repeat the analysis without the influential observation and examine which estimates change as a result. As for the Prestige data, although a couple of points have rather high leverage, their actual influence on the fit is not unduly large as all points fall below the 0.5 contour lines in their Cook's distance.

5.3.3 Test for outliers

The jackknife residual discussed in Section 5.1.3 motivates a formal test for outliers. We start from the model $\mathsf{E}(Y_\ell) = \mu_\ell = \mathbf{x}_\ell^\top \boldsymbol{\beta}_0$, and add a dummy variable which allows for a

location shift for the i-th observation only, as

$$\mathsf{E}(Y_\ell) = \mu_\ell = \mathbf{x}_\ell^\top \boldsymbol{\beta}_0 + \gamma z_\ell, \quad \text{where} \tag{5.11}$$

$$z_\ell = \mathbb{I}_{\{\ell=i\}} = \begin{cases} \underline{\quad 1 \quad} & \text{when } \ell = i, \\ \underline{\quad 0 \quad} & \text{otherwise.} \end{cases} \tag{5.12}$$

In this model, γ represents the extent to which the i-th response Y_i differs from what would be expected on the basis of its covariate values $\mathbf{x}_i$ and the regression coefficients $\boldsymbol{\beta}_0$. A formal test of the hypothesis $\mathcal{H}_0 : \gamma = 0$, can therefore be interpreted as a test that the i-th observation follows the same model as the rest of the data (i.e. it is __not an outlier__ under $\mathcal{H}_0$). The test statistic for the Wald test is

$$t = \frac{\widehat{\gamma}}{\sqrt{\widehat{\mathsf{Var}}(\widehat{\gamma})}},$$

A large value of $|t|$ indicates that γ differs significantly from zero, and hence that the i-th observation is an outlier. Formally, we compare $|t|$ against a critical value from the __t_{n-p-1}__-distribution. In fact, t is the jackknife residual. To see this, by letting the i-th observation have its own parameter γ,

- the fitted value of Y_i under (5.11) coincides with __Y_i__ itself, and
- we effectively estimate $\boldsymbol{\beta}$ from the $(n-1)$ observations excluding $(Y_i, \mathbf{x}_i)$ such that the least squares estimator coincides with __$\widehat{\boldsymbol{\beta}}_{(i)}$__ (using the notation from Section 5.1.3).

From these, we have

$$\widehat{\gamma} = Y_i - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_{(i)} = Y_i - \widehat{\mu}_i^{(i)} = e_i^{(i)}$$

and the rest follows from (5.4). [5]

In adopting the jackknife residual to test for outliers, we need to be cautious about *multiplicity*. If the suspect observation had been picked in advance *without looking at the data*, then the test would be valid without any adjustment. However, if the suspect observation has been selected from looking at the data e.g. by screening all n observations, the nominal significance level is not valid because we have implicitly conducted the hypothesis test

[5] See problem sheet Q1 for step-by-step arguments leading to this conclusion.

$\frac{n}{}$ times. Simply put, if you conduct 100 (independent) tests at the 5% significance level, you would expect to reject the null hypothesis five times just by chance alone *even when the corresponding null hypotheses are true*. When the aim is to conduct n independent tests at a given significance level α simultaneously, one can control the family-wise error rate by adopting the *Bonferroni* correction, whereby each test is conducted at the significance level α/n. This procedure tends to be conservative and the true significance level may be considerably less than α. There are other ways of controlling for the Type I error in multiple testing, such as controlling for the false discovery rate.

5.4 Example: Prestige data (cont'd)

We can compute the standardised and jackknife residuals as well as Cook's distance, both manually and using `R` functions. Recall `pres.fit2` from page 31.

```
e <- resid(pres.fit2) # or pres.fit2$residuals
Q <- qr.Q(qrx)
H <- Q%*%t(Q) # hat matrix
h <- diag(H) # leverage
s <- e/sqrt(sigma.hat*(1-h)) # standardised residuals
t <- s*sqrt((n-p-1)/(n-p-s^2)) # jackknife residuals

rstandard(pres.fit2) # compare with s
rstudent(pres.fit2) # compare with t

cookD <- s^2*h/(1-h)/p
cooks.distance(pres.fit2) # compare with cookD
```

In Figure 5.1, some observations of large residuals are labelled, which are general managers, ministers and physicians. We have a closer look at these observations, as well as testing for the outliers.

```
cbind(h, e, s, t, cookD)[c(2, 20, 24),]

            h         e         s         t     cookD
2  0.03041948 14179.984  5.596197  6.733913 0.3275160
```

```
20 0.03602570 -7620.947 -3.016375 -3.149307 0.1133434
24 0.06523105 10923.086  4.390383  4.867651 0.4483669

cooks.distance(pres.fit2)[c(2, 20, 24)]

        2        20        24
0.3275160 0.1133434 0.4483669

which(2*(1-pt(abs(t), n-p-1)) < 0.05/n)
# outlier testing with jackknife residuals and Bonferroni correction

general.managers       physicians
               2               24
```

6 Transformations

When any violation of the assumptions (i)–(iv) in Section 5 is detected e.g. from plotting the residuals, we may transform the response and/or explanatory variables to remedy them.

6.1 Box-Cox transform

When $\mathsf{E}(Y_i) = \mathbf{x}_i^\top \boldsymbol{\beta}_0$ is not a valid assumption, nor the true relationship between Y_i and $\mathbf{x}_i$ is known, we may resolve this issue by identifying a transformation $g(Y_i)$ that satisfies

$$\mathsf{E}(g(Y_i)) = \mathbf{x}_i^\top \boldsymbol{\beta}_0.$$

Box and Cox (1964) proposed a family of transformations that can be used when Y_i are positive, which includes commonly used transforms as special cases, such as reciprocal, logarithm and square root transformations. Specifically, it considers a family of power transforms as candidates:

$$Y(\lambda) = \begin{cases} \frac{Y^\lambda - 1}{\lambda} & \text{when } \lambda \neq 0, \\ \log(Y) & \text{when } \lambda = 0. \end{cases}$$

Note that this transformation is essentially Y^λ for $\lambda \neq 0$, but it takes the above form so that $\lim_{\lambda \to 0} Y(\lambda) = \log(Y)$. Useful values of λ are often found in the range $[-2, 2]$. Except for the scaling terms, $\lambda = -1$ leads to the reciprocal, $\lambda = 0$ to the logarithm, $\lambda = 1/2$ to the square root, $\lambda = 1$ to the identity and $\lambda = 2$ to the square transforms.
Then, we fit a linear model $\mathsf{E}[\mathbf{Y}(\lambda)] = \mathbf{X}\boldsymbol{\beta}_0(\lambda)$ under the assumption

$$\mathbf{Y}(\lambda) \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0(\lambda), \sigma^2(\lambda)\mathbf{I}_n)$$

for a grid of values for λ, and find λ that results in the maximum likelihood numerically. Residual sums of squares is not a suitable criterion because $Y(\lambda)$ are in different scales, depending on the values of λ.
From the normality assumption on $\mathbf{Y}(\lambda)$, we obtain the likelihood given $\mathbf{Y}$ via change of variables as

$$L(\boldsymbol{\beta}(\lambda), \sigma^2(\lambda), \lambda | \mathbf{Y}) = \left(\frac{1}{2\pi\sigma^2(\lambda)}\right)^{\frac{n}{2}} \exp\left[-\frac{\sum_{i=1}^n (Y_i(\lambda) - \mathbf{x}_i^\top \boldsymbol{\beta}(\lambda))^2}{2\sigma^2(\lambda)}\right] \cdot \prod_{i=1}^n Y_i^{\lambda - 1},$$

where $\prod_{i=1}^n Y_i^{\lambda-1}$ in the first equation is from the Jacobian of the transformation. Taking logarithm gives

$$\ell(\boldsymbol{\beta}(\lambda), \sigma^2(\lambda), \lambda|\mathbf{Y}) = -\frac{n}{2}\log(\sigma^2(\lambda)) - \frac{1}{2\sigma^2(\lambda)}\sum_{i=1}^n (Y_i(\lambda) - \mathbf{x}_i^\top \boldsymbol{\beta}(\lambda))^2 + (\lambda - 1)\sum_{i=1}^n \log(Y_i),$$

so that the maximum likelihood estimator (MLE) given λ is

$$\widehat{\boldsymbol{\beta}}(\lambda) = \underline{\quad (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}(\lambda) \quad} \text{ and}$$
$$\widehat{\sigma}^2(\lambda) = \underline{\quad \frac{1}{n}\sum_{i=1}^n (Y_i(\lambda) - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}(\lambda))^2 \quad} = \frac{\text{RSS}(\lambda)}{n}.$$

Plugging in the MLE produces the profile likelihood to be examined over a grid of values for λ:

$$\ell(\lambda) \equiv \ell(\widehat{\boldsymbol{\beta}}(\lambda), \widehat{\sigma}^2(\lambda), \lambda|\mathbf{Y}) = \text{constant} - \frac{n}{2}\log(\text{RSS}(\lambda)) + \lambda\sum_{i=1}^n \log(Y_i).$$

Then, λ that maximises $\log L(\lambda)$, or any value that is close to such λ and has some meaningful interpretation (e.g. $0, \pm 0.5, \pm 1$, etc.) is selected.

6.1.1 Example: Prestige data (cont'd)

We continue with the analysis of `Prestige` data frame from Section 3.8.2. The residual plots in Figure 5.1 indicate that the linearity assumption may not be valid for this dataset. To visualise the relationship between `income`, `prestige` and `women`, we plot a three-dimensional interactive graph with both regressors and the response, superimposed with the linear surface fitted from `pres.fit2` from page 31. From Figure 6.1, we can observe non-linear patterns between the regressors and response, and also three or four outliers more distinctively.

We can try to transform both the regressors and response and see how this improves the model fit. Since the average income of any given profession is positive, a logarithmic transformation is applicable to `income`. Also, as the relationship between `income` and large values of `prestige` or `women` appears to be different from that between `income` and

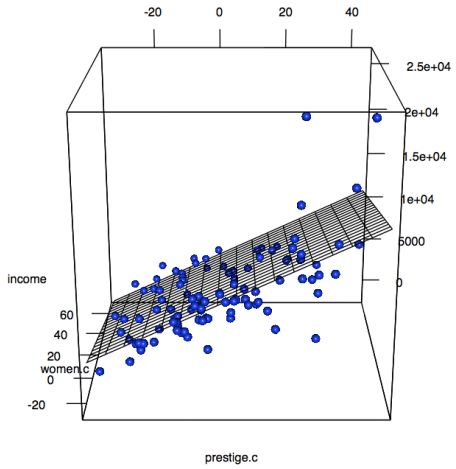


Figure 6.1: A snap shot of the three-dimensional interactive graph plotting the relationship between `income`, `prestige` and `women`, superimposed with the regression surface fitted from `pres.fit2`.

`prestige` or `women` at their small values, we could try to include squares of both regressors.

```
pres.fit3 <- lm(log(income) ~ prestige.c + women.c
          + I(prestige.c^2) + I(women.c^2), data = newdata)
# I(x^2) indicates that x^2 should be treated 'as is'
# check what happens if the model is fitted without I()
summary(pres.fit3)

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      8.809e+00  5.944e-02 148.188  < 2e-16 ***
prestige.c       2.518e-02  1.787e-03  14.096  < 2e-16 ***
women.c         -6.306e-03  1.476e-03  -4.271 4.53e-05 ***
I(prestige.c^2) -2.605e-04  9.396e-05  -2.773  0.00666 **
I(women.c^2)    -7.194e-05  4.014e-05  -1.792  0.07620 .

Residual standard error: 0.293 on 97 degrees of freedom
Multiple R-squared:  0.7643,Adjusted R-squared:  0.7546
F-statistic: 78.64 on 4 and 97 DF,  p-value: < 2.2e-16
```

Since the response variables are in different scales, we cannot directly compare the output from `pres.fit3` with that from `pres.fit2` using `anova` test. However, R^2_{adj} has been improved considerably. Most regressors appear to be significant except for `women^2`, so we re-fit the data after excluding this variable.

```
pres.fit4 <- lm(log(income) ~ prestige.c + women.c +
          + I(prestige.c^2), data = newdata)
summary(pres.fit4)

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)      8.729e+00  4.023e-02 216.995  < 2e-16 ***
prestige.c       2.499e-02  1.803e-03  13.858  < 2e-16 ***
women.c         -8.365e-03  9.376e-04  -8.922 2.64e-14 ***
I(prestige.c^2) -2.351e-04  9.392e-05  -2.503    0.014 *

Residual standard error: 0.2963 on 98 degrees of freedom
Multiple R-squared:  0.7565,Adjusted R-squared:  0.7491
F-statistic: 101.5 on 3 and 98 DF,  p-value: < 2.2e-16

anova(pres.fit4, pres.fit3)

Model 1: log(income) ~ prestige.c + women.c + I(prestige.c^2)
Model 2: log(income) ~ prestige.c + women.c + I(prestige.c^2)
          + I(women.c^2)
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     98 8.6027
2     97 8.3270  1   0.27576 3.2124 0.0762 .
```

The ANOVA table supports the reduced model `pres.fit4` since adding `I(women.c^2)` does not bring in significant reduction in the residual sum of squares, as indicated by the p-value $= 0.0762 > 0.05$.

The function `boxcox` from the package `MASS` performs the Box-Cox transform from which we can select the most suitable transform of the response.

```
library(MASS)
bc <- boxcox(lm(income ~ prestige.c + women.c + I(prestige.c^2),
data=newdata), lambda = seq(-2, 2, length.out=10))
bc$x[which.max(bc$y)]
```

```
[1] 0.1818182
```

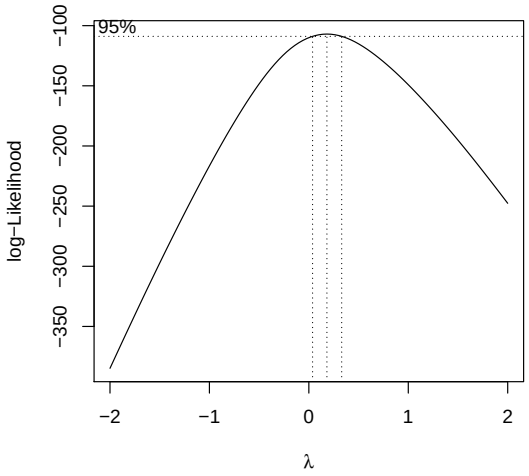


Figure 6.2: Profile log-likelihoods for the parameter of the Box-Cox power transformation applied to Prestige data.

Figure 6.2 is an outcome returned by the first command, where it appears that the maximum log-likelihood is attained at λ close to zero. Indeed, the maximum is attained at $\lambda = 0.181$, which justifies the model fitted by `pres.fit4`.
Compared to plots in Figure 5.1, the residual plots in Figure 6.3 indicate that patterns shown in the residuals are somewhat lessened after transformations. Observations associated with different professions appear to be outliers under the new model, but their influence does not seem to be unduly high as their Cook's distance are still less than one.

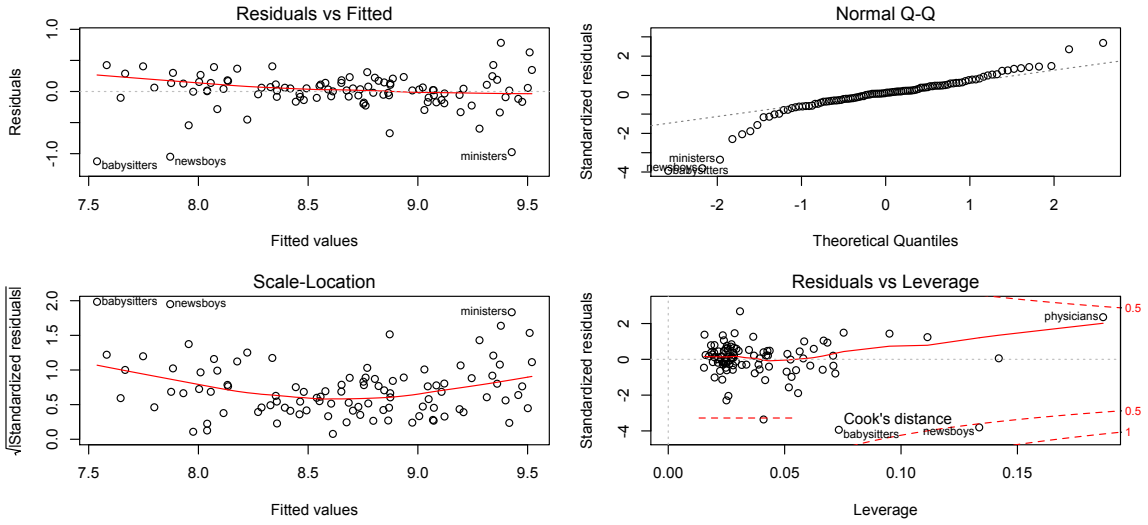


Figure 6.3: Residual plots from `pres.fit4`. Top left: e_i vs. $\widehat{\mu}_i$; bottom left: $\sqrt{|s_i|}$ vs. $\widehat{\mu}_i$; top right: normal Q-Q plot of s_i; bottom right: s_i vs. h_{ii} along with contours of Cook's distance.

6.2 Generalised least squares estimation

Consider the situation where the assumptions of equal variance and/or uncorrelatedness across ε_i, $1 \leq i \leq n$, no longer hold. That is, $\boldsymbol{\varepsilon} \sim (0, \sigma^2\mathbf{V})$ with $\mathbf{V} \neq \mathbf{I}_n$. When $\mathbf{V}$ is diagonal, the response variables Y_i, $1 \leq i \leq n$, are uncorrelated but may have unequal variance. If off-diagonal elements of $\mathbf{V}$ are non-zero, the observations are correlated and no longer independent. Then, $\widehat{\boldsymbol{\beta}}^{\text{ols}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ (OLS stands for ordinary least squares) is still an *unbiased* estimator, but it is not guaranteed to be optimal in the sense that it is no longer BLUE (best linear unbiased estimator), i.e. it is no longer guaranteed to attain the minimum variance among all unbiased, linear estimators (see Gauss-Markov theorem in Section 3.5).

Since $\mathbf{V}$ is the covariance matrix of $\sigma^{-1}\boldsymbol{\varepsilon}$, it is positive definite and symmetric, and admits the eigendecomposition

$$\mathbf{V} = \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^\top \quad \text{where} \quad \mathbf{E}\mathbf{E}^\top = \mathbf{I}_n \text{ and } \boldsymbol{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_n),$$

see Sections 2.1.1 and 2.2. Letting $\boldsymbol{\Gamma} = \mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top$ with $\boldsymbol{\Lambda}^{-1/2} = \text{diag}(\lambda_1^{-1/2}, \ldots, \lambda_n^{-1/2})$, we have

$$\boldsymbol{\Gamma}\boldsymbol{\Gamma}^\top = \underline{\mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top\mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top = \mathbf{E}\boldsymbol{\Lambda}^{-1}\mathbf{E}^\top = \mathbf{V}^{-1},}$$

and moreover, $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}^\top$ so that we can regard $\boldsymbol{\Gamma}$ as $\mathbf{V}^{-1/2}$. Then, by left-multiplying $\mathbf{Y}$, $\mathbf{X}$ and $\boldsymbol{\varepsilon}$ with $\boldsymbol{\Gamma}$ to obtain $\widetilde{\mathbf{Y}}$, $\widetilde{\mathbf{X}}$ and $\widetilde{\boldsymbol{\varepsilon}}$, we have

$$\begin{aligned}
\widetilde{\mathbf{Y}} &= \widetilde{\mathbf{X}}\boldsymbol{\beta}_0 + \widetilde{\boldsymbol{\varepsilon}}, \quad \text{where} \\
\mathsf{E}(\widetilde{\boldsymbol{\varepsilon}}) &= \boldsymbol{\Gamma}\mathsf{E}(\boldsymbol{\varepsilon}) = \mathbf{0} \quad \text{and} \\
\mathsf{Cov}(\widetilde{\boldsymbol{\varepsilon}}) &= \underline{\sigma^2\boldsymbol{\Gamma}\mathbf{V}\boldsymbol{\Gamma}^\top = \sigma^2\mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top\mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^\top\mathbf{E}\boldsymbol{\Lambda}^{-1/2}\mathbf{E}^\top = \sigma^2\mathbf{I}_n.}
\end{aligned}$$

Therefore, the OLS estimator of $\boldsymbol{\beta}_0$ under the model $\widetilde{\mathbf{Y}} \sim (\widetilde{\mathbf{X}}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$ is given by

$$\begin{aligned}
\widehat{\boldsymbol{\beta}}^{\text{gls}} &= (\widetilde{\mathbf{X}}^\top\widetilde{\mathbf{X}})^{-1}\widetilde{\mathbf{X}}^\top\widetilde{\mathbf{Y}} = \underline{(\mathbf{X}^\top\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}\mathbf{Y}} \\
&= \underline{(\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{Y}}
\end{aligned}$$

(GLS stands for generalised least squares). By Gauss-Markov theorem, $\widehat{\boldsymbol{\beta}}^{\text{gls}}$ is the linear

unbiased estimator of $\boldsymbol{\beta}_0$ with the minimum variance, i.e. for any $\mathbf{c} \in \mathbb{R}^p$,

$$\mathsf{Var}(\mathbf{c}^\top \widehat{\boldsymbol{\beta}}^{\text{gls}}) \leq \mathsf{Var}(\mathbf{c}^\top \widehat{\boldsymbol{\beta}}^{\text{ols}}).$$

When $\mathbf{V}$ is diagonal, say $\mathbf{V} = \text{diag}(1/w_1, \ldots, 1/w_n)$, $\boldsymbol{\Gamma}$ takes a simpler form

$$\boldsymbol{\Gamma} = \mathbf{V}^{-1/2} = \text{diag}(\sqrt{w_1}, \ldots, \sqrt{w_n}),$$

with which it is easily seen that

$$\widetilde{\mathbf{Y}} = \boldsymbol{\Gamma}\mathbf{Y} = \begin{bmatrix} \sqrt{w_1} Y_1 \\ \vdots \\ \sqrt{w_n} Y_n \end{bmatrix} \quad \text{and} \quad \widetilde{\mathbf{X}} = \boldsymbol{\Gamma}\mathbf{X} = \begin{bmatrix} \sqrt{w_1} \mathbf{x}_1^\top \\ \vdots \\ \sqrt{w_n} \mathbf{x}_n^\top \end{bmatrix}.$$

Then, the least squares estimator of $\boldsymbol{\beta}_0$ under $\widetilde{\mathbf{Y}} \sim (\widetilde{\mathbf{X}}\boldsymbol{\beta}_0, \sigma^2 \mathbf{I}_n)$ is the minimiser of

$$\widetilde{S}(\boldsymbol{\beta}) = \|\widetilde{\mathbf{Y}} - \widetilde{\mathbf{X}}\boldsymbol{\beta}\|^2 = \sum_{i=1}^n (\widetilde{Y}_i - \widetilde{\mathbf{x}}_i^\top \boldsymbol{\beta})^2 = \sum_{i=1}^n (\sqrt{w_i} Y_i - \sqrt{w_i} \mathbf{x}_i^\top \boldsymbol{\beta})^2$$
$$= \sum_{i=1}^n w_i (Y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{W} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

where $\mathbf{W} = \text{diag}(w_1, \ldots, w_n) = \mathbf{V}^{-1}$. The minimiser of $\widetilde{S}(\boldsymbol{\beta})$, referred to as weighted least squares (WLS) estimator of $\boldsymbol{\beta}_0$, is given by $\widehat{\boldsymbol{\beta}}^{\text{wls}} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{Y}$.

6.2.1 Example: Cars data

The `cars` data frame from the `car` package has 50 observations on the speed of cars and the distances taken to stop, which were recorded in the 1920s. Figure 6.4 plots the relationship between `dist` and `speed`.

`speed`	numeric, speed (mph)
`dist`	numeric, stopping distance (ft)

Theoretically, the stopping distance for a car is made up of a driver reaction distance and a braking distance. The former comes from the fixed length of time taken for the driver to respond to the stop signal and apply the brakes, so this distance is proportional to `speed`.

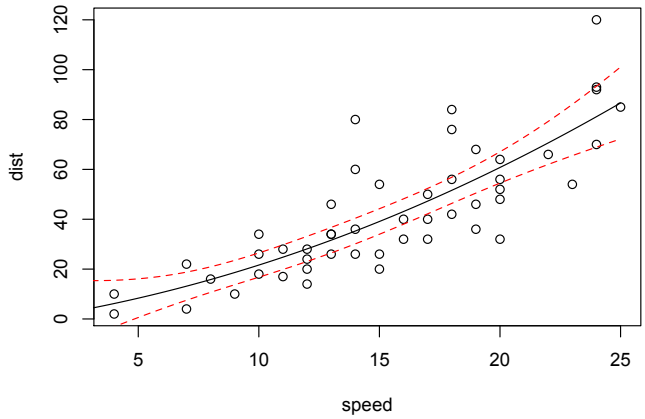


Figure 6.4: `dist` vs. `speed` from `cars` data frame, superimposed by the predicted values from the fitted model and the 95% confidence intervals.

Once the brakes are applied, they spend up the car's kinetic energy in direct proportion to the distance travelled, and the total amount of energy to dissipate is proportional to $\texttt{speed}^2$, so the braking distance should be proportional to the square of the initial speed. These observations suggest a model

$$\texttt{dist}_i = \beta_0 + \beta_1 \texttt{speed}_i + \beta_2 \texttt{speed}_i^2 + \varepsilon_i.$$

We fit this model to the data and examine the resulting residuals.

```
car.fit1 <- lm(dist ~ speed + I(speed^2), data = cars)
summary(car.fit1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.47014   14.81716   0.167    0.868
speed        0.91329    2.03422   0.449    0.656
I(speed^2)   0.09996    0.06597   1.515    0.136

Residual standard error: 15.18 on 47 degrees of freedom
```

```
Multiple R-squared:  0.6673,Adjusted R-squared:  0.6532
F-statistic: 47.14 on 2 and 47 DF,  p-value: 5.852e-12
```

Figure 6.5 contains diagnostic figures generated by `plot` method.

```
plot(car.fit1)
```

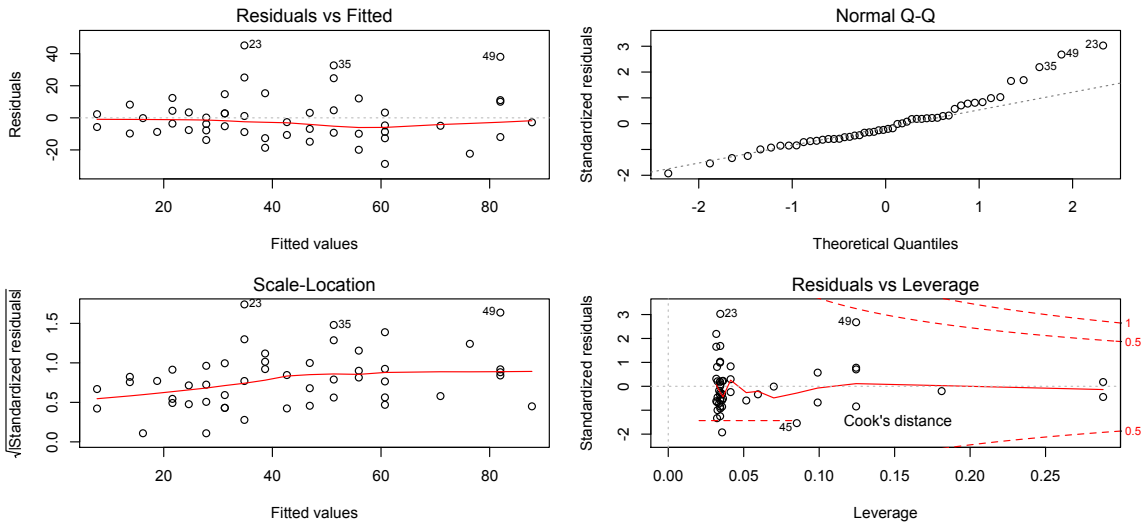


Figure 6.5: Residual plots from `car.fit1`. Top left: e_i vs. $\widehat{\mu}_i$; bottom left: $\sqrt{|s_i|}$ vs. $\widehat{\mu}_i$; top right: normal Q-Q plot of s_i; bottom right: s_i vs. h_{ii} along with contours of Cook's distance.

The two left-hand plots of Figure 6.5 indicate possible heteroscedasticity of Y_i. The megaphone-shaped increase of the variability in the residuals imply that the variance $\mathsf{Var}(Y_i) = \sigma_i^2$ increases with the mean $\mathsf{E}(Y_i) = \mu_i$, which violates the assumption of constant variance. On the other hand, the average of e_i as a function of $\widehat{\mu}_i$ does not seem to show any discernible pattern. The normal Q-Q plot suggests that the distribution of standardised residuals may be skewed to the right. In the bottom right plot, we do not detect any observation with excessively large influence according to Cook's distance.

Based on the above observations, it seems reasonable that we allow the variability in Y_i to increase with its mean. Since the mean is modelled as a function of `speed`, we first consider $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 \texttt{speed}_i^2)$.

```
car.fit2 <- lm(dist ~ speed + I(speed^2),
               weights = 1/(speed^2), data = cars)
summary(car.fit2)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.48008    5.93655  -0.249    0.804
speed        1.53205    1.16773   1.312    0.196
I(speed^2)   0.07887    0.04858   1.624    0.111

Residual standard error: 0.9782 on 47 degrees of freedom
Multiple R-squared:  0.752,Adjusted R-squared:  0.7414
F-statistic: 71.26 on 2 and 47 DF,  p-value: 5.883e-15
```

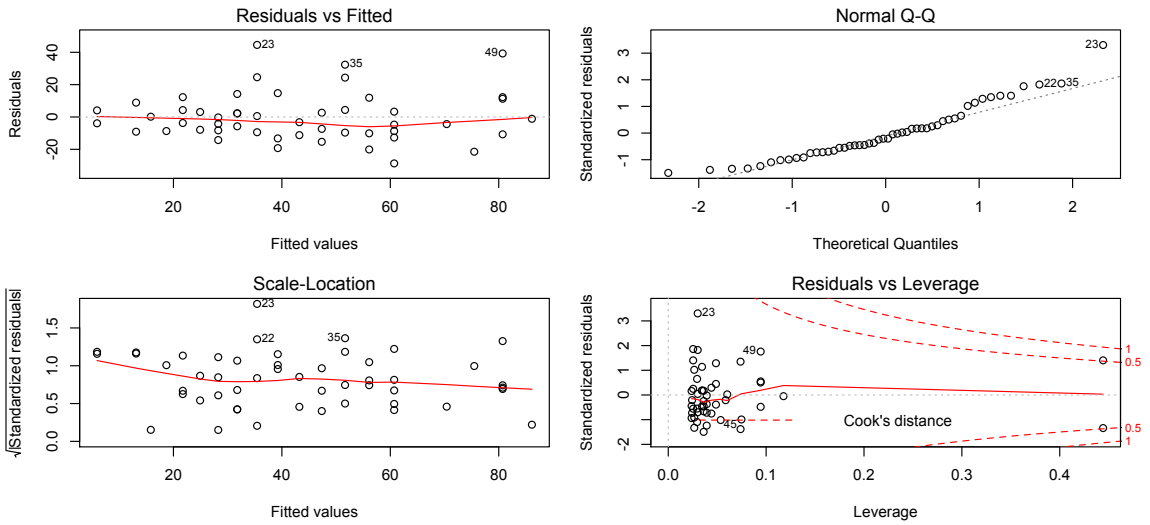


Figure 6.6: Residual plots from `car.fit2`. Top left: e_i vs. $\widehat{\mu}_i$; bottom left: $\sqrt{|s_i|}$ vs. $\widehat{\mu}_i$; top right: normal Q-Q plot of s_i; bottom right: s_i vs. h_{ii} along with contours of Cook's distance.

The residual plots in Figure 6.6 show that violations of assumptions observed previously are somewhat remedied by fitting the weighted least squares estimator. More specifically, we no longer observe any trend in the variability of s_i (bottom left), and the distribution

of s_i appears closer to to the standard normal distribution. Also, we notice improvement in both the F-statistic and R^2_{adj} as well as the increase of individual Wald test statistics. The solid and dotted lines superimposed on the observations from `cars` data frame in Figure 6.4 are predicted values based on `car.fit2` and the associated confidence intervals, and are generated as below.

```
with(cars, plot(speed, dist))
dat <- data.frame(speed = seq(0, 25, length=100))
pp <- predict(car.fit2, newdata=dat, se=TRUE)
lines(dat$speed, pp$fit)
lines(dat$speed, pp$fit+1.96*pp$se.fit, lty=2, col=2)
lines(dat$speed, pp$fit-1.96*pp$se.fit, lty=2, col=2)
```

7 Model selection

Suppose the data contains $X_1, \ldots, X_p$ as candidate variables to be included in a regression model for the given response Y. So far we have assumed that all the explanatory variables given in the dataset are relevant in modelling $\mathsf{E}(Y)$. However, in many situations, selecting a subset of $(X_1, \ldots, X_p)$ that explains the mean response well is an important part of the analysis. This, in addition to enhancing the interpretability of the model, can improve the predictive performance of the regression model and reduce the variance of the estimated coefficients.

In what follows, we denote by $\mathcal{M}$ a model in consideration which is usually represented by the index set of the variables included in the model, such that $\mathcal{M} \subset \{1, \ldots, p\}$. Accordingly, the design matrix $\mathbf{X}(\mathcal{M})$ and the vector of linear coefficients $\boldsymbol{\beta}_0(\mathcal{M})$ are determined.

7.1 Stepwise model building methods

Model comparison methods are often adopted in order to find a model that is 'best' according to some criterion among all 2^p candidate models. When the number of candidate variables p is small, it is possible to fit all the candidate models exhaustively and select the one that meets your chosen criterion. This, however, is time-consuming or even impossible when p is large. The following *stepwise* approaches provide a computationally feasible method for systematically searching for a model.

(i) **Backward selection:** Starting with the largest plausible model, it repeatedly deletes the variables from the current model one at a time, and re-fits the model with the remaining variables, until a certain criterion is met. Traditionally, p-value from the Wald test has been adopted as a criterion, by which the variable with the largest p-value is repeatedly removed until all p-values are below some threshold.

Drawback: It may not be possible to fit the largest plausible model when p is large.

(ii) **Forward selection:** Starting from the simplest null model (with the intercept only), it adds a single variable to the current model at each iteration for which there is the most evidence until no more variables would lead to significant improvement of the fit. Again, the p-value from the F-test between nested models can be adopted as a criterion.

Drawback: Early in the process, it is likely that both models in consideration are 'wrong' (too small). Then, the residual variance in the denominator of the F-test statistic

in (3.11) at early stages may be seriously inflated (too few variables in the model missing important variables), which results in premature termination of the selection process.

(iii) **Backward-forward selection:** Cycles of backward and forward selection are alternatively repeated until convergence, in order to allow variables that were dropped earlier to re-enter the current model.

Model selection methods based on the Wald or F-tests select the simplest model compatible with the observations, where compatibility is determined at some arbitrary significance level. In Section 7.2, we discuss information criteria that can be used instead as a model selection criterion in combination with the step-wise model building strategy.

Remark 7.1. When a model selection procedure is performed, care should be taken in interpreting the final model since it is not valid to interpret Wald or F-test results as if the current model was the only model we considered. Rather, the same dataset has been repeatedly used in order to fit several models and select the final model.
Figure 7.1 illustrates this point through an experiment. In a simulation study with $p = 75$ and all $\beta_j = 0$, a model of size $q = 4$ was selected according to the associated p-values. Repeating the experiments multiple times and plotting the distribution of $\widehat{\beta}_1$ in Figure 7.1, it has a point mass at zero, reflecting the situations that the variable was not selected. In addition, there is a continuous mass distribution over those times when the variable showed up in the final four-variable model. This distribution is far from a normal distribution, and its relation to the original coefficients is obscure.
In summary, after the above model selection procedures are performed, the p-value associated with $\widehat{\beta}_j$ in the final model can no longer be treated as we discussed in Section 3.6.2.

7.2 Information criteria

If we select the model with the minimum RSS among the subsets of $\{1, 2, \ldots, p\}$, the one with the most variables (referred to as the full model) is always selected since adding variables monotonically decreases the RSS. In order to select a *parsimonious* model that does well without including too many variables, we adopt another approach minimising some score or criterion, such as Akaike information criterion (AIC) and Mallow's C_p. They

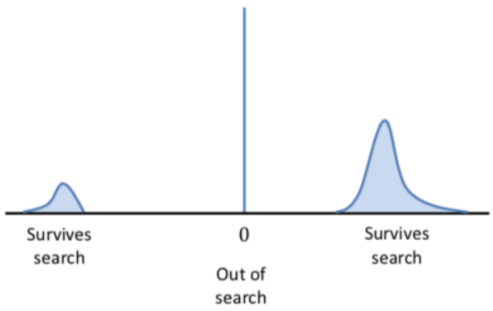


Figure 7.1: Taken from Breiman (1992). The distribution of $\widehat{\beta}_1$ after model selection procedure. See the text for more descriptions.

take the following general form:

$$\text{goodness-of-fit} \ + \ \text{penalty on model complexity.}$$

Information criteria such as AIC have been applied for model selection in a wider context with the negative log-likelihood as a goodness-of-fit measure. This is closely related to the RSS in the context of linear modelling. The model complexity is typically proportional to the number of variables included in the model. Compared to the selection procedures based on F-statistics, which aims at finding the model that gets as close as possible to the observed Y_i's, the aim of above criteria is to select the model that does well in predicting $\mu_i = \mathsf{E}(Y_i)$.

7.2.1 Akaike information criterion

The Akaike information criterion (AIC) is defined as

$$\text{AIC}(\mathcal{M}) = -2\ell(\widehat{\boldsymbol{\beta}}(\mathcal{M}), \widehat{\sigma}^2(\mathcal{M})) + 2|\mathcal{M}| = n \log\left(\frac{\text{RSS}(\mathcal{M})}{n}\right) + 2|\mathcal{M}|$$

for Gaussian linear models, where $|\mathcal{M}|$ denotes the number of variables (cardinality) included in the current model $\mathcal{M}$. Below, we discuss how the criterion is derived.

Under the normality assumption, [6]

$$\begin{aligned} &\text{maximises} \quad L(\mathcal{M}, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{\|\mathbf{Y} - \mathbf{X}(\mathcal{M})\boldsymbol{\beta}(\mathcal{M})\|^2}{2\sigma^2}\right) \\ \iff &\text{minimises} \quad S(\mathcal{M}) = \|\mathbf{Y} - \mathbf{X}(\mathcal{M})\boldsymbol{\beta}(\mathcal{M})\|^2 \end{aligned} \tag{7.1}$$

Generally, the more parameters are added to the model, the more closely it fits Y_i, $1 \leq i \leq n$, in terms of reducing the RSS even if that means fitting the *noise* component of Y_i. This *over-fitting* problem would be resolved if we could find the model maximising

$$\widetilde{L}(\mathcal{M}, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{\|\boldsymbol{\mu} - \mathbf{X}(\mathcal{M})\boldsymbol{\beta}(\mathcal{M})\|^2}{2\sigma^2}\right), \tag{7.2}$$

where $\boldsymbol{\mu} = \mathsf{E}(\mathbf{Y})$. Obviously we cannot find $\mathcal{M}$ maximising (7.2), since $\boldsymbol{\mu}$ is not observable, and thus we adopt a series of approximation. Recall that $\mathbf{X}(\mathcal{M})\widehat{\boldsymbol{\beta}}(\mathcal{M}) = \mathbf{H}(\mathcal{M})\mathbf{Y}$ (see page 15) and denote by $q = |\mathcal{M}|$ = number of columns of $\mathbf{X}(\mathcal{M})$. For notational convenience, the dependence on $\mathcal{M}$ is omitted. Then,

$$\begin{aligned} \|\boldsymbol{\mu} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2 &= \|\boldsymbol{\mu} - \mathbf{H}\mathbf{Y}\|^2 = \|\mathbf{Y} - \mathbf{H}\mathbf{Y} - \boldsymbol{\varepsilon}\|^2 \\ &= \|\mathbf{Y} - \mathbf{H}\mathbf{Y}\|^2 + \boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon} - 2\boldsymbol{\varepsilon}^\top(\mathbf{Y} - \mathbf{H}\mathbf{Y}) \\ &= \|\mathbf{Y} - \mathbf{H}\mathbf{Y}\|^2 + \boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon} - 2\boldsymbol{\varepsilon}^\top(\mathbf{Y} - \boldsymbol{\mu} - \mathbf{H}\boldsymbol{\varepsilon}) \\ &= \|\mathbf{Y} - \mathbf{H}\mathbf{Y}\|^2 + \boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon} - 2\boldsymbol{\varepsilon}^\top(\mathbf{I}_n - \mathbf{H})\boldsymbol{\varepsilon} \\ &= \|\mathbf{Y} - \mathbf{H}\mathbf{Y}\|^2 - \boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon} + 2\boldsymbol{\varepsilon}^\top\mathbf{H}\boldsymbol{\varepsilon}. \end{aligned}$$

Then, $\mathsf{E}(\boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon}) =$ $n\sigma^2$, and

$$\mathsf{E}(\boldsymbol{\varepsilon}^\top\mathbf{H}\boldsymbol{\varepsilon}) = \mathsf{E}(\mathsf{tr}(\mathbf{H}\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top)) = \mathsf{tr}(\mathbf{H} \cdot \sigma^2\mathbf{I}_n) = \sigma^2 \sum_{i=1}^{n} h_{ii} = q\sigma^2. \tag{7.3}$$

Hence,

$$\begin{aligned} &\mathsf{E}\|\boldsymbol{\mu} - \mathbf{X}(\mathcal{M})\widehat{\boldsymbol{\beta}}(\mathcal{M})\|^2 = \mathsf{E}\|\mathbf{Y} - \mathbf{H}(\mathcal{M})\mathbf{Y}\|^2 - (n - 2q)\sigma^2 \quad \text{and} \\ &\|\boldsymbol{\mu} - \mathbf{X}(\mathcal{M})\widehat{\boldsymbol{\beta}}(\mathcal{M})\|^2 \approx \|\mathbf{Y} - \mathbf{H}(\mathcal{M})\mathbf{Y}\|^2 - (n - 2q)\widehat{\sigma}^2. \end{aligned} \tag{7.4}$$

[6] See problem sheet Section 3 Q3.

Therefore, $\widetilde{\ell} = \log(\widetilde{L})$ is approximated by

$$\begin{aligned}\widehat{\widetilde{\ell}} &= -\frac{n}{2}\log(\widehat{\sigma}^2) - \frac{\|\mathbf{Y} - \mathbf{X}(\mathcal{M})\widehat{\boldsymbol{\beta}}(\mathcal{M})\|^2}{2\widehat{\sigma}^2} + \frac{n-2q}{2} + \text{constant} \\ &= \ell(\widehat{\boldsymbol{\beta}}(\mathcal{M}), \widehat{\sigma}^2) - q + \text{constant},\end{aligned}$$

where with some abuse of notation, $\ell(\widehat{\boldsymbol{\beta}}(\mathcal{M}), \widehat{\sigma}^2)$ denotes the log-likelihood from (7.1) evaluated at the MLE $\widehat{\boldsymbol{\beta}}(\mathcal{M})$ and

$$\widehat{\sigma}^2 = \widehat{\sigma}^2(\mathcal{M}) = \frac{1}{n}\|\mathbf{Y} - \mathbf{X}(\mathcal{M})\widehat{\boldsymbol{\beta}}(\mathcal{M})\|^2 = \frac{1}{n}\text{RSS}(\mathcal{M}).$$

Any model that maximises $\widehat{\widetilde{\ell}}$ minimises

$$\text{AIC}(\mathcal{M}) = -2\ell(\widehat{\boldsymbol{\beta}}(\mathcal{M}), \widehat{\sigma}^2(\mathcal{M})) + 2q = n\log\left(\frac{\text{RSS}(\mathcal{M})}{n}\right) + 2q.$$

In `R`, AIC of the fitted model can be extracted by the method `AIC`.

7.2.2 Mallows's C_p

Mallows's C_p is based on minimising the following mean squared error (MSE)

$$\mathsf{E}\|\widehat{\boldsymbol{\mu}}(\mathcal{M}) - \boldsymbol{\mu}\|^2.$$

Following the identical steps leading to (7.4),

$$\|\widehat{\boldsymbol{\mu}}(\mathcal{M}) - \boldsymbol{\mu}\|^2 \approx \text{RSS}(\mathcal{M}) - (n - 2|\mathcal{M}|)\sigma^2,$$

where $2|\mathcal{M}|$ represents the penalty on the model complexity. Again, denoting by $q = |\mathcal{M}|$, minimising the above is approximated by minimising

$$C_p(\mathcal{M}) = \frac{\text{RSS}(\mathcal{M})}{\widetilde{\sigma}^2} - n + 2q.$$

Here, $\widetilde{\sigma}^2$ is an estimate of σ^2, which is typically fixed at the estimate given by the fit of the largest candidate model (unless σ^2 is actually known). [7]

[7] See Q1.

7.3 Cross-validation

Cross-validation (CV) is a model validation technique which (randomly) partitions the data into training and test sets, uses the training data to fit a model and tests its validity on the test data. It is another general model selection method aimed at estimating how accurately a model will perform in prediction, as different parts of the data are used for model fitting and testing.

The K-fold cross-validation randomly partitions the observations into K groups where, each of the K groups serving as test data once, fits the model using the observations from $K-1$ groups as the training data and tests its validity on the remaining test data. For simplicity, suppose that $n = KN$ and denote by Y_{ki} the i-th observation in the k-th group, and $\widehat{\mu}_i^{(k)}(\mathcal{M})$ the prediction of $\mu_{ki} = \mathsf{E}(Y_{ki})$ based on the model $\mathcal{M}$ fitted to $Y_{k'i}$, $1 \le i \le N$, $k' \in \{1, \ldots, K\} \setminus \{k\}$. Then, measuring the test error by the mean cross-validated error:

$$\mathrm{CV}(\mathcal{M}) = \sum_{k=1}^{K} \sum_{i=1}^{N} (Y_{ki} - \widehat{\mu}_i^{(k)}(\mathcal{M}))^2,$$

we select the model $\mathcal{M}$ returning the smallest $\mathrm{CV}(\mathcal{M})$.

Often, $K = 5$ or 10 are used but other choices of K are possible. When $K = n$, we repeatedly leave a single observation out as a test set where the model fitted with $(n-1)$ observations is evaluated, until all n observations serve as a test set once. Also known as the leave-one-out CV, we have

$$\mathrm{CV}(\mathcal{M}) = \sum_{i=1}^{n} (Y_i - \widehat{\mu}_i^{(i)}(\mathcal{M}))^2,$$

where $\widehat{\mu}_i^{(i)}(\mathcal{M})$ denotes the predicted value of $\mu_i = \mathsf{E}(Y_i)$ from the model $\mathcal{M}$ fitted without the i-th observation. From the results derived for the jackknife residuals (Section 5.1.3), we derive a short cut to calculating the leave-one-out CV. Omitting $\mathcal{M}$ for brevity, recall that $e_i = Y_i - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}$ denote the residuals and $e_i^{(i)} = Y_i - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_{(i)}$ denote the residuals from the model fitted without the i-th observation. Then,

$$\mathrm{CV}(\mathcal{M}) = \sum_{i=1}^{n} (e_i^{(i)})^2 = \sum_{i=1}^{n} \frac{e_i^2}{(1-h_{ii})^2},$$

by (5.7), i.e. the leave-one-out CV can be performed without having to fit the model $\mathcal{M}$ n

times – it suffices to fit it with all n observations once.

7.4 R examples

7.4.1 Example: Cars data (cont'd)

Recall from the output from `summary(car.fit2)` in Section 6.2.1, none of the fitted parameters appear to be significant according to the Wald test. Indeed, the p-values for all the model terms are very high, even though

- Figure 6.4 shows that the model does reasonably well in fitting the data, and
- F-test clearly rejects the null model with the intercept only, in favour of the model fitted in `car.fit2`.

So should we drop all the terms from `car.fit2`?
The answer is no. The Wald test is designed to test whether β_j could really be zero given that *all the other terms remain in the model* ($\beta_{j'} \neq 0$ for all $j' \neq j$). As the coefficient estimates are typically not independent (recall that $\widehat{\boldsymbol{\beta}} \sim \mathcal{N}_p(\boldsymbol{\beta}_0, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$ where $\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$ is not necessarily diagonal), dropping one variable (= setting the corresponding coefficient to zero) will change the estimates of the other coefficients and hence their p-values. Therefore, if we are to drop variables from the model, we should drop only one at a time and re-fit the reduced model after each drop.
Since the intercept had the largest p-value, we first remove the intercept from the model.

```
car.fit3 <- lm(dist ~ 0 + speed + I(speed^2),
         weights = 1/(speed^2), data = cars)
# alternatively, dist ~ speed + I(speed^2) - 1
summary(car.fit3)

Coefficients:
          Estimate Std. Error t value Pr(>|t|)
speed       1.26136    0.42563   2.963  0.00472 **
I(speed^2)  0.08904    0.02617   3.402  0.00136 **

Residual standard error: 0.9686 on 48 degrees of freedom
Multiple R-squared:  0.8881,Adjusted R-squared:  0.8834
```

```
F-statistic: 190.5 on 2 and 48 DF,  p-value: < 2.2e-16
```

This brings in an immediate improvement of the fit not only according to F-statistic and R^2_{adj}, but also with regards to the p-values associated with individual coefficient estimates. We also apply the `AIC` method to compare the two models, which favours `car.fit3` over `car.fit2`.

```
AIC(car.fit3, car.fit2)
```

```
         df      AIC
car.fit3  3 408.8608
car.fit2  4 410.7947
```

7.4.2 Example: Housing data

The dataset `Housing` from the `R` package `Ecdat` contains sales prices of houses in the city of Windsor, Canada in 1987, with $n = 546$. Eleven variables were observed about various aspects about the houses, including the size of a property, the number of bedrooms/bathrooms and so forth; details of the explanatory variables can be found in `help(Housing)`.

```
library(Ecdat)
data(Housing)
head(Housing)
```

```
  price lotsize bedrooms bathrms stories driveway recroom ...
1 42000    5850        3       1       2      yes      no      ...
2 38500    4000        2       1       1      yes      no       ...
3 49500    3060        3       1       1      yes      no       ...
4 60500    6650        3       1       2      yes     yes       ...
5 61000    6360        2       1       1      yes      no       ...
```

When the number of candidate variables q is small, it may be possible to fit all 2^q submodels and select the one that meets your chosen criterion. The function `regsubsets` in the package `leaps` implements regression subset selection and allows one to view the ranked models according to different scoring criteria by plotting the results of `regsubsets`.

```
library(leaps)
subset.fit <- regsubsets(price ~ ., data = Housing, nbest = 10)
plot(subset.fit, scale = 'adjr2')
plot(subset.fit, scale = 'Cp')
```

Figure 7.2 illustrates the ranked models according to the adjusted R-squared criteria and Mallow's C_p, respectively. According to both criteria, the model containing `lotsize`,

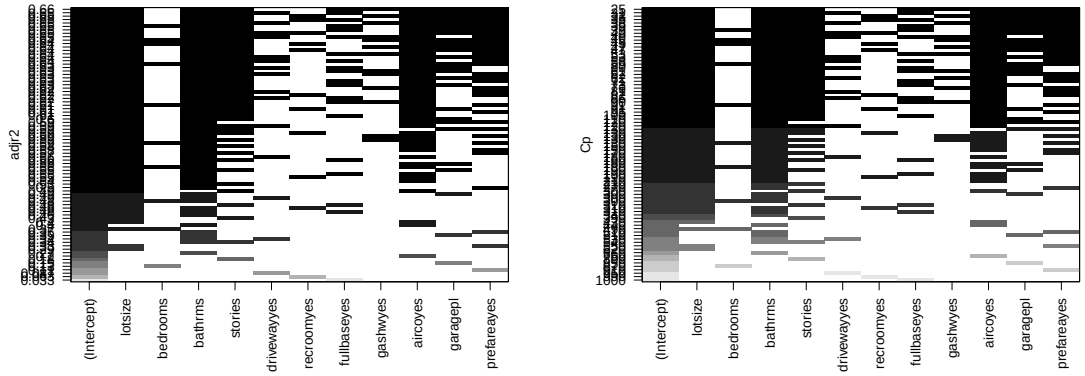


Figure 7.2: Best subsets of varying size ranked according to R^2_{adj} (left) and Mallow's C_p (right) obtained from plotting the `regsubsets` object `subset.fit`.

`bathrms`, `stories`, `fullbase`, `gashw`, `airco`, `garagepl` and `prefarea` are selected.
The exhaustive search performed by `regsubsets` becomes quickly impossible with increasing q. The `step` function in the package `MASS` automates the process of backward, forward, or backward-forward selection based on AIC. To perform forward selection, we need to specify a starting model and the range of models which we want to examine in the search. The following code tells `R` to start with the null model and search through models lying in the range between the null and full models using the forward selection algorithm.

```
library(MASS)
null <- lm(price ~ 1, data = Housing)
full <- lm(price ~ ., data = Housing)
step(null, scope = list(lower = null, upper = full), direction = "forward")
```

It gives rise to the following output.

```
Start:  AIC=11131.23
price ~ 1

           Df  Sum of Sq        RSS   AIC
+ lotsize   1 1.1156e+11 2.7704e+11 10948
+ bathrms   1 1.0376e+11 2.8485e+11 10964
+ airco     1 7.9867e+10 3.0874e+11 11008
.
.
.
<none>                   3.8860e+11 11131

(truncated for presentation)

Step:  AIC=10543.83
price ~ lotsize + bathrms + airco + stories + prefarea + garagepl +
    fullbase + gashw + driveway + recroom

           Df Sum of Sq        RSS   AIC
+ bedrooms  1 728294831 1.2703e+11 10543
<none>                  1.2775e+11 10544

Step:  AIC=10542.71
price ~ lotsize + bathrms + airco + stories + prefarea + garagepl +
    fullbase + gashw + driveway + recroom + bedrooms
```

According to this procedure, the best model is the one that includes all eleven variables. We can perform backward elimination or stepwise regression on the same data set using the commands:

```
step(full, direction = "backward")
step(full, scope = list(lower = null), direction = "both")
```

8 High-dimensional linear modelling

8.1 Least square estimation in high dimensions

The least squares estimation enjoys a number of desirable properties; it has an analytic solution and, under some assumptions, the ordinary least squares (OLS) estimator is the best linear unbiased estimator of $\boldsymbol{\beta}_0$. However, it also suffers from several shortcomings particularly when faced with *high dimensionality* of the datasets where p, the number of variables, is large relative to the number of observations, n. One typical such example is found in genetic applications: In genome wide associations studies, it is of interest to identify genetic variants across many genomes, which are in the range of hundreds of thousands, that are associated with a specific trait or disease, with the sample size in the range of a few thousands.

When $p > n$, the OLS estimator is not defined, since the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$ is not invertible. Even if $p \leq n$, if p is close to n, inverting $\mathbf{X}^\top \mathbf{X}$ brings large variability to $\widehat{\boldsymbol{\beta}}$, i.e. the OLS estimator suffers from large variance, which deteriorates the prediction accuracy. To see this, let $\mathbf{x}_*$ denote a newly observed vector of covariates for which we wish to predict Y_*, and $\widehat{\mu}_*$ an estimator of $\mathsf{E}(Y_*) = \mu_*$ returned by a fitted model. Then, the mean squared prediction error of $\widehat{\mu}_*$ is decomposed as

$$\begin{aligned}\mathsf{E}[(Y_* - \widehat{\mu}_*)^2] &= \mathsf{E}[(\mu_* + \varepsilon_* - \widehat{\mu}_*)^2] = \sigma^2 + \underbrace{\mathsf{E}[(\widehat{\mu}_* - \mu_*)^2]}_{\text{MSE}} \\ &= \sigma^2 + \underbrace{(\mathsf{E}(\widehat{\mu}_*) - \mu_*)^2}_{\text{bias}} + \underbrace{\mathsf{E}[(\widehat{\mu}_* - \mathsf{E}(\widehat{\mu}_*))^2]}_{\text{variance}}.\end{aligned}$$

Previously we showed that provided that it exists, the OLS estimator is unbiased and so is $\widehat{\mu}_*^{\text{ols}} = \mathbf{x}_*^\top \widehat{\boldsymbol{\beta}}^{\text{ols}}$, i.e. $\mathsf{E}(\widehat{\mu}_*^{\text{ols}}) = \mu_*$. In other words, apart from the irreducible σ^2, the prediction error of the OLS estimator is entirely due to the variance of $\widehat{\mu}_*^{\text{ols}}$. Sometimes, prediction accuracy can be improved by sacrificing a small amount of bias in order to decrease the variance, which is referred to as the bias-variance trade-off.

Shrinking small coefficients towards zero, or setting them to be exactly zero, may introduce some ___bias___ but can reduce the ___variance___ and potentially result in smaller prediction error. An additional benefit is that setting some coefficients to zero and using a small set of important variables improves the interpretation of the model.

Penalised least squares estimation has been proposed in this context, which solve

$$\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \|\mathbf{Y}-\mathbf{X}\boldsymbol{\beta}\|^2 + \lambda g(\boldsymbol{\beta}) \qquad \text{for some} \quad \lambda > 0,$$

where λ is a tuning parameter that controls the amount of penalisation, and $g(\boldsymbol{\beta})$ is a function of $\boldsymbol{\beta}$ that is to be shrunk towards zero with increasing λ. Common choices of $g(\boldsymbol{\beta})$ lead to the ridge regression and the Lasso (least absolute shrinkage and selection operator), which are discussed in the following sections.
Prior to performing penalised least squares estimation, it is a common practice to center and standardise the variables appropriately, since:

- It is not easy to interpret the intercept shrunk towards zero. By centring both the response and covariates as $Y_i \leftarrow Y_i - \bar{Y}$ and $x_{ij} \leftarrow x_{ij} - \bar{X}_j$ with $\bar{Y} = n^{-1}\sum_{i=1}^n Y_i$ and $\bar{X}_j = n^{-1}\sum_{i=1}^n x_{ij}$, the model has zero intercept. Once the model is fitted, we can shift the mean response by $\bar{Y} - \sum_{j=1}^p \widehat{\beta}_j \bar{X}_j$.
- The penalisation on $g(\boldsymbol{\beta})$ is not fair when the regressors are measured in different units, such as Centigrade/Fahrenheit or gram/ounce. We remedy this by standardising the columns of the (centerd) $\mathbf{X}$ as $x_{ij} \leftarrow x_{ij}/s_j$ with $s_j^2 = n^{-1}\sum_{i=1}^n x_{ij}^2$, so that all the columns of the standardised model matrix have equal norm.

8.2 Ridge regression (Hoerl and Kennard, 1970)

Ridge regression penalises $g(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|^2 = \sum_j \beta_j^2$ in order to shrink small coefficients towards zero:

$$\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \|\mathbf{Y}-\mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2.$$

For a given λ, there exists $t > 0$ such that the ridge regression problem can be formulated alternatively as

$$\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \|\mathbf{Y}-\mathbf{X}\boldsymbol{\beta}\|^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \le t.$$

When $\lambda = 0$, we have $\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) =$ $\widehat{\boldsymbol{\beta}}^{\text{ols}}$, whereas when $\lambda = \infty$, it returns $\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) =$ $\mathbf{0}$. For $\lambda \in (0, \infty)$, we fit a linear regression between $\mathbf{Y}$ and $\mathbf{X}$ while introducing some

bias to the linear coefficients by shrinking them towards zero.
Let us write the objective function of ridge regression as

$$\begin{aligned} S^{\text{ridge}}(\boldsymbol{\beta}) &= (\mathbf{Y}-\mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y}-\mathbf{X}\boldsymbol{\beta}) + \lambda\boldsymbol{\beta}^\top\boldsymbol{\beta} \\ &= \underline{\qquad \mathbf{Y}^\top\mathbf{Y} - 2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{Y} + \boldsymbol{\beta}^\top(\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}_p)\boldsymbol{\beta}. \qquad} \end{aligned}$$

Since $S^{\text{ridge}}(\boldsymbol{\beta})$ is convex in $\boldsymbol{\beta}$, there exists a unique minimiser that solves

$$\begin{aligned} \frac{\partial}{\partial\boldsymbol{\beta}} S^{\text{ridge}}(\boldsymbol{\beta}) &= \underline{\qquad 2(\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}_p)\boldsymbol{\beta} - 2\mathbf{X}^\top\mathbf{Y} = 0, \qquad} \\ \therefore\ \widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) &= \underline{\qquad (\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}_p)^{-1}\mathbf{X}^\top\mathbf{Y}. \qquad} \end{aligned} \tag{8.1}$$

The name 'ridge' regression comes from the fact that $\widehat{\boldsymbol{\beta}}^{\text{ridge}}$ is obtained by adding a ridge to the diagonal of $\mathbf{X}^\top\mathbf{X}$ as seen in (8.1). Note that the inclusion of $\lambda\mathbf{I}_p$ with $\lambda > 0$, enables the computation of $\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)$ even if $\mathbf{X}^\top\mathbf{X}$ is not invertible, which was the original motivation behind the proposal of the ridge regression. In general, larger λ brings in ___greater___ bias in $\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)$, but its variance ___decreases.___ This is easily seen by considering the special case where the columns of $\mathbf{X}$ are orthogonal to each other, i.e. $\mathbf{X}^\top\mathbf{X} = \mathbf{I}_p$. Then,

$$\begin{aligned} \widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) &= \underline{\qquad \frac{1}{1+\lambda}\mathbf{X}^\top\mathbf{Y}, \quad \text{so that} \qquad} \\ \left\| \mathsf{E}(\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)) - \boldsymbol{\beta}_0 \right\| &= \underline{\qquad \frac{\lambda}{1+\lambda}\|\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}_0\| \qquad} = \frac{\lambda}{1+\lambda}\|\boldsymbol{\beta}_0\|, \\ \mathsf{Cov}(\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)) &= \underline{\qquad \frac{\sigma^2}{(1+\lambda)^2}\mathbf{X}^\top\mathbf{X} \qquad} = \frac{\sigma^2}{(1+\lambda)^2}\mathbf{I}_p. \end{aligned}$$

Compared to $\widehat{\boldsymbol{\beta}}^{\text{ols}} =$ ___$\mathbf{X}^\top\mathbf{Y}$___, the ridge regression estimator incurs non-zero bias but also attains a smaller variance by a multiplicative factor of ___$(1+\lambda)^2$___.
The choice of the penalty parameter λ is crucial in ridge regression. A common practice is to select the parameter using K-fold cross-validation over a grid of values for λ.

8.3 Lasso (Tibshirani, 1996)

In high-dimensional settings, it is often reasonable to assume that only a small subset of variables contribute to the response variable, i.e. the vector of coefficients $\boldsymbol{\beta}_0$ is assumed to be ___*sparse*___ with many zeros. Variable selection is performed in order to select a small

subset of relevant or significant variables from $X_1, \ldots, X_p$, whose indices are $\{j : \beta_j \neq 0\}$ or $\{j : |\beta_j| > c_*\}$ for some small $c_* > 0$. The ridge regression does not set any β_j exactly to ___zero___, and thus does not perform any variable selection.
The Lasso solves a penalised least squares problem with its penalty imposed on the ℓ_1-norm $\|\boldsymbol{\beta}\|_1 =$ ___$\sum_{j=1}^p |\beta_j|$:___

$$\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda) = \underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\arg\min} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1.$$

For a given $\lambda > 0$, there exists $t > 0$ such that the Lasso problem is formulated as

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq t.$$

Again, when $\lambda = 0$, we have $\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda) = \widehat{\boldsymbol{\beta}}^{\text{ols}}$, whereas when $\lambda = \infty$, it returns $\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda) = \mathbf{0}$.

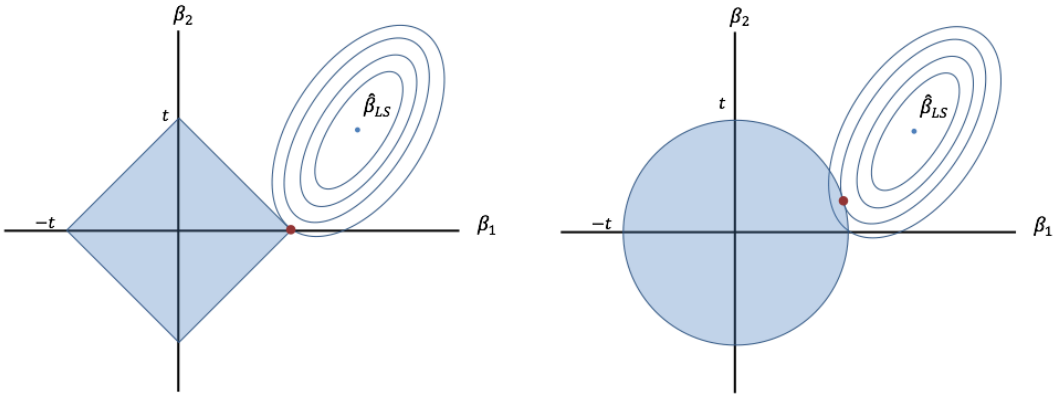


Figure 8.1: Visualisation of Lasso (left) and the ridge regression (right) in when $p = 2$ (taken from Figure 3.11 of Friedman et al. (2001)). Contours represent the least squares error $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ which increases as $\boldsymbol{\beta}$ moves away from $\widehat{\boldsymbol{\beta}}^{\text{ols}}$. The shaded areas represent the constraints $|\beta_1| + |\beta_2| \leq t$ (Lasso) and $\beta_1^2 + \beta_2^2 \leq t^2$ (ridge regression). Here, the Lasso returns an estimator with $\widehat{\beta}_2^{\text{lasso}} = 0$.

Figure 8.1 compares the Lasso and ridge regression estimators. It demonstrates that due to the penalty imposed on $\|\boldsymbol{\beta}\|_1$, the Lasso sets some of the coefficients exactly to zero with increasing λ, since the ℓ_1-penalty produces a 'pointy' constraint. In other words, the Lasso is able to perform variable selection depending on the choice of the penalty parameter λ.

Again, K-fold cross-validation is often adopted for this purpose.
Unlike the ridge regression, there is no closed form expression for the solution of the Lasso estimation problem in general, except for some special cases. Let us suppose that $\mathbf{X}^\top\mathbf{X} = \mathbf{I}_p$, and we consider minimising the following objective function (the factor $1/2$ is introduced to simplify the arguments):

$$S^{\text{lasso}}(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1.$$

As in this setting, the OLS estimator $\widehat{\boldsymbol{\beta}}^{\text{ols}}$ exists so we can write

$$\begin{aligned} S^{\text{lasso}}(\boldsymbol{\beta}) &= \frac{1}{2}\|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}^{\text{ols}}\|^2 + \frac{1}{2}\|\mathbf{X}\widehat{\boldsymbol{\beta}}^{\text{ols}} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1 \\ &= \frac{1}{2}\|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}^{\text{ols}}\|^2 + \frac{1}{2}(\widehat{\boldsymbol{\beta}}^{\text{ols}} - \boldsymbol{\beta})^\top\mathbf{X}^\top\mathbf{X}(\widehat{\boldsymbol{\beta}}^{\text{ols}} - \boldsymbol{\beta}) + \lambda\|\boldsymbol{\beta}\|_1 \\ &= \frac{1}{2}\|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}^{\text{ols}}\|^2 + \sum_{j=1}^{p} \underbrace{\left\{\frac{1}{2}(\widehat{\beta}_j^{\text{ols}} - \beta_j)^2 + \lambda|\beta_j|\right\}}_{S_j^{\text{lasso}}(\beta_j)}, \end{aligned}$$

where

- the first equality follows from that $\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}^{\text{ols}} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}$, and that $(\mathbf{I}_n - \mathbf{H})\mathbf{X} = \mathbf{0}$,
- the third equality follows from that $\mathbf{X}^\top\mathbf{X} = \mathbf{I}_p$, and that the squared Euclidean norm and the ℓ_1-norm can be separated over $j = 1, \ldots, p$.

From the above, we can find $\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda)$ by individually finding the minimiser of each $S_j^{\text{lasso}}(\beta_j)$, which is convex in β_j. While $f(x) = |x|$ is not differentiable at $x = 0$, we have

$$\frac{dS_j^{\text{lasso}}}{d\beta_j} = \beta_j - \widehat{\beta}_j^{\text{ols}} + \begin{cases} \lambda & \text{if } \beta_j > 0, \\ -\lambda & \text{if } \beta_j < 0. \end{cases}$$

Solving the equation for each j leads to the following solution

$$\widehat{\beta}_j^{\text{lasso}}(\lambda) = \begin{cases} \widehat{\beta}_j^{\text{ols}} - \lambda & \text{if } \widehat{\beta}_j^{\text{ols}} > \lambda, \\ \widehat{\beta}_j^{\text{ols}} + \lambda & \text{if } \widehat{\beta}_j^{\text{ols}} < -\lambda, \\ 0 & \text{otherwise.} \end{cases}$$

That is, the Lasso estimator $\widehat{\beta}_j^{\text{lasso}}(\lambda)$ is obtained by *soft-thresholding* $\widehat{\beta}_j^{\text{ols}}$ with λ as the

threshold, an operation that shrinks $\widehat{\beta}_j^{\text{ols}}$ towards zero by λ if $|\widehat{\beta}_j^{\text{ols}}| > \lambda$, while setting it to zero if towards zero by λ in the absolute values if $|\widehat{\beta}_j^{\text{ols}}| \leq \lambda$.

There are several `R` packages that implement Lasso, such as `lars` and `glmnet`. Both produce the entire solution path for the Lasso estimator with a range of values for λ within a fraction of time.

8.4 R example: Hitters data

`Hitters` data frame from the `ISLR` package has 263 observations on the salary and 19 variables associated with Major League baseball players collected during the 1986/1987 season. Look up `help(Hitters)` for details of the data frame.

```
library(ISLR)
help(Hitters)
head(Hitters)
```

```
                  AtBat Hits HmRun Runs RBI Walks Years CAtBat
-Andy Allanson      293   66     1   30  29    14     1    293
-Alan Ashby         315   81     7   24  38    39    14   3449
-Alvin Davis        479  130    18   66  72    76     3   1624
-Andre Dawson       496  141    20   65  78    37    11   5628
                  CHits CHmRun CRuns CRBI CWalks League Division
-Andy Allanson       66      1    30   29     14      A        E
-Alan Ashby         835     69   321  414    375      N        W
-Alvin Davis        457     63   224  266    263      A        W
-Andre Dawson      1575    225   828  838    354      N        E
                  PutOuts Assists Errors Salary NewLeague
-Andy Allanson        446      33     20     NA          A
-Alan Ashby           632      43     10  475.0          N
-Alvin Davis          880      82     14  480.0          A
-Andre Dawson         200      11      3  500.0          N
```

Before analysing the data frame, we omit those players whose salary is unknown from our consideration. Also, factor variables such as `League`, `Division` and `NewLeague` are manually changed to numeric variables taking values 0/1.

```
new.hitters <- Hitters[!is.na(Hitters$Salary), ]
```

```
y <- new.hitters[, 19]
X0 <- new.hitters[, -19]
X0$League <- as.numeric(X0$League)-1
X0$Division <- as.numeric(X0$Division)-1
X0$NewLeague <- as.numeric(X0$NewLeague)-1
```

Recall that for the penalised least squares estimators to have a meaningful interpretation, we need to standardise the regressors so that $\sum_{i=1}^{n} x_{ij} = 0$ and $\sum_{i=1}^{n} x_{ij}^2 = 1$. This task can easily be handled using the function `scale`.

```
X <- scale(X0)
# all columns of X0 are both centerd and standardised
```

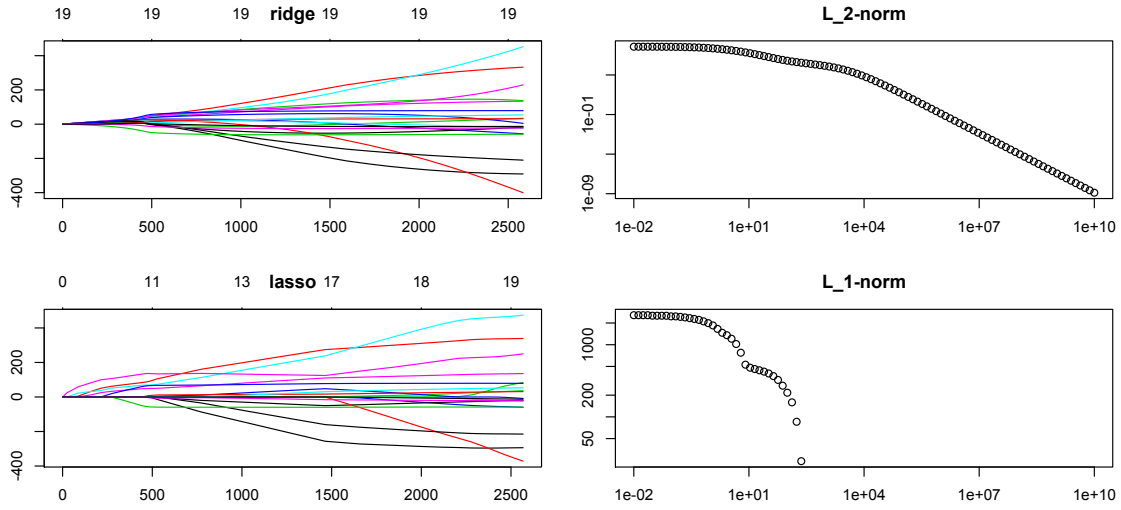


Figure 8.2: Left: solution paths for $\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)$ (top) and $\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda)$ (bottom) over a grid of λ, where the x-axis represents the ℓ_1-norm of the estimator; right: $\|\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)\|^2$ (top) and $\|\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda)\|_1$ against λ (note that both x and y axes are in logarithmic scales).

The function `glmnet` from the package of the same name fits a generalised linear model via penalised maximum likelihood. It can deal with the response variable of a variety of types, as those we'll discuss in Part 2 of this unit, but we focus on numeric and continuous Y_i.

The regularisation path is computed for the following penalty (called *elastic net*) at a grid of values for the penalty parameter λ,

$$g_{\lambda,\alpha}(\boldsymbol{\beta}) = \lambda \left(\frac{(1-\alpha)}{2} \|\boldsymbol{\beta}\|^2 + \alpha \|\boldsymbol{\beta}\|_1 \right)$$

so that when $\alpha =$ 0, it becomes the ridge regression and when $\alpha =$ 1, the Lasso. We fit the ridge regression and Lasso estimators over a grid of penalty parameters λ and plot their solution paths ($\widehat{\beta}_j(\lambda)$ for each $j = 1, \ldots, p$, over varying λ) in Figure 8.2. From the plots in the right, it is easily seen that the ridge and the Lasso penalty functions evaluated at corresponding estimators, $\|\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)\|^2$ and $\|\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda)\|_1$, decrease with increasing λ.

```
library(glmnet)
grid <- 10^seq(10, -2, length.out=100)
# a grid of lambda values
# from 10^10 to 10^-2 of length 100
ridge.fit <- glmnet(X, y, lambda = grid, standardize = TRUE, alpha = 0)
lasso.fit <- glmnet(X, y, lambda = grid, standardize = TRUE, alpha = 1)

plot(ridge.fit, main = 'ridge')
plot(grid, apply(ridge.fit$beta^2, 2, sum), log = 'xy',
xlab = 'log(lambda)', ylab = '', main='L_2-norm')
plot(lasso.fit, main = 'lasso')
plot(grid, apply(abs(lasso.fit$beta), 2, sum), log = "xy',
xlab = 'log(lambda)', ylab = '', main='L_1-norm')
```

Remark 8.1. `glmnet` internally scales the variables, setting its argument `standardize = TRUE`. Therefore, in fact, the following two commands return the identical results.

```
ridge.fit <- glmnet(X0, y, lambda = grid, standardize = TRUE, alpha = 0)
ridge.fit <- glmnet(X, y, lambda = grid, standardize = TRUE, alpha = 0)
```

Cross-validation is adopted for selecting the amount of penalisation, λ. `cv.glmnet` performs 10-fold CV by default (`nfolds = 10`). `cv.glmnet` object reports two values for the penalty parameter λ, namely `lambda.min` ($\lambda_{\min}$) that returns the minimum mean cross-validated error, and `lambda.1se` ($\lambda_{1\text{se}}$) that is the largest value of λ such that the cross-validated error is within one standard error (across the partitions of the data) of the minimum; we always have $\lambda_{1\text{se}} \geq \lambda_{\min}$. Plotting a `cv.glmnet` object, $\lambda_{\min}$ and $\lambda_{1\text{se}}$ are indicated by the broken vertical lines, see Figure 8.3.

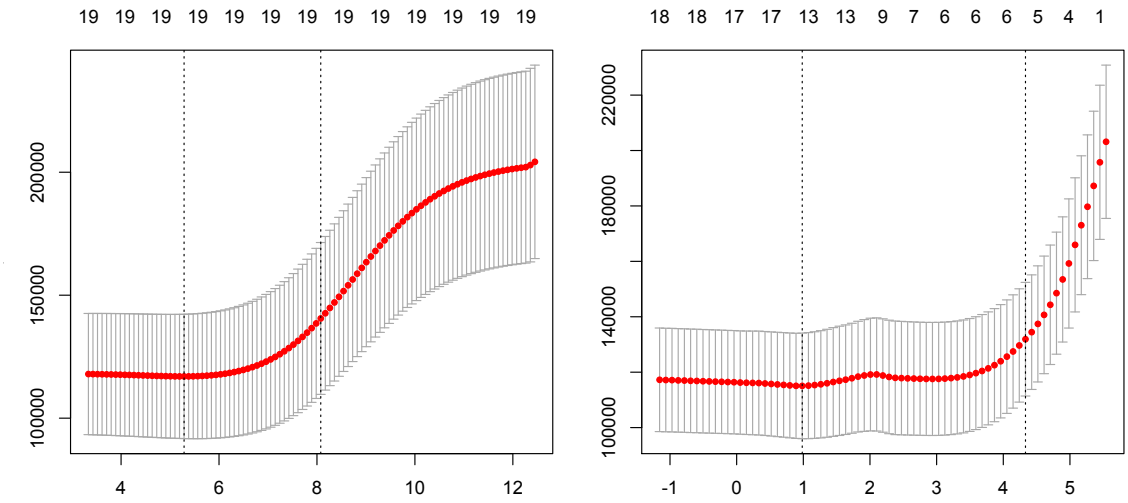


Figure 8.3: Mean cross-validated errors against $\log(\lambda)$ from `cv.ridge.fit` (left) and `cv.lasso.fit` (right), along with the upper and lower standard deviation curves.

Looking at the regression coefficients estimated with these choices of λ in Figure 8.4, if you were coaching a young aspiring baseball player, which handful of attributes would you tell this player to focus on?

```
cv.ridge.fit <- cv.glmnet(X, y, standardize = TRUE, alpha = 0)
cv.lasso.fit <- cv.glmnet(X, y, standardize = TRUE, alpha = 1)

plot(cv.ridge.fit)
plot(cv.lasso.fit)
coef(cv.ridge.fit, s=c(cv.ridge.fit$lambda.min, cv.ridge.fit$lambda.1se))
coef(cv.lasso.fit, s=c(cv.lasso.fit$lambda.min, cv.lasso.fit$lambda.1se))

matplot(cbind(coef(cv.ridge.fit, s=c(cv.ridge.fit$lambda.min,
    cv.ridge.fit$lambda.1se)),
    coef(cv.lasso.fit, s=c(cv.lasso.fit$lambda.min, cv.lasso.fit$lambda.1se))),
    pch = 1:4, pty = 1:4, xlab = '', ylab = '', xaxt = 'n')
axis(1, at = 1:(1 + ncol(X)), labels = c('intercept', colnames(X)), las = 2)
legend("bottomright", pch=1:4, col=1:4, cex = .5,
    legend=c("ridge-min", "ridge-1se", "lasso-min", "lasso-1se"))
```

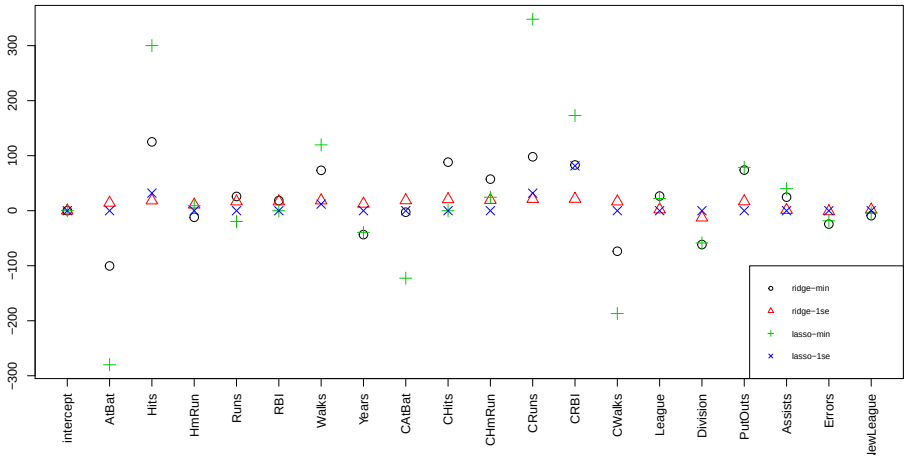


Figure 8.4: Estimated coefficients $\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda^{\text{ridge}}_{\text{min}})$, $\widehat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda^{\text{ridge}}_{\text{1se}})$, $\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda^{\text{lasso}}_{\text{min}})$ and $\widehat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda^{\text{lasso}}_{\text{1se}})$.

9 Introduction to generalised linear models

Linear models are one of the most widely used examples of parametric statistical models. In their simplest form, linear modelling often assumes $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_n)$ and consequently, is applicable when the response variable is continuous, normally distributed and has constant variance. This makes its application to discrete, non-normal responses with complex heteroscedasticity inappropriate.

Generalised linear models (GLM) extend linear modelling to situations where the distribution of the response is a member of a much broader class of distributions termed the *exponential family*, and provide both flexibility for applications and a unified framework for inference. As special cases, they embed linear models and analysis-of-variance (ANOVA) models, logit and probit models for binary responses, and log-linear models and multinomial response models for count data, which we will discuss in due course.

9.1 Limitations of linear models and its generalisation

We observe the following three components of the linear modelling we have studied so far.

(i) The linear predictor: $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0$.

(ii) The random component: $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$.

(iii) The linear predictor and the mean response are linked via the identity function, i.e. $\mu_i = \mathsf{E}(Y_i) =$ η_i.

GLM extends the second and the third component. For (ii), it allows the response variable to come from *any* distribution whose density function can be written as

$$f(y; \theta, \phi) = q(\phi, y) \exp\left[\frac{1}{\phi}(y\theta - K(\theta))\right].$$

We see later that the parameter θ is closely related to the mean $\mu = \mathsf{E}(Y)$ and ϕ to the variance of Y. For (iii), generalised linear modelling 'links' the linear predictor and the mean response via a monotone and differentiable *link* function g, i.e. $\eta = g(\mu)$, where g is not necessarily the identity function.

9.2 Examples of GLM

Example 9.1 (Binomial response: Pneumonia data)**.** The following experiment appears in 'The titration of antipneumococcus serum' by Smith (1932), published in the Journal of Pathology.

Given a sample consisting of five groups of forty mice, every mouse was injected with the same dose of a culture of pneumococci, and then one of five dose levels of an anti-pneumococcus serum where the dose level depended on the group. That is, the dose of anti-pneumococcus serum was the same for each mouse within the same group, but varied between groups. For all mice that died during a seven-day period following the inoculation, a blood sample was taken to check whether the pneumococci culture was present or not. Mice that lived beyond the seventh day were regarded as survivors and were not examined. The following is the data collected.

Dose of serum	Number of mice	Number of deaths from pneumonia
0.0028	40	35
0.0056	40	21
0.0112	40	9
0.0225	40	6
0.0450	40	1

We are interested in examining the relationship between the dose level and death rate. Clearly, the proportion of deaths decreases as the dose level increases.

Let Y_i be the number of deaths from pneumonia from the i-th group, and let x_i be the corresponding dose level (dose of serum). We assume that at each dose level (within each group), the mice die independently of one another and they all have the same probability of dying. At each dose level, $i = 1, \ldots, 5$, we may model the number of deaths Y_i, out of 40 mice, using a binomial distribution, where π_i denotes the probability of a mouse receiving dose level i dying. In summary, Y_i is modelled such that

$$Y_i \sim \underline{\quad \text{Binomial}(40, \pi_i) \quad} \quad \text{and} \quad \mathsf{E}(Y_i) = \underline{\quad 40\pi_i. \quad}$$

Alternatively, we may consider the random variable Z_i, which takes the value $Z_i = 1$ when a randomly chosen mouse from group i dies and $Z_i = 0$ if not, so that

$$Z_i \sim \underline{\quad \text{Bernoulli} \quad}(\pi_i) \quad \text{and} \quad \mu_i = \mathsf{E}(Z_i) = \pi_i.$$

Would it be sensible to model $\mu_i = \pi_i$ as a linear function of x_i, i.e. $\pi_i = \eta_i = \beta_1 + \beta_2 x_i$? Since π_i is a probability, it must lie in the range $[0, 1]$, which is not guaranteed if a linear function of x_i is used. With such a restriction, a commonly adopted model is

$$\pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)} = \frac{\exp(\beta_1 + \beta_2 x_i)}{1 + \exp(\beta_1 + \beta_2 x_i)},$$

which ensures that $\pi_i \in [0, 1]$, see Figure 9.1 for its shape.

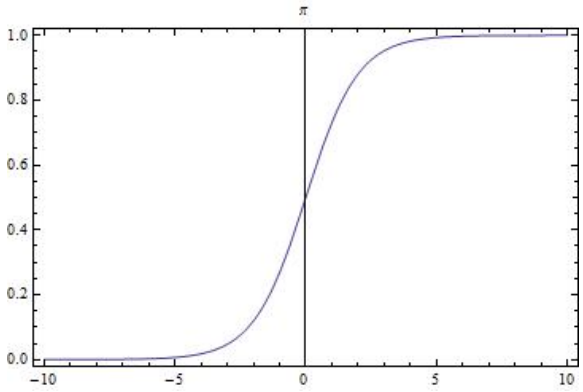


Figure 9.1: Plot of $\pi = e^\eta/(1 + e^\eta)$ against η.

Rearranging, we have the *logistic regression* model:

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \eta_i = \beta_1 + \beta_2 x_i,$$

which models the logarithm of the *odds* (log-odds) of death ($= \text{logit}(\pi_i)$) as a linear function of x_i.
To sum up,

$$Z_i \sim \text{Bernoulli}(\pi_i), \quad \mu_i = \mathsf{E}(Z_i) = \pi_i, \quad \text{and} \quad \text{logit}(\pi_i) = \beta_1 + \beta_2 x_i.$$

This model is one example of a generalised linear model.

Example 9.2 (Poisson response: Moth data)**.** The number of moths caught overnight in a trap is recorded on 20 different days, along with the average night time temperature.

No. caught	4	6	2	7	9	5	5	8	7	1
Temp (°C)	5	10	4	6	7	3	7	4	3	3
No. caught	0	1	2	4	10	11	7	7	4	4
Temp (°C)	2	4	6	8	13	11	6	3	3	3

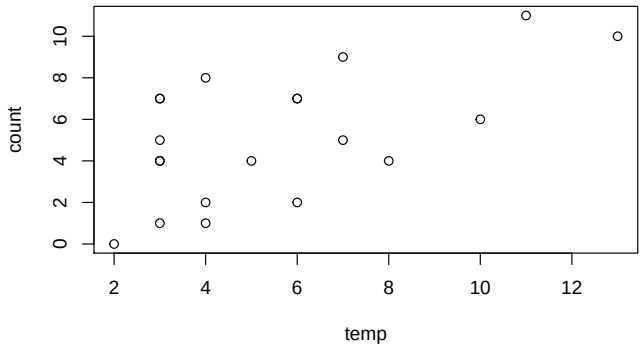


Figure 9.2: The scatter plot of the number of moths caught against the temperature.

We want to propose a model for the expectation of Y_i, the number of moths caught overnight, using x_i, the average night time temperature, as a covariate. The number of moths caught overnight (count) can naturally be thought to follow a Poisson distribution, i.e. $Y_i \sim \text{Poisson}(\mu_i)$, where the associated parameter $\mu_i > 0$ satisfies $\mu_i = \mathsf{E}(Y_i)$. Modelling a linear relationship between μ_i and x_i as $\mu_i = \eta_i = \beta_1 + \beta_2 x_i$ is inappropriate, as μ_i is restricted to be positive. As an alternative, consider modelling $\mu_i = \exp(\eta_i)$ or, equivalently,

$$\log(\mu_i) = \eta_i = \beta_1 + \beta_2 x_i.$$

To sum up, we have

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \mathsf{E}(Y_i), \quad \text{and} \quad \log(\mu_i) = \beta_1 + \beta_2 x_i.$$

9.3 Components of the GLM

Classical linear regression is based on the assumption of normality (at least approximately) for the distribution of the responses. The response variable can take any value in $\mathbb{R}$, and the mean and the variance are unrelated. As seen in the previous section, the linear regression model is not always appropriate.
GLM is a model equipped with the following three components.

(i) ___Systematic___ component: Covariates furnish a linear predictor

$$\eta_i = \sum_{j=1}^{p} \beta_j x_{ij} = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

(ii) ___Random___ component: $Y_i,\ 1 \leq i \leq n$, are distributed independently with distribution F, where F belongs to the *exponential family*.

(iii) ___Link___ between random and systematic parts: g is the link function which relates the linear predictor η_i to the expected value of the response variable $\mathsf{E}(Y_i) = \mu_i$, as

$$\underline{\quad \eta_i = g(\mu_i). \quad}$$

For the standard linear model, F is the ___normal___ distribution and $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta} = \mu_i$, i.e. g is the ___identity___ function.

9.4 Exponential family

The *exponential family* is a class of distributions, continuous or discrete, whose probability density/mass functions are of the form

$$f(y; \theta, \phi) = q(\phi, y) \exp\left[\frac{1}{\phi}(\theta y - K(\theta))\right]. \tag{9.1}$$

(i) $\theta = \theta(\mu)$ is termed the *natural parameter* supported on $\theta \in \Theta \subseteq \mathbb{R}$, and is a function of $\underline{\quad \mu = \mathsf{E}(Y). \quad}$

(ii) ϕ is the *dispersion parameter* supported on $\phi \in \Phi \subseteq \mathbb{R}$.

(iii) $K(\theta)$ is a known function of θ.

(iv) $q(\phi, y)$ is a known positive function of $y \in \mathcal{Y} \subseteq \mathbb{R}$.

Example 9.3. The normal distribution $\mathcal{N}(\mu, \sigma^2)$ belongs to the exponential family. Its density function admits the following decomposition

$$\begin{aligned} f(y; \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) \\ &= \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y^2}{2\sigma^2}\right)}_{q(\phi,y)} \exp\left(\frac{1}{\sigma^2}\left(\mu y - \frac{\mu^2}{2}\right)\right) \end{aligned}$$

with $\theta(\mu) = \mu$, $\phi = \sigma^2$ and $K(\theta) = \theta^2/2$.

9.4.1 Properties of the exponential family

Components of (9.1) are related to the moments of Y. In what follows, we demonstrate this in two different ways.

Using the cumulant generating function. For Y whose distribution belongs to the exponential family, its cumulant generating function $c_Y(t) = \log(\mathsf{E}(e^{tY}))$, the logarithm of the moment generating function, is given by [8]

$$c_Y(t; \theta, \phi) = \frac{1}{\phi}(K(\theta + \phi t) - K(\theta)).$$

As the name suggests, the k-th cumulant of Y is given by $c_Y^{(k)}(0)$, where $c_Y^{(k)}$ denotes the k-times differentiated c_Y. Most notably, we have $\mathsf{E}(Y) = c_Y^{(1)}(0) = c_Y'(0)$ and $\mathsf{Var}(Y) = c_Y^{(2)}(0) = c_Y''(0)$, such that

$$\begin{aligned} \mathsf{E}_{\theta,\phi}(Y) = c_Y'(0) &= \underline{\quad K'(\theta) \quad \text{and}} \\ \mathsf{Var}_{\theta,\phi}(Y) = c_Y''(0) &= \underline{\quad \phi K''(\theta) = \phi V(\mu), \quad} \end{aligned}$$

where $V(\mu) = K''(\theta(\mu))$ is called the *variance function*. Here, we use the expressions $\mathsf{E}_{\theta,\phi}(\cdot)$ and $\mathsf{Var}_{\theta,\phi}(\cdot)$ to highlight their dependence on the parameters θ and ϕ.

[8] Verify this in problem sheet Q1.

Using the likelihood function. An alternative way is to use the following well-known results which hold for likelihoods of any distribution besides those belonging to the exponential family.
We first introduce the notations

$$\begin{aligned} L &= L(y;\nu) = L(\nu|y) = f(y;\nu) = f_\nu(y) \quad \text{and} \\ \ell &= \ell(y;\nu) = \ell(\nu|y) = \log(L) = \log(f(y;\nu)) = \log(f_\nu(y)), \end{aligned}$$

where ν denotes an arbitrary parameter. We have

$$\frac{\partial \ell}{\partial \nu} = \frac{1}{L}\frac{\partial L}{\partial \nu}, \tag{9.2}$$

$$\begin{aligned} \frac{\partial^2 \ell}{\partial \nu^2} &= \left(\frac{1}{L}\frac{\partial L}{\partial \nu}\right)' = \left(\frac{1}{L}\right)'\left(\frac{\partial L}{\partial \nu}\right) + \left(\frac{\partial L}{\partial \nu}\right)'\left(\frac{1}{L}\right) \\ &= -\frac{1}{L^2}\left(\frac{\partial L}{\partial \nu}\right)^2 + \frac{1}{L}\frac{\partial^2 L}{\partial \nu^2} = \underline{\qquad \frac{1}{L}\frac{\partial^2 L}{\partial \nu^2} - \left(\frac{\partial \ell}{\partial \nu}\right)^2 .} \end{aligned} \tag{9.3}$$

Using these equations, we can show that

$$\mathsf{E}_\nu\left(\frac{\partial \ell}{\partial \nu}(Y;\nu)\right) = 0. \tag{9.4}$$

Proof of (9.4). By (9.2),

$$\begin{aligned} \mathsf{E}_\nu\left(\frac{\partial \ell}{\partial \nu}(Y;\nu)\right) &= \int_{\mathcal{Y}} \left(\frac{\partial \ell}{\partial \nu}(y;\nu)\right) f_\nu(y)dy = \int_{\mathcal{Y}} \left(\frac{1}{L}\frac{\partial L}{\partial \nu}\right)(y;\nu) f_\nu(y)dy \\ &= \underline{\qquad \int_{\mathcal{Y}} \frac{\partial}{\partial \nu} L(y;\nu)\cdot \frac{f_\nu(y)}{f_\nu(y)}dy = \frac{\partial}{\partial \nu}\int_{\mathcal{Y}} f_\nu(y)dy \qquad} = 0 \end{aligned}$$

where the penultimate equality holds from that $L = f_\nu(y)$ and the last equality holds because $\int_{\mathcal{Y}} f_\nu(y)dy = 1$. □

Next, we show that

$$\mathsf{Var}_\nu\left(\frac{\partial \ell}{\partial \nu}(Y;\nu)\right) = \mathsf{E}_\nu\left(-\frac{\partial^2 \ell}{\partial \nu^2}(Y;\nu)\right). \tag{9.5}$$

Proof of (9.5).

$$\mathsf{Var}_\nu\left(\frac{\partial \ell}{\partial \nu}\right) = \mathsf{E}_\nu\left(\left(\frac{\partial \ell}{\partial \nu}\right)^2\right) - \left(\mathsf{E}_\nu\left(\frac{\partial \ell}{\partial \nu}\right)\right)^2 = \mathsf{E}_\nu\left(\left(\frac{\partial \ell}{\partial \nu}\right)^2\right),$$

where the last equality follows from (9.4). Then, (9.5) follows if

$$\mathsf{E}_\nu\left(\left(\frac{\partial \ell}{\partial \nu}\right)^2\right) = \mathsf{E}_\nu\left(-\frac{\partial^2 \ell}{\partial \nu^2}(Y;\nu)\right).$$

Note that from (9.3),

$$\mathsf{E}_\nu\left(\frac{\partial^2 \ell}{\partial \nu^2}(Y;\nu)\right) = \mathsf{E}_\nu\left(\frac{1}{L}\frac{\partial^2 L}{\partial \nu^2}\right) - \mathsf{E}_\nu\left(\left(\frac{\partial \ell}{\partial \nu}\right)^2\right)$$

and hence it boils down to showing

$$\underline{\mathsf{E}_\nu\left(\frac{1}{L}\frac{\partial^2 L}{\partial \nu^2}\right) = 0,}$$

which follows as

$$\begin{aligned}\mathsf{E}_\nu\left(\frac{1}{L}\frac{\partial^2 L}{\partial \nu^2}\right) &= \int_{\mathcal{Y}}\left(\frac{1}{L}\right)\frac{\partial^2 L}{\partial \nu^2} f_\nu(y)dy \\ &= \underline{\int_{\mathcal{Y}}\frac{\partial^2 L}{\partial \nu^2}\frac{f_\nu(y)}{f_\nu(y)}dy = \frac{\partial^2}{\partial \nu^2}\int_{\mathcal{Y}} f_\nu(y)dy} \quad = 0.\end{aligned}$$

□

We now apply these general results to the exponential family, where

$$\ell(\theta,\phi|y) = \log q(\phi,y) + \frac{1}{\phi}(\theta y - K(\theta)).$$

Equation (9.4) gives

$$\mathsf{E}_{\theta,\phi}\left(\frac{\partial \ell}{\partial \theta}\right) = \underline{\mathsf{E}_{\theta,\phi}\left[\frac{1}{\phi}(Y - K'(\theta))\right] = 0,} \quad \text{hence } \mathsf{E}_{\theta,\phi}(Y) = K'(\theta) = \mu,$$

and equation (9.5) gives

$$\mathsf{E}_{\theta,\phi}\left(-\frac{\partial^2 \ell}{\partial \theta^2}\right) = \frac{K''(\theta)}{\phi} = \mathsf{Var}_{\theta,\phi}\left(\frac{\partial \ell}{\partial \theta}\right) = \mathsf{Var}_{\theta,\phi}\left[\frac{1}{\phi}(Y - K'(\theta))\right],$$

$$\text{hence } \mathsf{Var}_{\theta,\phi}(Y) = \phi K''(\theta). \tag{9.6}$$

We sometimes write $\theta = \theta(\mu)$ and hence $K''(\theta(\mu)) = V(\mu)$, i.e. the natural parameter depends on the mean μ and the variance may vary with the mean, as is the case for the binomial and Poisson distributions. Sometimes we write $Y \sim \text{ED}(\mu, \phi V(\mu))$ to indicate that Y is of exponential family with mean μ and variance $\phi V(\mu)$.

9.5 Link functions

Link functions are required to be twice differentiable and strictly monotonic. Some of widely-used link functions are:

(i) Identity link: $g(\mu) = \mu$.

(ii) Logit link: $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$.

(iii) Log link: $g(\mu) = \log(\mu)$.

(iv) Reciprocal link: $g(\mu) = 1/\mu$.

(v) Inverse square link: $g(\mu) = 1/\mu^2$.

9.5.1 Canonical link functions

For each exponential family distribution, one particular choice of link function has a special mathematical status. Recall the mappings between the three key parameters θ, μ and η:

$$\eta = g(\mu) \quad \text{and} \quad \mu = K'(\theta).$$

Suppose that g is the inverse of K', then

$$\eta = g(K'(\theta)) = \theta,$$

i.e. the linear predictor is set to be identical to the natural parameter. This choice of $g(\mu) = \theta(\mu)$ is called the *canonical link function* for the distribution.

Example 9.4. For $Y \sim \text{Binomial}(m, \pi)$, we have $\theta = \log(\pi/(1-\pi))$ and $\mu = \mathsf{E}(Y) = m\pi$. Therefore,

$$g(\mu) = \log\left(\frac{\mu}{m-\mu}\right)$$

is the canonical link function for the binomial distribution. In the special case when $Y \sim \text{Binomial}(1, \pi) = \text{Bernoulli}(\pi)$, the canonical link function is $g(\pi) = \log(\pi/(1-\pi)) =: \text{logit}(\pi)$.

10 Fitting generalised linear models

Suppose that we have independent observations $(Y_i, \mathbf{x}_i)$, $1 \leq i \leq n$ satisfying

$$Y_i \sim \text{ED}(\mu_i, \phi V(\mu_i)) \quad \text{and} \qquad g(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}_0.$$

Option 1: Adopt the method of least squares estimation by identifying the minimiser of the residual sum of squares as

$$\underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\arg\min} \sum_{i=1}^{n} (Y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2.$$

Problem: This ignores the presence of ___link function___ which places μ_i (hence Y_i) and $\mathbf{x}_i^\top \boldsymbol{\beta}_0$ on different scales.

Option 2: Adopt the least squares estimation after transforming Y_i via g, identifying

$$\underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\arg\min} \sum_{i=1}^{n} (g(Y_i) - \mathbf{x}_i^\top \boldsymbol{\beta})^2.$$

Problem: Neither $\mathsf{Var}(Y_i)$ nor $\mathsf{Var}(g(Y_i))$ are constant since they depend on $\mathbf{x}_i^\top \boldsymbol{\beta}_0$ in general (recall that ___$\mathsf{Var}(Y_i) = \phi V(\mu_i)$___). If the variance is heterogeneous, observations should contribute differently to the estimation of the parameters: Recall the weighted least squares estimator discussed in page 65, where observations of high variance contribute ___less___ than those of small variance to the estimation of $\boldsymbol{\beta}_0$.

Option 3: Adopt the weighted least squares estimation after transforming Y_i, identifying

$$\underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\arg\min} \sum_{i=1}^{n} \frac{(g(Y_i) - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{\phi V(g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta}))}$$

Problem: The objective function is no longer ___quadratic___ in $\boldsymbol{\beta}$, and thus requires numerical methods to find a solution and does not enjoy the main benefit of adopting the least squares estimation.

10.1 Revision of maximum likelihood estimation

Suppose that $y_1, \ldots, y_n$ are realisations of random variables $Y_1, \ldots, Y_n$, that are independently distributed with the density function $f_{\boldsymbol{\beta}_0}$, where $\boldsymbol{\beta}_0 \in \mathbb{R}^p$ denotes the unknown parameter to be estimated. The likelihood is the joint probability density function of $Y_1, \ldots, Y_n$ evaluated at their realisations $y_1, \ldots, y_n$, and is viewed as a function of $\boldsymbol{\beta} \in \mathbb{R}^p$. Thanks to the assumption of independence, the likelihood and the log-likelihood can be written as

$$L(\boldsymbol{\beta}) = L(\boldsymbol{\beta}; y_1, \ldots, y_n) = \prod_{i=1}^{n} f_{\boldsymbol{\beta}}(y_i), \quad \text{and}$$
$$\ell(\boldsymbol{\beta}) = \log L(\boldsymbol{\beta}) = \sum_{i=1}^{n} \log f_{\boldsymbol{\beta}}(y_i) = \sum_{i=1}^{n} \ell_i(\boldsymbol{\beta}).$$

Then, the maximum likelihood estimation gives an estimator $\widehat{\boldsymbol{\beta}}$ of $\boldsymbol{\beta}_0$ as

$$\widehat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta} \in \Theta} \ell(\boldsymbol{\beta}).$$

For linear models with independently and normally distributed Y_i, $1 \le i \le n$, the MLE of $\boldsymbol{\beta}_0$ is the least squares estimator $\widehat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$. For GLM, suppose that $Y_i \sim \mathrm{ED}(\mu_i, \phi K(\mu_i))$, independently, with $\mu_i = K'(\theta_i)$. Recall that from (9.6), we have

$$-\frac{\partial^2 \ell_i}{\partial \theta_i^2} = \frac{K''(\theta_i)}{\phi} = \frac{1}{\phi^2} \mathsf{Var}_{\theta_i, \phi}(Y_i) > 0,$$

which indicates that $\ell_i(\theta_i)$ is *concave* in θ_i and thus ensures that $\ell_i(\theta_i)$ has a unique maximiser. Combined with requirements on g (twice differentiable and strictly monotonic), we can find an MLE of $\boldsymbol{\beta}_0$.

Example 10.1 (Pneumonia data)**.** For the pneumonia example studied in Example 9.1, there are two parameters β_1 and β_2. The model is

$$Y_i \sim \mathrm{Binomial}(m_i, \pi_i) \quad \text{with} \quad m_i = 40,$$

for $i = 1, \ldots, n = 5$, with the logit link function

$$\pi_i = \frac{\exp(\beta_1 + \beta_2 x_i)}{1 + \exp(\beta_1 + \beta_2 x_i)},$$

where $x_i = \log(\text{dose given to group } i)$.
The log-likelihood for the unknown parameters $\boldsymbol{\beta}_0 = (\beta_1, \beta_2)^\top$ given the data is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \ell_i(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left(\log \binom{m_i}{y_i} + y_i \log(\pi_i) + (m_i - y_i) \log(1 - \pi_i) \right).$$

Then the derivatives of $\ell(\boldsymbol{\beta})$ is given by

$$\begin{aligned} \frac{\partial \ell(\boldsymbol{\beta})}{\partial \beta_1} &= \sum_{i=1}^{n} \frac{\partial \ell_i}{\partial \pi_i} \frac{\partial \pi_i}{\partial \beta_1} = \sum_{i=1}^{n} \left(\frac{y_i}{\pi_i} - \frac{m_i - y_i}{1 - \pi_i} \right) \pi_i (1 - \pi_i) = \sum_{i=1}^{n} (y_i - m_i \pi_i) \quad , \\ \frac{\partial \ell(\boldsymbol{\beta})}{\partial \beta_2} &= \sum_{i=1}^{n} \frac{\partial \ell_i}{\partial \pi_i} \frac{\partial \pi_i}{\partial \beta_2} = \sum_{i=1}^{n} \left(\frac{y_i}{\pi_i} - \frac{m_i - y_i}{1 - \pi_i} \right) x_i \pi_i (1 - \pi_i) = \sum_{i=1}^{n} (y_i x_i - m_i x_i \pi_i). \end{aligned}$$

Here, we use that

$$\frac{\partial \pi_i}{\partial \beta_1} = \pi_i(1 - \pi_i) \quad \text{and} \quad \frac{\partial \pi_i}{\partial \beta_2} = x_i \pi_i (1 - \pi_i).$$

By equating $\frac{\partial \ell(\widehat{\boldsymbol{\beta}})}{\partial \beta_j} = 0$ for $j = 1, 2$, with the MLE $\widehat{\boldsymbol{\beta}} = (\widehat{\beta}_1, \widehat{\beta}_2)^\top$, we obtain the *likelihood equations*

$$\sum_{i=1}^{n} y_i = \sum_{i=1}^{n} m_i \widehat{\pi}_i \quad \text{and} \quad \sum_{i=1}^{n} y_i x_i = \sum_{i=1}^{n} m_i \widehat{\pi}_i x_i, \tag{10.1}$$

where $\widehat{\pi}_i$ is given by $\widehat{\beta}_1$ and $\widehat{\beta}_2$, as $\widehat{\pi}_i = \frac{\exp(\mathbf{x}_i^\top \widehat{\boldsymbol{\beta}})}{1 + \exp(\mathbf{x}_i^\top \widehat{\boldsymbol{\beta}})}$, and itself is the MLE of π_i. We cannot analytically solve (10.1) for $\widehat{\beta}_1$ and $\widehat{\beta}_2$; numerical calculations yield $\widehat{\beta}_1 = -9.1894$ and $\widehat{\beta}_2 = -1.8296$.

Often the solution of the likelihood equations

$$\frac{\partial \ell(\widehat{\boldsymbol{\beta}})}{\partial \beta_j} = 0, \qquad j = 1, \ldots, p$$

cannot be obtained analytically, and we need to adopt a numerical method such as the Newton-Raphson algorithm.

10.2 Iterative methods

To obtain an estimate of $\boldsymbol{\beta}_0$ numerically, we start with an initial estimate $\widehat{\boldsymbol{\beta}}^{(0)}$ and successively update it until the difference between two consecutive estimates

$$\|\widehat{\boldsymbol{\beta}}^{(t+1)} - \widehat{\boldsymbol{\beta}}^{(t)}\|,$$

is sufficiently small so that we can consider that convergence has occurred.

10.2.1 The Newton-Raphson algorithm for GLM

The Newton-Raphson (NR) algorithm is an iterative procedure for finding the root (zero) of a function f. Starting with $x^{(0)}$, an initial guess of a root, the NR algorithm approximates the function with its tangent line. Then, the root of the tangent line iteratively replaces the guess as

$$x^{(t+1)} = x^{(t)} - \frac{f(x^{(t)})}{f'(x^{(t)})}, \tag{10.2}$$

and this procedure is repeated until convergence. We see that if $x^{(t+1)} = x^{(t)}$, then we have $f(x^{(t)}) = 0$. The rationale behind the NR algorithm is that the derivative at a point is the slope of a tangent at that point:

$$f'(x^{(t)}) = \frac{\Delta y}{\Delta x}(x^{(t)}) = \frac{f(x^{(t)}) - 0}{x^{(t)} - x^{(t+1)}}$$

where $x^{(t+1)}$ is the root of the tangent line, see Figure 10.1. Re-arranging, we obtain the equation in (10.2).
We adopt this numerical method to solve the likelihood equations

$$\frac{\partial \ell}{\partial \beta_j}(\widehat{\boldsymbol{\beta}}) = 0, \qquad j = 1, \ldots, p.$$

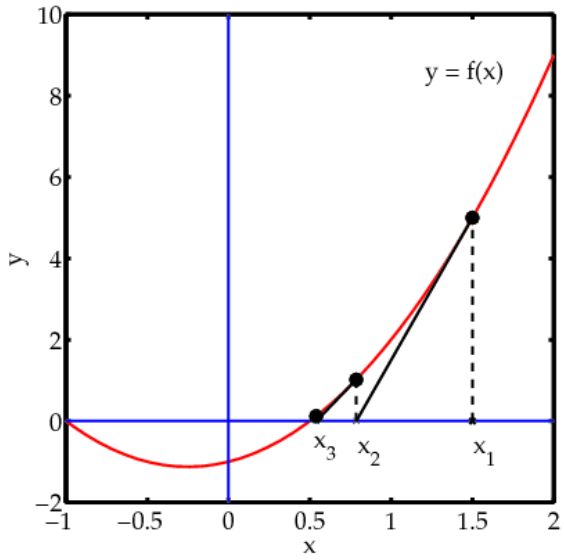


Figure 10.1: The aim is to identify the root of the function f. The initial guess is x_1. Then the NR algorithm approximates the function with its tangent line at $x = x_1$ and obtains a better guess for the root of the function, x_2. The procedure is repeated and the new guess of x_3 appears to have converged to the root of the function.

At the t-th iteration, we update the current solution $\widehat{\boldsymbol{\beta}}^{(t)}$ as

$$\widehat{\boldsymbol{\beta}}^{(t+1)} = \widehat{\boldsymbol{\beta}}^{(t)} - (\mathbf{H}^{(t)})^{-1}\mathbf{u}^{(t)},$$

where

$$\mathbf{u}^{(t)} = \mathbf{u}(\widehat{\boldsymbol{\beta}}^{(t)}) = \begin{bmatrix} \frac{\partial \ell}{\partial \beta_1}(\widehat{\boldsymbol{\beta}}^{(t)}) \\ \vdots \\ \frac{\partial \ell}{\partial \beta_p}(\widehat{\boldsymbol{\beta}}^{(t)}) \end{bmatrix} \quad \text{and}$$

$$\mathbf{H}^{(t)} = \mathbf{H}(\widehat{\boldsymbol{\beta}}^{(t)}) = \begin{bmatrix} \frac{\partial^2 \ell}{\partial \beta_1 \partial \beta_1}(\widehat{\boldsymbol{\beta}}^{(t)}) & \frac{\partial^2 \ell}{\partial \beta_1 \partial \beta_2}(\widehat{\boldsymbol{\beta}}^{(t)}) & \cdots & \frac{\partial^2 \ell}{\partial \beta_1 \partial \beta_p}(\widehat{\boldsymbol{\beta}}^{(t)}) \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 \ell}{\partial \beta_p \partial \beta_1}(\widehat{\boldsymbol{\beta}}^{(t)}) & \frac{\partial^2 \ell}{\partial \beta_p \partial \beta_2}(\widehat{\boldsymbol{\beta}}^{(t)}) & \cdots & \frac{\partial^2 \ell}{\partial \beta_p \partial \beta_p}(\widehat{\boldsymbol{\beta}}^{(t)}) \end{bmatrix}$$

are referred to as the vector of score statistics and the Hessian matrix, respectively, both

of which are evaluated at the current estimate $\widehat{\boldsymbol{\beta}}^{(t)}$. Notice that

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^{n} \frac{\partial \ell_i}{\partial \beta_j} = \sum_{i=1}^{n} \frac{d\ell_i}{d\eta_i}\frac{\partial \eta_i}{\partial \beta_j} = \underline{\sum_{i=1}^{n} \frac{d\ell_i}{d\eta_i} x_{ij}} = \mathbf{X}_{(j)}^{\top} \begin{bmatrix} \frac{d\ell_1}{d\eta_1} \\ \vdots \\ \frac{d\ell_n}{d\eta_n} \end{bmatrix},$$

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = \sum_{i=1}^{n} \frac{\partial^2 \ell_i}{\partial \beta_j \partial \beta_k} = \sum_{i=1}^{n} \frac{\partial}{\partial \beta_k}\left(\frac{\partial \ell_i}{\partial \beta_j}\right)$$

$$= \underline{\sum_{i=1}^{n} \frac{d}{d\eta_i}\left(\frac{d\ell_i}{d\eta_i} x_{ij}\right) x_{ik} = \sum_{i=1}^{n} \frac{d^2\ell_i}{d\eta_i^2} x_{ij} x_{ik}} = \mathbf{X}_{(j)}^{\top} \mathrm{diag}\left(\tfrac{d^2\ell_1}{d\eta_1^2}, \ldots, \tfrac{d^2\ell_n}{d\eta_n^2}\right) \mathbf{X}_{(j)},$$

where $\mathbf{X}_{(j)} = (x_{1j}, \ldots, x_{nj})^{\top}$ denotes the j-th column of $\mathbf{X} = [x_{ij}]$. Let $\mathbf{s}^{(t)} = \mathbf{s}(\widehat{\boldsymbol{\beta}}^{(t)})$ denote the n-vector with its elements

$$s_i^{(t)} = s_i(\widehat{\boldsymbol{\beta}}^{(t)}) = \frac{d\ell_i}{d\eta_i}(\widehat{\boldsymbol{\beta}}^{(t)}), \quad i = 1, \ldots, n,$$

and $\widetilde{\mathbf{W}}^{(t)} = \widetilde{\mathbf{W}}(\widehat{\boldsymbol{\beta}}^{(t)})$ denote the $n \times n$ diagonal matrix with its diagonal elements

$$\widetilde{W}_{ii}^{(t)} = \widetilde{W}(\widehat{\boldsymbol{\beta}}^{(t)}) = -\frac{d^2\ell_i}{d\eta_i^2}(\widehat{\boldsymbol{\beta}}^{(t)}), \quad i = 1, \ldots, n.$$

Then, the above calculations give

$$\mathbf{u}^{(t)} = \underline{\mathbf{X}^{\top}\mathbf{s}^{(t)}} \quad \text{and} \quad \mathbf{H}^{(t)} = \underline{-\mathbf{X}^{\top}\widetilde{\mathbf{W}}^{(t)}\mathbf{X}}.$$

Hence, the Newton-Raphson iteration step can be written as

$$\widehat{\boldsymbol{\beta}}^{(t+1)} = \widehat{\boldsymbol{\beta}}^{(t)} - (\mathbf{H}^{(t)})^{-1}\mathbf{u}^{(t)} = \widehat{\boldsymbol{\beta}}^{(t)} + (\mathbf{X}^{\top}\widetilde{\mathbf{W}}^{(t)}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{s}^{(t)}. \tag{10.3}$$

10.2.2 Fisher's scoring algorithm

Replacing $\widetilde{\mathbf{W}}^{(t)}$ by

$$\mathbf{W}^{(t)} = \mathbf{W}(\widehat{\boldsymbol{\beta}}^{(t)}) = \mathrm{diag}\left(\mathsf{E}\left(-\frac{d^2\ell_i}{d\eta_i^2}\right)(\widehat{\boldsymbol{\beta}}^{(t)}),\, i = 1, \ldots, n\right)$$

in the iteration step (10.3) leads to Fisher's scoring algorithm, where the following updating step is repeated until convergence:

$$\widehat{\boldsymbol{\beta}}^{(t+1)} = \widehat{\boldsymbol{\beta}}^{(t)} + (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{s}^{(t)}. \tag{10.4}$$

One criticism of the Newton-Raphson algorithm is that the iteration has a tendency to be unstable as $\widetilde{\mathbf{W}}^{(t)}$ may have negative diagonals unless $\widehat{\boldsymbol{\beta}}^{(t)}$ is already very close to the MLE. Also, $\widetilde{\mathbf{W}}^{(t)}$ may be harder to calculate. In contrary, $\mathbf{W}^{(t)}$ is easier to calculate and its diagonals are always positive since from (9.5), we have the i-th diagonal element of $\mathbf{W}$ satisfies

$$W_{ii} = \mathsf{E}\left(-\frac{d^2\ell_i}{d\eta_i^2}\right) = \underline{\qquad \mathsf{Var}\left(\frac{d\ell_i}{d\eta_i}\right) = \mathsf{Var}(s_i).}$$

This generally stabilises the iterative algorithm.

The iteration step in (10.4) resembles the weighted least squares (WLS) estimation in linear models: For some weight matrix $\mathbf{W} = \mathrm{diag}(w_1, \ldots, w_n)$, the weighted residual sum of squares

$$\sum_{i=1}^{n} w_i (Y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{W} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}),$$

is minimised at $\widehat{\boldsymbol{\beta}}^{\mathrm{wls}} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{Y}$, see Section 6.2 for further details. By writing

$$\mathbf{W} = \mathbf{W}^{(t)} \quad \text{and} \quad \mathbf{Y} = \mathbf{v}^{(t)} = \underline{\qquad \mathbf{X}\widehat{\boldsymbol{\beta}}^{(t)} + (\mathbf{W}^{(t)})^{-1} \mathbf{s}^{(t)},}$$

we obtain

$$\widehat{\boldsymbol{\beta}}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{v}^{(t)}$$

coinciding with (10.4). For this reason, Fisher's scoring algorithm is sometimes referred to as the Iteratively Re-weighted Least Squares (IRLS) algorithm. [9]

[9] Problem sheet Q2 invites you to use the `weights` argument in the `lm` function to perform the IRLS and update $\widehat{\boldsymbol{\beta}}^{(t)}$ till convergence, for the pneumonia data in Example 9.1.

10.3 Estimation of ϕ

The dispersion parameter ϕ is typically estimated separately from $\boldsymbol{\beta}_0$. From that $\mathsf{Var}(Y_i) = \phi V(\mu_i)$ with $\mu_i = \mathsf{E}(Y_i)$, we can write

$$\frac{\mathsf{E}(Y_i - \mu_i)^2}{V(\mu_i)} = \phi.$$

Hence, a simple estimator for ϕ is

$$\widehat{\phi} = \frac{1}{n-p} \sum_{i=1}^{n} \frac{(Y_i - \widehat{\mu}_i)^2}{V(\widehat{\mu}_i)},$$

where $\widehat{\mu}_i = g^{-1}(\mathbf{x}_i^\top \widehat{\boldsymbol{\beta}})$ are the fitted values for $\mu_i = \mathsf{E}(Y_i)$ at the MLE. Scaling by $(n-p)$ in the denominator is similarly motivated as that in the unbiased estimator of σ^2 in linear models, see Section 3.2.3.

11 Inference for generalised linear models

In linear modelling, *exact* distribution theory is available for inference thanks to the normality assumption. In generalised linear models, we make use of the *asymptotic* arguments available from the central limit theorem instead, to understand the behaviour of estimators and test statistics as $n \to \infty$.

11.1 Asymptotic distribution of the MLE

We first introduce the multivariate central limit theorem (CLT).

Theorem 11.1. Let $(\mathbf{Z}_i)_{i=1}^n$ be independent random p-vectors satisfying $\mathsf{E}(\mathbf{Z}_i) = \mathbf{0}$ and $\mathsf{E}(\mathbf{Z}_i\mathbf{Z}_i^\top) = \boldsymbol{\Sigma}_i$ and let $\boldsymbol{\Sigma} = n^{-1}\sum_{i=1}^n \boldsymbol{\Sigma}_i$ is positive definite. Then, under some conditions on the tail of $\mathbf{Z}_i$ (known as Lindeberg condition),

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \mathbf{Z}_i \to_d \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma}) \quad \text{as } n \to \infty,$$

where $\to_d$ implies the convergence in distribution.

In GLM, the assumption that Y_i are *identically distributed* does not hold since $\mathsf{E}(Y_i) = \mu_i$ varies according to $g(\mu_i) = \mathbf{x}_i^\top\boldsymbol{\beta}_0$ and so does $\mathsf{Var}(Y_i)$. For CLT, it is sufficient to have Y_i's independent with finite $\mathsf{Var}(Y_i)$ and some additional conditions on the tail. In what follows, we do not verify whether these conditions hold but simply take them as valid.

Exercise 11.1. Consider independent random variables $Y_1, \ldots, Y_n$ from some distribution $F_{\boldsymbol{\beta}_0}$ with $\boldsymbol{\beta}_0$ denoting the true parameter value. Also, denote the associated log-likelihood function by $\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \ell_i(\boldsymbol{\beta})$. As $n \to \infty$,

$$\frac{1}{\sqrt{n}}\frac{\partial \ell}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0) \to_d \mathcal{N}_p(\mathbf{0}, I_1(\boldsymbol{\beta}_0)) \tag{11.1}$$

where I_1 is the scaled *Fisher information matrix*

$$I_1(\boldsymbol{\beta}) = \left[\frac{1}{n}\mathsf{E}\left(-\frac{\partial^2 \ell}{\partial\beta_j\partial\beta_k}(\boldsymbol{\beta})\right)\right]_{j,k=1}^p = \frac{1}{n}I(\boldsymbol{\beta}).$$

Proof of (11.1). We use the multivariate CLT in Theorem 11.1 and the following results

from Section 9.4.1.

$$\mathsf{E}_{\boldsymbol{\beta}_0}\left(\frac{\partial \ell_i}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0)\right) = \mathbf{0}, \tag{11.2}$$

$$\mathsf{Cov}_{\boldsymbol{\beta}_0}\left(\frac{\partial \ell_i}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0)\right) = \left[\mathsf{E}_{\boldsymbol{\beta}_0}\left(-\frac{\partial^2 \ell_i}{\partial \beta_j \partial \beta_k}(\boldsymbol{\beta}_0)\right)\right]_{j,k=1}^{p} \tag{11.3}$$

Since $\{Y_i\}_{i=1}^n$ are independent, so are $\{\frac{\partial \ell_i}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0)\}_{i=1}^n$. Let us write $\mathbf{Z}_i = \frac{\partial \ell_i}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0)$. Then, by (11.2)–(11.3),

$$\mathsf{E}_{\boldsymbol{\beta}_0}(\mathbf{Z}_i) = \mathbf{0} \quad \text{and} \quad \frac{1}{n}\sum_{i=1}^n \mathsf{Cov}_{\boldsymbol{\beta}_0}(\mathbf{Z}_i) = \frac{1}{n}\sum_{i=1}^n \left[\mathsf{E}_{\boldsymbol{\beta}_0}\left(-\frac{\partial^2 \ell_i}{\partial \beta_j \partial \beta_k}(\boldsymbol{\beta}_0)\right)\right]_{j,k=1}^{p} = I_1(\boldsymbol{\beta}_0).$$

Then applying the multivariate CLT to $\{\mathbf{Z}_i\}_{i=1}^n$ gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \mathbf{Z}_i = \frac{1}{\sqrt{n}}\sum_{i=1}^n \frac{\partial \ell_i}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0) = \underline{\qquad \frac{1}{\sqrt{n}}\frac{\partial \ell}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0) \qquad} \to_d \mathcal{N}_p\left(\mathbf{0}, I_1(\boldsymbol{\beta}_0)\right).$$

□

Exercise 11.2. Let $\widehat{\boldsymbol{\beta}}$ be the maximum likelihood estimator of $\boldsymbol{\beta}_0$, which satisfies

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}}(\widehat{\boldsymbol{\beta}}) = \mathbf{0}.$$

Use the result from Exercise 11.1 and show that, as $n \to \infty$,

$$\sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \to_d \mathcal{N}_p(\mathbf{0}, (I_1(\boldsymbol{\beta}_0))^{-1}). \tag{11.4}$$

Hint: You can assume that, as $n \to \infty$, the matrix with its (j,k)-th element defined as

$$-\frac{1}{n}\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k}(\bar{\boldsymbol{\beta}}) = -\frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell_i}{\partial \beta_j \partial \beta_k}(\bar{\boldsymbol{\beta}}),$$

converges in probability to $I_1(\boldsymbol{\beta}_0)$. Here, $\bar{\boldsymbol{\beta}}$ is a vector in the convex hull of $\widehat{\boldsymbol{\beta}}$ and $\boldsymbol{\beta}_0$.

Proof of (11.4). We use the fact that the likelihood equation is solved by $\widehat{\boldsymbol{\beta}}$, along with the

mean value theorem, which leads to

$$\mathbf{0} = \frac{\partial \ell}{\partial \boldsymbol{\beta}}(\widehat{\boldsymbol{\beta}}) = \underline{\qquad \frac{\partial \ell}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0) + \mathbf{H}(\bar{\boldsymbol{\beta}})^\top (\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0),}$$

where $\bar{\boldsymbol{\beta}} \in$ convex hull$(\widehat{\boldsymbol{\beta}}, \boldsymbol{\beta}_0)$ and

$$\mathbf{H}(\bar{\beta}) = \left[\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k}(\bar{\boldsymbol{\beta}}) \right]_{j,k=1}^{p} = \left[\sum_{i=1}^{n} \left(\frac{\partial^2 \ell_i}{\partial \beta_j \partial \beta_k}(\bar{\boldsymbol{\beta}}) \right) \right]_{j,k=1}^{p}.$$

By the hint, we have $-\frac{1}{n}\mathbf{H}(\bar{\beta}) \to_P I_1(\boldsymbol{\beta}_0)$. Multiplying the above equation by $n^{-1/2}$, we have

$$\mathbf{0} = \frac{1}{\sqrt{n}} \frac{\partial \ell}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0) + \left(\frac{1}{n} \mathbf{H}(\bar{\boldsymbol{\beta}}) \right)^\top \sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0).$$

Re-arranging the above,

$$\begin{aligned} \sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) &= \underline{\qquad \left(-\frac{1}{n} \mathbf{H}(\bar{\boldsymbol{\beta}}) \right)^{-1} \frac{1}{\sqrt{n}} \frac{\partial \ell}{\partial \beta}(\boldsymbol{\beta}_0) \qquad} \\ &\to_d \underline{\qquad \mathcal{N}_p\left(\mathbf{0}, (I_1(\boldsymbol{\beta}_0))^{-1} I_1(\boldsymbol{\beta}_0) (I_1(\boldsymbol{\beta}_0))^{-1}\right) \qquad} = \underline{\qquad \mathcal{N}_p(\mathbf{0}, (I_1(\boldsymbol{\beta}_0))^{-1}), \qquad} \end{aligned} \tag{11.5}$$

which follows from:

(i) $\frac{1}{\sqrt{n}} \frac{\partial \ell}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_0) \to_d \mathcal{N}_p(\mathbf{0}, I_1(\boldsymbol{\beta}_0))$ (Exercise 11.1).

(ii) $(-\frac{1}{n}\mathbf{H}(\bar{\boldsymbol{\beta}}))^{-1} \to_p (I_1(\boldsymbol{\beta}_0))^{-1}$ from the hint, Slutsky's theorem (continuous mapping theorem) and the continuity of the inverse function.

(iii) If $\mathbf{x} \in \mathbb{R}^m$ is a normal random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and if $\mathbf{A}$ is a $n \times m$ matrix, then $\mathbf{A}\mathbf{x} \sim \mathcal{N}_n(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$ (see Section 2.3).

□

Since $I_1(\boldsymbol{\beta}_0)$ depends on the unknown $\boldsymbol{\beta}_0$, we estimate the variance of $\widehat{\boldsymbol{\beta}}$ by

$$\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}}) = (nI_1(\widehat{\boldsymbol{\beta}}))^{-1} = (I(\widehat{\boldsymbol{\beta}}))^{-1}.$$

The standard error of the j-th element of $\widehat{\boldsymbol{\beta}}$, is given by the square root of the j-th diagonal

element of $\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}})$, i.e.

$$\widehat{\text{s.e.}}(\widehat{\beta}_j) = \sqrt{[(I(\widehat{\boldsymbol{\beta}}))^{-1}]_{j,j}}.$$

From the asymptotic normality of the MLE, we obtain the (approximate) $100(1-\alpha)\%$ confidence interval of β_j as

$$\widehat{\beta}_j \pm z_{\alpha/2}\, \widehat{\text{s.e.}}(\widehat{\beta}_j),$$

where $z_{\alpha/2}$ denotes the upper $\alpha/2$-quantile of the standard normal distribution. If the confidence interval for β_j contains zero, we would not reject the null hypothesis $\mathcal{H}_0 : \beta_j = 0$ at the significance level $\alpha = 0.05$.

Example 11.1 (Pneumonia data)**.** Recall that from Example 10.1, the MLE of $\boldsymbol{\beta}_0 = (\beta_1, \beta_2)^\top$ are numerically calculated as $\widehat{\beta}_1 = -9.1894$ and $\widehat{\beta}_2 = -1.8296$. The MLE $\widehat{\boldsymbol{\beta}}$ has (asymptotically) a bivariate normal distribution

$$\widehat{\boldsymbol{\beta}} \sim_{\text{approx.}} \mathcal{N}_2(\boldsymbol{\beta}_0, (I(\boldsymbol{\beta}_0))^{-1}).$$

Noting that $\pi_i = \frac{\exp(\beta_1+\beta_2 x_i)}{1+\exp(\beta_1+\beta_2 x_i)}$, we have

$$\frac{\partial \pi_i}{\partial \beta_1} = \pi_i(1-\pi_i) \quad \text{and} \quad \frac{\partial \pi_i}{\partial \beta_2} = x_i \pi_i (1-\pi_i),$$

and recall from Example 10.1 that

$$\begin{aligned}
\ell(\boldsymbol{\beta}) &= \sum_{i=1}^{n} \left(\binom{m_i}{Y_i} + Y_i(\beta_1 + \beta_2 x_i) + m_i \log(1-\pi_i) \right), \\
\frac{\partial \ell}{\partial \beta_1} &= \sum_{i=1}^{n} (Y_i - m_i \pi_i), \\
\frac{\partial \ell}{\partial \beta_2} &= \sum_{i=1}^{n} (Y_i x_i - m_i x_i \pi_i).
\end{aligned}$$

Hence, $I(\boldsymbol{\beta}_0) = [I_{jk}]_{j,k=1}^2 = [\mathsf{E}(-\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k})(\boldsymbol{\beta}_0)]_{j,k=1}^2$ is calculated by obtaining the matrix of

second cross-partial derivatives

$$\begin{aligned}
\frac{\partial^2 \ell}{\partial \beta_1^2} &= -\sum_{i=1}^n m_i \frac{\partial \pi_i}{\partial \beta_1} = -\sum_{i=1}^n m_i \pi_i (1-\pi_i) \ , \\
\frac{\partial^2 \ell}{\partial \beta_2^2} &= -\sum_{i=1}^n m_i x_i \frac{\partial \pi_i}{\partial \beta_2} = -\sum_{i=1}^n m_i x_i^2 \pi_i (1-\pi_i) \ , \\
\frac{\partial^2 \ell}{\partial \beta_1 \partial \beta_2} &= -\sum_{i=1}^n m_i x_i \frac{\partial \pi_i}{\partial \beta_1} = -\sum_{i=1}^n m_i x_i \pi_i (1-\pi_i) \quad .
\end{aligned}$$

The elements of the Fisher information matrix are thus computed as

$$\begin{aligned}
I_{11} &= \sum_{i=1}^n m_i \pi_i (1-\pi_i) \ , \\
I_{22} &= \sum_{i=1}^n m_i x_i^2 \pi_i (1-\pi_i) \ , \\
I_{12} = I_{21} &= \sum_{i=1}^n m_i x_i \pi_i (1-\pi_i). \\
\end{aligned}$$

Since $I(\boldsymbol{\beta}_0)$ depends on the unknown parameters $\boldsymbol{\beta}_0$, we estimate $\mathsf{Cov}(\widehat{\boldsymbol{\beta}})$ by $\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}}) = (I(\widehat{\boldsymbol{\beta}}))^{-1}$. For the pneumonia data, we obtain

$$\widehat{\mathsf{Cov}}(\widehat{\boldsymbol{\beta}}) = \begin{pmatrix} 1.5753 & 0.3158 \\ 0.3158 & 0.0648 \end{pmatrix},$$

from which we derive that

$$\begin{aligned}
\widehat{\text{s.e.}}(\widehat{\beta}_1) &= \sqrt{1.5753} = 1.2551 \ , \\
\widehat{\text{s.e.}}(\widehat{\beta}_2) &= \sqrt{0.0648} = 0.2546 \ , \\
\widehat{\text{corr}}(\widehat{\beta}_1, \widehat{\beta}_2) &= 0.3158/(1.2551 \cdot 0.2546) = 0.9882.
\end{aligned}$$

Then (approximate) 95% confidence intervals for β_1 and β_2 are, respectively,

$$\begin{aligned}
-9.1894 \pm 1.96 \cdot 1.2551 &= (-11.649, -6.729) \ , \\
-1.8296 \pm 1.96 \cdot 0.2546 &= (-2.329, -1.331) \ .
\end{aligned}$$

Since the confidence interval of β_2 does not contain zero, we ___reject___ the null hypothesis that the serum dose (in logarithm) does not have an effect on the probability of a mouse dying from pneumonia.

11.2 Prediction

After conducting an experiment, it is often of interest to calculate a range of plausible values for a future outcome. For example, in the case of the pneumonia data, the scientist may want to know the probability of survival for a group of mice given a dose of the serum that was not considered in the original study.

Let $\widehat{\eta}_* = \mathbf{x}_*^\top \widehat{\boldsymbol{\beta}}$ denote the estimator of the linear predictor $\eta_* =$ ___$\mathbf{x}_*^\top \boldsymbol{\beta}_0$___ at a newly obtained vector of covariates $\mathbf{x}_*$. From the asymptotic normality of $\widehat{\boldsymbol{\beta}}$, for n large enough,

$$\begin{aligned}
&\widehat{\eta}_* \sim_{\text{approx}} \mathcal{N}(\eta_*, \mathsf{Var}(\widehat{\eta}_*)), \quad \text{where} \\
&\mathsf{Var}(\widehat{\eta}_*) = \mathsf{Var}(\mathbf{x}_*^\top \widehat{\boldsymbol{\beta}}) = \mathbf{x}_*^\top \mathsf{Cov}(\widehat{\boldsymbol{\beta}}) \mathbf{x}_* = \mathbf{x}_*^\top (I(\boldsymbol{\beta}_0))^{-1} \mathbf{x}_*.
\end{aligned}$$

A natural estimator of $\mathsf{Var}(\widehat{\eta}_*)$ is given by $\widehat{\mathsf{Var}}(\widehat{\eta}_*) = \mathbf{x}_*^\top (I(\widehat{\boldsymbol{\beta}}))^{-1} \mathbf{x}_*$, and a $100(1-\alpha)\%$ confidence interval for η_* is derived as

$$(\widehat{\eta}_*^L, \widehat{\eta}_*^R) = (\widehat{\eta}_* - z_{\alpha/2} \sqrt{\widehat{\mathsf{Var}}(\widehat{\eta}_*)},\ \widehat{\eta}_* + z_{\alpha/2} \sqrt{\widehat{\mathsf{Var}}(\widehat{\eta}_*)}).$$

Since $\mu = g^{-1}(\eta)$ is a monotonic function of η, a $100(1-\alpha)\%$ confidence interval for $\mathsf{E}(Y_*) = \mu_* = g^{-1}(\eta_*)$, the mean of Y_* associated with $\mathbf{x}_*$, is

$$\left(g^{-1}(\widehat{\eta}_*^L), g^{-1}(\widehat{\eta}_*^R)\right). \tag{11.6}$$

11.3 Deviance

11.3.1 Goodness-of-fit

In linear modelling, the goodness-of-fit is measured by the coefficient R^2 or its adjusted version R^2_{adj} which, involving the residual sum of squares $\|\mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2$, account for the proportion of the variability in Y_i explained by $\widehat{\eta}_i = \widehat{\mu}_i = \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}$. In the GLM framework, we cannot use such a measure since Y_i and η_i are in different scales.

The procedure of fitting a model to data can be thought of as replacing a set of observations ___$y_1, \ldots, y_n$___ by their fitted values ___$\widehat{\mu}_1, \ldots, \widehat{\mu}_n$.___ Consider two extreme model fits:

(i) Under-fit: The simplest model, the *null model*, has only ___one___ parameter and fits a single value for all i, i.e. $\widehat{\mu}_i = \widehat{\mu}$, which is too simple to be useful.

(ii) Over-fit: At the other extreme, the *full model* uses ___n___ parameters and obtains an exact fit, i.e. ___$\widehat{\mu}_i = y_i$___; it is too specific to the observed data (a single realisation of the random variables $Y_1, \ldots, Y_n$) to make meaningful inference, and does not generalise to other datasets that are generated in the same way.

A good fit lies somewhere in-between, where a *small* set of parameters achieve a reasonable accord between observed and predicted values.

Example 11.2. Given a dataset of four observations (y_i, x_i), $i = 1, \ldots, n = 4$, we consider four linear models: A null model (one parameter only, $p = 1$), a straight line (two parameters, $p = 2$), a quadratic model (three parameters, $p = 3$) and a cubic model (four parameters, $p = n = 4$). Below is the `R` code used to fit these four models.

```
y <- c(34, 26, 45, 25)
x <- c(5, 6, 7, 8)
mod.null <- lm(y ~ 1)
mod.straight <- lm(y ~ x)
mod.quad <- lm(y ~ x + I(x^2))
mod.full <- lm(y ~ x + I(x^2) + I(x^3))
```

The four fitted models are as below.

$$\begin{aligned}
\mathsf{E}(Y_i) &= \beta_0 = 32.5, \\
\mathsf{E}(Y_i) &= \beta_0 + \beta_1 x_i = 37.7 - 0.8 x_i, \\
\mathsf{E}(Y_i) &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 = -85.3 + 38.2 x_i - 3 x_i^2, \\
\mathsf{E}(Y_i) &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 = 2789 - 1333.5 x_i + 211.5 x_i^2 - 11 x_i^3.
\end{aligned}$$

All the models are linear in that they are linear in the parameters. Figure 11.1 demonstrates that when $n = p$, the fitted values exactly coincide with the observed values which, however, does not necessarily perform well in prediction.

If the discrepancies between observed and fitted values lie within acceptable limits, the model can be said to provide a good fit to the data, with the full model used as a benchmark for assessing the relative fit.

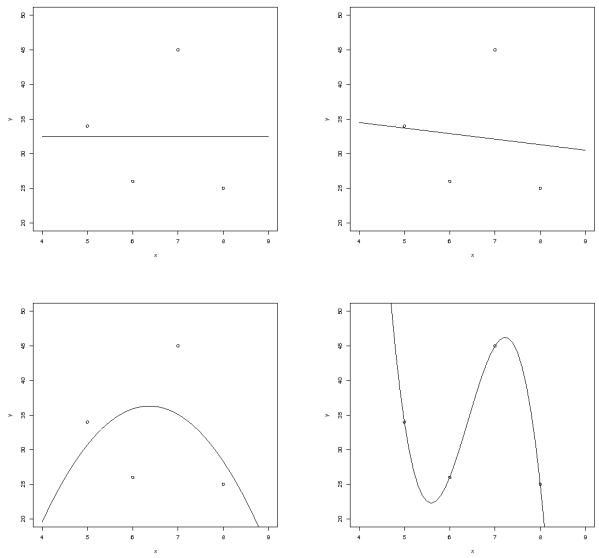


Figure 11.1: The four points represent the observed data points (y_i, x_i), $i = 1, \ldots, 4$, while the lines represent the fitted model for $\mathsf{E}(Y_i) = \beta_0$ (top left), $\mathsf{E}(Y_i) = \beta_0 + \beta_1 x_i$ (null model, top right), $\mathsf{E}(Y_i) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$ (bottom left) and $\mathsf{E}(Y_i) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$ (full mode, bottom right).

11.3.2 Definition of deviance

In what follows, we use $\ell(\widehat{\boldsymbol{\mu}}|\mathbf{y})$ to denote the value of the log-likelihood given the observations $\mathbf{y} = (y_1, \ldots, y_n)^\top$, evaluated at the fitted values $\widehat{\boldsymbol{\mu}} = (\widehat{\mu}_1, \ldots, \widehat{\mu}_n)$ obtained as $\widehat{\mu}_i = g^{-1}(\mathbf{x}_i^\top \widehat{\boldsymbol{\beta}})$ with the MLE $\widehat{\boldsymbol{\beta}}$. In doing so, we view model fitting as replacing the observed values with their fitted counterparts. The set of fitted values for the full (saturated) model is just the data values $y_1, \ldots, y_n,$ and the log-likelihood of the full model is given by $\ell(\mathbf{y}|\mathbf{y})$, which is in fact the maximum achievable log-likelihood.
We adopt the *scaled deviance* to compare different model fits, which is defined as

$$D^*(\mathbf{y}; \widehat{\boldsymbol{\mu}}) = 2(\ell(\mathbf{y}|\mathbf{y}) - \ell(\widehat{\boldsymbol{\mu}}|\mathbf{y})).$$

Let $\boldsymbol{\beta}_{0,F}$ be the n-vector of parameters which maximise the expected log-likelihood under the full model ('F' stands for 'full') using n parameters. Then, since $\widehat{\mu}_i = g^{-1}(\mathbf{x}_i^\top \widehat{\boldsymbol{\beta}})$, we

may also write

$$D^*(\mathbf{y};\widehat{\boldsymbol{\mu}}) = 2(\ell(\widehat{\boldsymbol{\beta}}_F|\mathbf{y}) - \ell(\widehat{\boldsymbol{\beta}}_R|\mathbf{y})),$$

where $\widehat{\boldsymbol{\beta}}_F$ is the n-vector of parameter estimates for the full model and $\widehat{\boldsymbol{\beta}}_R$ is the p-vector ($p < n$) of parameter estimates. Here, 'R' stands for 'restricted' and this subscript is introduced to contrast $\boldsymbol{\beta}_0 = \boldsymbol{\beta}_{0,R} \in \mathbb{R}^p$ against $\boldsymbol{\beta}_{0,F} \in \mathbb{R}^n$ estimated by $\widehat{\boldsymbol{\beta}}_F$. We also define the unscaled deviance as

$$D(\mathbf{y};\widehat{\boldsymbol{\mu}}) = \phi D^*(\mathbf{y};\widehat{\boldsymbol{\mu}}),$$

where ϕ is the dispersion parameter.

Example 11.3. Consider a multiple linear regression model with independent errors from $\mathcal{N}(0,\sigma^2)$ with known σ^2. Suppose that using p regressors, we obtain fitted values $\widehat{\mu}_1,\ldots,\widehat{\mu}_n$ for the responses $y_1,\ldots,y_n$. The log-likelihood is

$$\begin{aligned} 2\ell(\boldsymbol{\mu}|\mathbf{y}) &= 2\sum_{i=1}^n \log f(y_i;\mu_i) = \sum_{i=1}^n \left(-\log(2\pi\sigma^2) - \frac{(y_i-\mu_i)^2}{\sigma^2}\right) \\ &= \underline{\qquad -n\log(2\pi\sigma^2) - \frac{1}{\sigma^2}\sum_{i=1}^n (y_i-\mu_i)^2. \qquad} \end{aligned}$$

Therefore

$$D^*(\mathbf{y};\widehat{\boldsymbol{\mu}}) = 2(\ell(\mathbf{y}|\mathbf{y}) - \ell(\widehat{\boldsymbol{\mu}}|\mathbf{y})) = \underline{\qquad \frac{1}{\sigma^2}\sum_{i=1}^n (y_i-\widehat{\mu}_i)^2. \qquad}$$

Here, the dispersion parameter is $\phi =$ $\underline{\quad \sigma^2, \quad}$ and thus the unscaled deviance is

$$D(\mathbf{y};\widehat{\boldsymbol{\mu}}) = \sigma^2 D^*(\mathbf{y};\widehat{\boldsymbol{\mu}}) = \sum_{i=1}^n (y_i-\widehat{\mu}_i)^2.$$

Notice that D is simply the $\underline{\quad \text{residual sum of squares,} \quad}$ a familiar measure of goodness-of-fit in linear modelling. The scaled deviance is a scaled version of the RSS relative to the error variance.

Example 11.4. Consider a generalised linear model where

$$Y_i \sim \text{Poisson}(\mu_i) \quad \text{where} \quad \log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}_0, \quad \text{for} \quad 1 \leq i \leq n.$$

This is the framework of the moth data analysis introduced in Example 9.2. The log-likelihood at the observed $\mathbf{y} = (y_1, \ldots, y_n)^\top$ gives

$$\ell(\boldsymbol{\mu}|\mathbf{y}) = \sum_{i=1}^{n} \log f(y_i|\mu_i) = \underline{\qquad \sum_{i=1}^{n} \left(y_i \log(\mu_i) - \mu_i - \log(y_i!)\right). \qquad}$$

Therefore

$$D^*(\mathbf{y}; \widehat{\boldsymbol{\mu}}) = 2(\ell(\mathbf{y}|\mathbf{y}) - \ell(\widehat{\boldsymbol{\mu}}|\mathbf{y})) = \underline{\qquad 2\sum_{i=1}^{n} \left(y_i \log\left(\frac{y_i}{\widehat{\mu}_i}\right) - (y_i - \widehat{\mu}_i)\right). \qquad}$$

In this model, the dispersion parameter is $\phi =$ ___1___ and thus the unscaled deviance is equal to the scaled deviance.

11.3.3 Asymptotic distribution of deviance

We are often faced with the following test:

$\mathcal{H}_0$: The model with p parameters provides a good fit to the data, vs.

$\mathcal{H}_1$: The model does not give a good fit.

If $D^*(\mathbf{y}; \widehat{\boldsymbol{\mu}})$ is 'small', it indicates that the model with p parameters provides a fit nearly as good as that of the saturated one. For a more precise statement, we derive the asymptotic distribution of the deviance $D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}})$ under $\mathcal{H}_0$, as

$$\underline{\qquad D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}}) \sim_{\mathcal{H}_0} \chi^2_{n-p} \text{ asymptotically,} \qquad}$$

where $\mathbf{Y} = (Y_1, \ldots, Y_n)^\top$ denotes the vector of random responses. This is not surprising since, in fact, $D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}})$ is the likelihood ratio test statistic for testing

$$\mathcal{H}_0 : \boldsymbol{\beta}_{0,F} = \begin{bmatrix} \boldsymbol{\beta}_{0,R} \}_p \\ \mathbf{0} \}_{n-p} \end{bmatrix}$$

against the alternative without such a restriction, i.e.

$$D^*(\mathbf{Y};\widehat{\boldsymbol{\mu}}) = 2(\ell(\widehat{\boldsymbol{\beta}}_F) - \ell(\widehat{\boldsymbol{\beta}}_R)) = -2\log\left(\frac{\max_{\boldsymbol{\beta}_F \text{ is restricted as in } \mathcal{H}_0} L(\boldsymbol{\beta}_F)}{\max_{\boldsymbol{\beta}_F \in \mathbb{R}^n} L(\boldsymbol{\beta}_F)}\right).$$

The following lemma plays an important role.

Lemma 11.1. Let $\mathbf{Y} = (Y_1, \ldots, Y_n)^\top$ be a vector of random variables generated from a generalised linear model. That is, all Y_i follow a distribution from the exponential family, whose mean values are related to those of the linear predictor $\mathbf{x}_i^\top \boldsymbol{\beta}_0$ via the link function g, as $g(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}_0$. With $\widehat{\boldsymbol{\beta}}$ denoting the MLE of $\boldsymbol{\beta}_0$,

$$2(\ell(\widehat{\boldsymbol{\beta}}|\mathbf{Y}) - \ell(\boldsymbol{\beta}_0|\mathbf{Y})) \to_d \chi^2_p \quad \text{as } n \to \infty,$$

where p is the dimension of $\boldsymbol{\beta}_0$.

Proof of Lemma 11.1. For simplicity, let us write $\ell(\boldsymbol{\beta}_0) = \ell(\boldsymbol{\beta}_0|\mathbf{Y})$. The mean value expansion of $\ell(\boldsymbol{\beta}_0)$ around $\widehat{\boldsymbol{\beta}}$ yields

$$\begin{aligned}\ell(\boldsymbol{\beta}_0) = &\quad \ell(\widehat{\boldsymbol{\beta}}) + (\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}})^\top \frac{\partial \ell}{\partial \boldsymbol{\beta}}(\widehat{\boldsymbol{\beta}}) + \frac{1}{2}(\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}})^\top \mathbf{H}(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}}) \\ = &\quad \ell(\widehat{\boldsymbol{\beta}}) + \frac{1}{2}(\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}})^\top \mathbf{H}(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}}),\end{aligned}$$

where the second equation follows from the definition of $\widehat{\boldsymbol{\beta}}$ as an MLE, $\bar{\boldsymbol{\beta}}$ is in the convex hull of $\widehat{\boldsymbol{\beta}}$ and $\boldsymbol{\beta}_0$ and $\mathbf{H}(\boldsymbol{\beta})$ is the Hessian matrix with its entries

$$[\mathbf{H}(\boldsymbol{\beta})]_{j,k} = \frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k}(\boldsymbol{\beta}).$$

Notice that $\mathsf{E}(-\mathbf{H}(\boldsymbol{\beta}_0))$ is the Fisher information matrix $\mathbf{I}(\boldsymbol{\beta}_0)$. Then

$$\begin{aligned}2(\ell(\widehat{\boldsymbol{\beta}}) - \ell(\boldsymbol{\beta}_0)) = &\quad -(\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}})^\top \mathbf{H}(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}}) = -(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \mathbf{H}(\bar{\boldsymbol{\beta}})(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \\ = &\quad \sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \left(-\frac{1}{n}\mathbf{H}(\bar{\boldsymbol{\beta}})\right) \sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0). \qquad (11.7)\end{aligned}$$

Since (see Exercise 11.2)

$$-\frac{1}{n}\mathbf{H}(\bar{\boldsymbol{\beta}}) \to_p I_1(\boldsymbol{\beta}_0) \quad \text{and} \quad \sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \to_d \mathcal{N}_p(\mathbf{0}, (I_1(\boldsymbol{\beta}_0))^{-1})$$

we have

$$2(\ell(\widehat{\boldsymbol{\beta}}) - \ell(\boldsymbol{\beta}_0)) \to_d \boldsymbol{\gamma}^\top I_1(\boldsymbol{\beta}_0)\boldsymbol{\gamma},$$

where $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)^\top \sim \mathcal{N}_p(\mathbf{0}, (I_1(\boldsymbol{\beta}_0))^{-1})$. Invoking the results on the quadratic form of multivariate normal random vector (see Section 2.3.2),

$$\boldsymbol{\gamma}^\top I_1(\boldsymbol{\beta}_0)\boldsymbol{\gamma} \sim \chi^2_p \quad \text{and therefore} \quad 2(\ell(\widehat{\boldsymbol{\beta}}) - \ell(\boldsymbol{\beta}_0)) \to_d \chi^2_p.$$

□

Theorem 11.2 (Wilks' theorem)**.** Under the same framework as in Lemma 11.1, we have

$$D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}}) \to_d \chi^2_{n-p}$$

where p is the dimension of $\boldsymbol{\beta}_0 = \boldsymbol{\beta}_{0,R}$ and $\widehat{\boldsymbol{\mu}} = (\widehat{\mu}_1, \ldots, \widehat{\mu}_n)$ with $\widehat{\mu}_i = g^{-1}(\mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_R)$.

Sketch of the proof (Wilks' theorem). Decompose the deviance as

$$\begin{aligned} D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}}) &= 2(\ell(\widehat{\boldsymbol{\beta}}_F) - \ell(\widehat{\boldsymbol{\beta}}_R)) \\ &= 2(\ell(\boldsymbol{\beta}_{0,F}) - \ell(\boldsymbol{\beta}_{0,R})) - 2(\ell(\widehat{\boldsymbol{\beta}}_R) - \ell(\boldsymbol{\beta}_{0,R})) + 2(\ell(\widehat{\boldsymbol{\beta}}_F) - \ell(\boldsymbol{\beta}_{0,F})) \\ &= T_1 - T_2 + T_3. \end{aligned}$$

Under $\mathcal{H}_0$, we have $T_1 = 0$. By Exercise 11.1,

$$\frac{1}{\sqrt{n}} \frac{\partial \ell}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}_{0,F}) \to_d \boldsymbol{\zeta} = \begin{bmatrix} \boldsymbol{\xi}\}_p \\ \boldsymbol{\eta}\}_{n-p} \end{bmatrix} \sim \mathcal{N}_n(\mathbf{0}, \mathbf{I}_1(\boldsymbol{\beta}_{0,F})).$$

Then by (11.5) and (11.7),

$$T_3 = 2(\ell(\widehat{\boldsymbol{\beta}}_F) - \ell(\boldsymbol{\beta}_{0,F})) \approx \boldsymbol{\zeta}^\top (\mathbf{I}_1(\boldsymbol{\beta}_{0,F}))^{-1} \boldsymbol{\zeta}.$$

Similarly, we can write

$$T_2 = 2(\ell(\widehat{\boldsymbol{\beta}}_R) - \ell(\boldsymbol{\beta}_{0,R})) \approx \boldsymbol{\xi}^\top \mathbf{A}^{-1} \boldsymbol{\xi} \quad \text{where} \quad \mathbf{I}_1(\boldsymbol{\beta}_{0,F}) = \begin{bmatrix} \mathbf{A}\}_{p\times p} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix}.$$

Then, using the Sherman-Morrison-Woodbury formula in (2.3), we yield

$$T_3 - T_2 \approx (\boldsymbol{\eta} - \mathbf{B}^\top \mathbf{A}^{-1}\boldsymbol{\xi})^\top (\mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1}\mathbf{B})^{-1} (\boldsymbol{\eta} - \mathbf{B}^\top \mathbf{A}^{-1}\boldsymbol{\xi}),$$
$$\text{with} \quad \boldsymbol{\eta} - \mathbf{B}^\top \mathbf{A}^{-1}\boldsymbol{\xi} \sim \mathcal{N}_{n-p}(\mathbf{0}, \mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1}\mathbf{B})$$

and hence $T_3 - T_2 \to_d \chi^2_{n-p}$. □

11.3.4 Usage of deviance for goodness-of-fit testing

Equipped with the asymptotic null distribution of $D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}})$, we are able to evaluate the goodness-of-fit of a given model to the data. To test the null hypothesis

$$\mathcal{H}_0 : \text{The model provides a good fit to the data,}$$

we adopt $D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}})$ as the test statistic and derive the p-value $\mathsf{P}(\quad Z > D^*(\mathbf{y}; \widehat{\boldsymbol{\mu}}) \quad)$, where Z denotes a random variable following the χ^2_{n-p} distribution. If the p-value is small (say, less than 0.05), we have enough evidence to reject $\mathcal{H}_0$ and conclude that the model does not provide a good fit to the data.

Example 11.5 (Pneumonia data)**.** For now, let `pneumoniaLogit` denote a `glm` object in `R`, fitting the logistic regression model discussed in Example 9.1; full details are given in Section 13.5.1.
The `R` command `summary(pneumoniaLogit)` produces summary statistics for inference using a particular generalised linear model. Included in these summary statistics are the values of the deviances.

```
    Null deviance: 90.0516 on 4 degrees of freedom
Residual deviance:  2.8089 on 3 degrees of freedom
```

The null model is $Y_i \sim \text{Binomial}(m_i, \pi_i)$ with $\quad \text{logit}(\pi_i) = \beta_1 \text{ for all } i. \quad$ The p-value is given by $\mathsf{P}(\quad chi^2_4 \quad > 90.0516) < 0.0005$, indicating that the null model should be rejected. For the model $\text{logit}(\pi_i) = \beta_1 + \beta_2 x_i$, the corresponding p-value is $\mathsf{P}(\quad \chi^2_3 \quad > 2.8089) = 0.45$, indicating that the model may be a good fit for the data, i.e. the model with the dose level (in logarithm) as a regressor explains the mortality rate of mice well.

11.4 Analysis of deviance

Suppose that Model A has p parameters and Model B has a subset of these parameters of size q (i.e. $q < p$), such that Model B is ___*nested*___ in Model A. Let us the scaled deviance of the two models by $D_A^* = D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}}_A)$ and $D_B^* = D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}}_B)$, respectively. If Model B, although having fewer variables, explains the data as well as Model A such that $D_B^* - D_A^*$ is small, we prefer Model B over Model A; if $D_B^* - D_A^*$ is large, we select Model A over Model B. Writing the null and alternative hypotheses as

$$\mathcal{H}_0 : \text{Model B} \quad \text{vs.} \quad \mathcal{H}_1 : \text{Model A},$$

the asymptotic distribution of $D_B^* - D_A^*$ *under Model B* is ___χ^2_{p-q},___ which follows from the arguments analogous to those adopted in the proof of Theorem 11.2. The p-value associated with this test is $\mathsf{P}(Z$ ___$> D_B^* - D_A^*$___$)$, where Z is a random variable following the χ^2_{p-q} distribution, and we reject $\mathcal{H}_0$ if the p-value is smaller than a given significance level (say 0.05). Referred to as the *analysis of deviance*, we can compare two nested models, sequentially building up the model from the null one by adding one parameter at a time.

Example 11.6 (Pneumonia data)**.** Continuing from Example 9.1, we model the mortality of mice as $Y_i \sim \text{Binomial}(m_i, \pi_i)$, $1 \le i \le n = 5$, with the link function $\text{logit}(\pi_i) = \beta_1 + \beta_2 x_i$, where x_i denotes the dose level (in logarithm). To test the significance of the dose level on mortality is equivalent to test

$$\mathcal{H}_0 : \text{logit}(\pi_i) = \beta_1 \quad \text{vs.} \quad \mathcal{H}_1 : \text{logit}(\pi_i) = \beta_1 + \beta_2 x_i.$$

Using the `glm` object `pneumoniaLogit` fitted on the pneumonia data, we produce the analysis of deviance table as below.

```
anova(pneumoniaLogit, test = 'Chisq')

Model: binomial, link: logit
Response: ymat
Terms added sequentially (first to last)

             Df   Deviance resid.   Df   Resid. Dev  P(>|Chi|)
    NULL                            4      90.052
```

```
log(dose)      1       87.243          3       2.809    9.6e-21
```

The very small p-value in the above display (`9.6e-21` $= 9.6 \times 10^{-21}$) indicates that we should reject $\mathcal{H}_0$ in favour of the current model.

In the example above, $\phi = 1$ and is known, such that the scaled and the unscaled deviances are identical. Generally ϕ may be unknown and has to be estimated. For instance, in Example 11.3, we showed that the scaled deviance for a normal distribution is

$$D^*(\mathbf{y}; \widehat{\boldsymbol{\mu}}) = 2(\ell(\mathbf{y}|\mathbf{y}) - \ell(\widehat{\boldsymbol{\mu}}|\mathbf{y})) = \frac{1}{\sigma^2} \sum_{i=1}^{n} (y_i - \widehat{\mu}_i)^2$$

with typically unknown σ^2, and it must be replaced by an estimate.
When we replace ϕ by its estimate $\widehat{\phi}$ from page 107 and yield $\widehat{D}_A^* = D_A/\widehat{\phi}$ and $\widehat{D}_B^* = D_B/\widehat{\phi}$, the additional uncertainty due to the estimation of $\widehat{\phi}$ needs to be reflected in the asymptotic null distribution of $\widehat{D}_B^* - \widehat{D}_A^*$. Without proof, we use the following result:

$$\frac{\widehat{D}_B^* - \widehat{D}_A^*}{p - q} \sim_{\text{approx.}} \underline{\quad F_{p-q,n-p} \quad} \quad \text{under Model B.}$$

In this case, we should use the F-distribution to perform analysis of deviance. In `R`, this can be achieved by setting `test = 'F'` in the function `anova`. When `test = 'Chisq'` is specified in the analysis of deviance for a GLM with unknown ϕ, the analysis of deviance is performed on the basis of the unscaled deviance.
In fact, the analysis of deviance exactly coincides with the F-test in the context of linear modelling when Gaussianity is assumed.

12 Model diagnosis for GLM

It is important to check the validity of the model fitted for the data whether the assumptions required for the model are appropriate, prior to performing any inference such as the analysis of deviance.

12.1 Residual analysis

In linear modelling, the first step in model diagnosis is to plot the residuals $e_i = Y_i - \widehat{\mu}_i$ and standardised residuals which contain information about the data not explained by the systematic part of the model. Examination of residuals is also the chief means for model checking in the case of GLMs. However, we cannot simply examine the raw residuals e_i as it is difficult to check the validity of the assumed mean-variance relationship ($\mathsf{Var}(Y_i) = \phi V(\mu_i)$ with $\mu_i = \mathsf{E}(Y_i)$) from e_i. Recall the example of the moth data (Example 9.2), where we model Y_i as following a Poisson distribution so that $\mathsf{Var}(Y_i) = \mathsf{E}(Y_i) = \mu_i$. However, it is challenging to judge whether the residual variability indeed follows such a relationship from the plot of e_i against $\widehat{\mu}_i$. For this reason it is typical to transform the raw residuals such that, if the model assumptions are appropriate, the transformed raw residuals should behave like the raw residuals from a Gaussian linear model.

12.1.1 Pearson residuals

One way to transform the residuals is to divide them by a quantity proportional to their standard deviation according to the fitted model. This gives rise to the Pearson residuals

$$e_i^p = \frac{Y_i - \widehat{\mu}_i}{\sqrt{V(\widehat{\mu}_i)}}.$$

The name 'Pearson residuals' relates to the fact that the sum of squares of the Pearson residuals, $\sum_{i=1}^{n}(e_i^p)^2$, gives the Pearson statistic. Asymptotically, when the modelling assumptions are valid and $\phi = 1$, the Pearson statistic follows the χ^2_{n-p} distribution, where p is the number of parameters involved, and we may use this to interpret the size of the Pearson residuals.

12.1.2 Deviance residuals

In practice, the distribution of the Pearson residuals tends to be asymmetric around zero, so that their behaviour is not as close to the residuals from linear modelling. The deviance residuals are preferable in this respect as shown below.

The deviance residuals are obtained by noting that the deviance plays much the same role for GLMs that the residual sum of squares plays for a Gaussian linear model. Indeed for a Gaussian linear model, the (unscaled) deviance is the residual sum of squares, i.e. the residuals are the square roots of the components of the (unscaled) deviance with the appropriate sign attached. Denoting by d_i the component of the deviance contributed by the i-th observation, we have

$$D(\mathbf{Y};\widehat{\boldsymbol{\mu}}) = 2\phi\sum_{i=1}^{n}(\ell(Y_i|Y_i) - \ell(\widehat{\mu}_i|Y_i)) = \sum_{i=1}^{n} d_i.$$

Comparing this to the RSS, $\sum_{i=1}^{n}(Y_i - \widehat{\mu}_i)^2 = \sum_{i=1}^{n} e_i^2$, we define

$$e_i^d = \underline{\qquad \text{sign}(Y_i - \widehat{\mu}_i)\sqrt{d_i} \qquad}.$$

By definition, the sum of squares of e_i^d gives the deviance itself. This comparison of the unscaled deviance and the RSS implies that the individual e_i^d are expected to behave like normal random variables when the fitted model is suitable for the data.

12.2 Influence and Cook's distance

As in linear modelling (see Section 3.2.4), we can define the influence or hat matrix $\mathbf{H}$ for GLMs. Recall the connection of Fisher's scoring (a.k.a. iteratively re-weighted least squares, IRLS) algorithm to the weighted least squares estimation (see Section 10.2.2):

$$\widehat{\boldsymbol{\beta}}^{(t+1)} = (\mathbf{X}^\top\mathbf{W}^{(t)}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{W}^{(t)}\mathbf{v}^{(t)},$$

where

$$\mathbf{W}^{(t)} = \mathbf{W}(\widehat{\boldsymbol{\beta}}^{(t)}) = \text{diag}\left(\mathsf{E}\left(-\frac{d^2\ell_i}{d\eta_i^2}\right)(\widehat{\boldsymbol{\beta}}^{(t)}),\, i = 1,\ldots,n\right).$$

From this, we define the hat matrix as

$$\mathbf{H} = \mathbf{W}^{1/2}\mathbf{X}(\mathbf{X}^\top\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{W}^{1/2},$$

with $\mathbf{W} = \mathbf{W}(\widehat{\boldsymbol{\beta}})$.
Unlike in linear modelling, in GLM, it generally does not hold that $\widehat{\boldsymbol{\mu}} = \mathbf{H}\mathbf{Y}$ since the estimators are often non-linear. However, $\mathbf{H}$ still shares many properties with the influence matrix in linear modelling.

(i) The trace of $\mathbf{H}$ is the number of identifiable parameters in the model,

$$\begin{aligned}\mathsf{tr}(\mathbf{H}) &= \mathsf{tr}(\mathbf{W}^{1/2}\mathbf{X}(\mathbf{X}^\top\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{W}^{1/2}) \\ &= \underline{\qquad \mathsf{tr}((\mathbf{X}^\top\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{W}\mathbf{X}) = \mathsf{tr}(\mathbf{I}_p) = p. \qquad}\end{aligned}$$

(ii) $\mathbf{H}$ is symmetric, i.e. $\underline{\quad \mathbf{H} = \mathbf{H}^\top \quad}$.

(iii) $\mathbf{H}$ is idempotent, i.e. $\underline{\quad \mathbf{H}^2 = \mathbf{H} \quad}$.

(iv) $\mathbf{H}\mathbf{W}^{1/2}\mathbf{X} = \mathbf{W}^{1/2}\mathbf{X}$ and $\mathbf{X}^\top\mathbf{W}^{1/2}\mathbf{H} = \mathbf{X}^\top\mathbf{W}^{1/2}$.

(v) $\mathbf{H}\mathbf{e}^p = \mathbf{0}$, where $\mathbf{e}^p = (e_1^p, \ldots, e_n^p)^\top$.

The diagonal elements h_{ii} of $\mathbf{H}$ are termed $\underline{\quad \textit{leverages} \quad}$ and measure the influence of each observation. The variance of the raw residual is, to a first-order approximation, $\mathsf{Var}(e_i) \simeq (1-h_{ii})\mathsf{Var}(Y_i)$. Therefore, if h_{ii} is close to one for some i, we have $\mathsf{Var}(e_i) \to 0$ and thus the i-th observation is deemed to have large influence on the model fit often due to an unusual combination of explanatory variables. Based on this, the residuals are standardised as

$$s_i^p = \frac{e_i^p}{\sqrt{1-h_{ii}}} \quad \text{and} \quad s_i^d = \frac{e_i^d}{\sqrt{1-h_{ii}}},$$

which are referred to as the standardised Pearson and deviance residuals, respectively.
A combination of large residuals and large leverage indicates a point with substantial influence on the fit. Cook's distance is a measure of influence based on the extent to which parameter estimates would change if one omitted the i-th observation, and is a function of the standardised residual and the leverage, see (5.10). Cook's distance in GLM is analo-

gously defined using Pearson residuals and leverages, as

$$D_i = (e_i^p)^2 \frac{h_{ii}}{(1-h_{ii})^2 \widehat{\phi} p} = (s_i^p)^2 \frac{h_{ii}}{(1-h_{ii}) \widehat{\phi} p}.$$

D_i over 0.5 are considered borderline problematic, whereas values over 1 are usually considered highly influential.

Example 12.1 (Moth data)**.** Continuing from Example 9.2, we model the number of moths caught overnight, Y_i, using the average night temperature x_i, by

$$Y_i \sim \text{Poisson}(\mu_i) \quad \text{where} \quad \log(\mu_i) = \beta_1 + \beta_2 x_i,$$

where $1 \leq i \leq n = 20$. Since $V(\mu_i) = \mu_i$, the Pearson residuals are given by

$$e_i^p = \frac{Y_i - \widehat{\mu}_i}{\sqrt{\widehat{\mu}_i}}, \qquad 1 \leq i \leq n.$$

Also, noting that $\ell(\mu_i|Y_i) = -\mu_i + Y_i \log(\mu_i) - \log(Y_i!)$, the deviance residuals are

$$e_i^d = \text{sign}(Y_i - \widehat{\mu}_i) \cdot \sqrt{2(Y_i \log(Y_i/\widehat{\mu}_i) - Y_i + \widehat{\mu}_i)}.$$

The following table contains the observations and the corresponding fitted values and residuals.

y_i	x_i	$\widehat{\mu}_i$	e_i	e_i^p	e_i^d
4	5	4.67	-0.67	-0.31	-0.32
6	10	7.81	-1.81	-0.65	-0.67
2	4	4.21	-2.21	-1.08	-1.20
7	6	5.18	1.82	0.80	0.76
9	7	5.74	3.26	1.36	1.26
⋮	⋮	⋮	⋮	⋮	⋮

The Pearson test statistic is $\sum_{i=1}^n (e_i^p)^2 = 25.54$, and comparing this to the $\chi^2_{n-p} = \chi^2_{18}$ distribution produces a p-value of 0.11, indicating that the model has a reasonable fit to the data. For a Poisson distribution, $D^* = D = \sum_{i=1}^n (e_i^d)^2 = 29.71$ and comparing this

to the χ^2_{18} distribution, we obtain the p-value $= 0.04$, from which we draw a contradicting conclusion.
In `R`, we fit this model by

```
moth.fit <- glm(moths ~ temp, family = poisson(link = log))
```

from which we can access the Pearson and deviance residuals

```
ep <- resid(moth.fit, 'pearson')
ed <- resid(moth.fit, 'deviance')
```

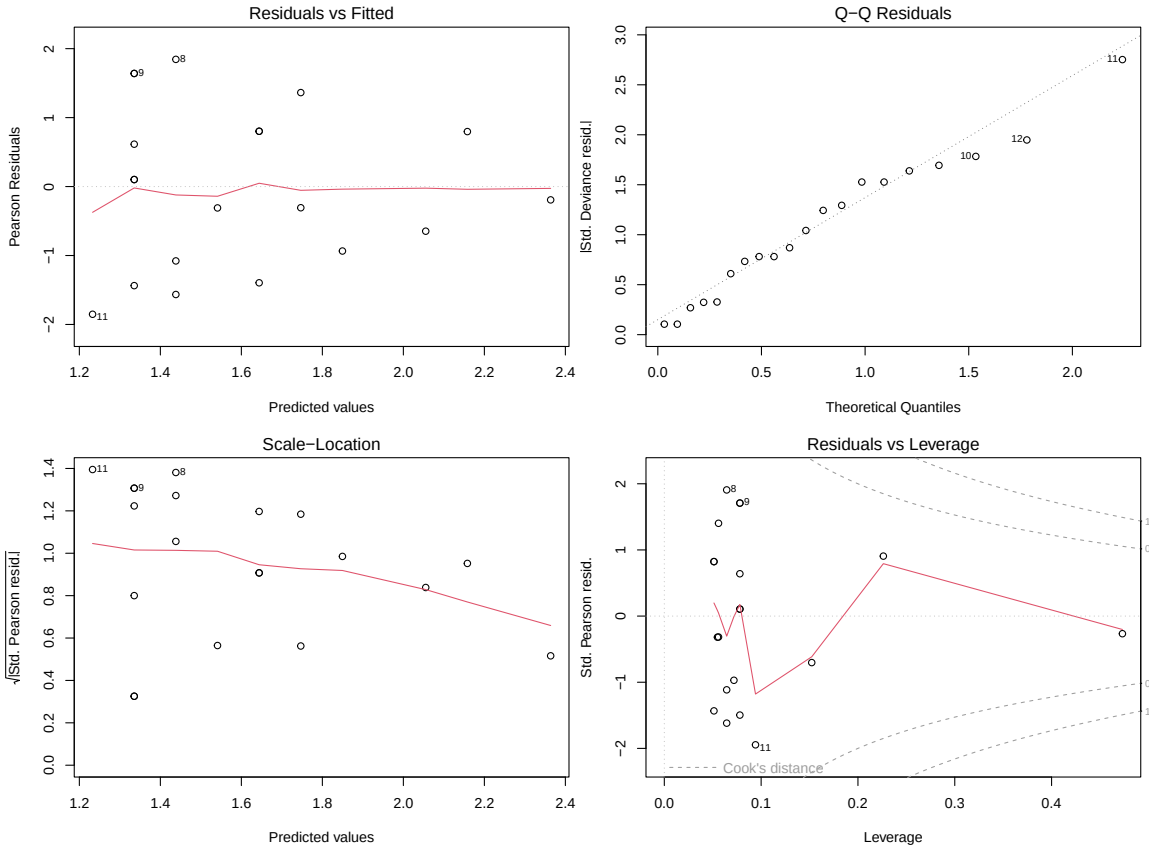


Figure 12.1: Residual plots from the moth data.

The command `plot(moth.fit)` plots the residual plots for model checking, which have similar interpretation as those plotted for model checking in linear modelling. Here, the

fitted (predicted) values are in the scale of the linear predictor η_i rather than the mean response μ_i. Also, some departure from a straight line relationship in the normal Q-Q plot is often to be expected. The bottom right plot of Figure 12.1 displays the Cook's distance of each observation where we can see that no observation is exercising unduly high influence on the model fit.

13 Binary and binomial responses

Binary responses taking values from {yes, no}, {affected, unaffected}, {presence, absence} and so on, are commonly observed; consider e.g. the responses collecting the presence or the absence of a disease or a symptom, or the failure or the success of mechanical parts. For the binary response Y_i, we are interested in modelling

$$\mathsf{E}(Y_i) = \mathsf{P}(Y_i = 1) = \pi_i$$

which, given covariates $\mathbf{x}_i$, can be written as a function of $\mathbf{x}_i$:

$$\mathsf{E}(Y_i|\mathbf{x}_i) = \mathsf{P}(Y_i = 1|\mathbf{x}_i) = \pi(\mathbf{x}_i)$$

The random variable Y_i follows the Binomial$(1, \pi_i)$ = Bernoulli(π_i) distribution with $\pi_i = \pi(\mathbf{x}_i)$. The probability mass function of Y_i is given as

$$f(y;\pi) = \pi^y(1-\pi)^{1-y} = \exp\left(y\log\left(\frac{\pi}{1-\pi}\right) + \log(1-\pi)\right),$$

which belongs to the exponential family with [10]

$$\phi = \underline{\quad 1, \quad} \quad \theta = \underline{\quad \log(\pi/(1-\pi)) = \text{logit}(\pi), \quad}$$
$$K(\theta) = \underline{\quad \log(1+e^{\theta}) \quad} \text{ and } q(\phi, y) = \underline{\quad 1. \quad}$$

Linear models are not appropriate for modelling Y_i since

(i) $\mathsf{E}(Y_i) = \pi_i = \pi(\mathbf{x}_i) \in$ $\underline{\quad [0,1]. \quad}$

(ii) $\mathsf{Var}(Y_i) =$ $\underline{\quad \pi(\mathbf{x}_i)(1-\pi(\mathbf{x}_i)) \quad}$ is not constant, but rather depends on the value of $\mathbf{x}_i$ through $\mathsf{E}(Y_i) = \pi(\mathbf{x}_i)$.

[10] See problem sheet Q2.

13.1 Odds ratio

13.1.1 Categorical covariates

A particular framework that arises frequently, especially in medical statistics, is the one in which the covariates are categorical. For example,

$$x_{ij} = \begin{cases} 1 & \text{if individual } i \text{ is exposed to allergen/treatment/etc. } j, \\ 0 & \text{otherwise .} \end{cases}$$

Let $x \in \{0, 1\}$ be the exposure indicator and let Y be a binary random variable that takes the value $Y = 1$ if a subject develops symptoms of some particular disease, and $Y = 0$ otherwise. We may model the distribution of Y given the exposure x, as

$$Y|x = 1 \sim \text{Binomial}(1, \pi_1) \quad \text{and} \quad Y|x = 0 \sim \text{Binomial}(1, \pi_0).$$

The odds of disease given exposure, $x = 1$, are defined as

$$\frac{\Pr(Y = 1|x = 1)}{\Pr(Y = 0|x = 1)} = \frac{\pi_1}{1 - \pi_1}.$$

Similarly, the odds of disease given no exposure, $x = 0$, are

$$\frac{\Pr(Y = 1|x = 0)}{\Pr(Y = 0|x = 0)} = \frac{\pi_0}{1 - \pi_0}.$$

The odds ratio (OR) is obtained by taking the ratio between the odds of the two groups:

$$\text{OR} = \frac{\pi_1/(1 - \pi_1)}{\pi_0/(1 - \pi_0)} = \frac{\pi_1(1 - \pi_0)}{\pi_0(1 - \pi_1)}.$$

- OR $= 1$ indicates that the even under study is equally likely to occur in both groups;
- OR > 1 indicates that the event is __more__ likely to occur in the exposed group than in the control group;
- OR < 1 indicates that the event is __less__ likely to occur in the exposed group.

13.1.2 Odds ratio in logistic regression

Logistic regression is often preferred to other forms of regression for binary response since, besides the logit function being the ___canonical___ link function, the parameter estimates have nice interpretations. Consider below the logistic regression model for the binary response Y_i and the vector of covariate information $\mathbf{x}_i$ which may include both categorical and continuous variables,

$$\mathsf{P}(Y_i = 1|\mathbf{x}_i) = \pi(\mathbf{x}_i) = \frac{\exp(\mathbf{x}_i^\top \boldsymbol{\beta}_0)}{1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta}_0)},$$
$$\log\left(\frac{\pi(\mathbf{x}_i)}{1 - \pi(\mathbf{x}_i)}\right) = \underline{\qquad \mathbf{x}_i^\top \boldsymbol{\beta}_0. \qquad}$$

The odds ratio for the two different sets of information $\mathbf{x}_1$ and $\mathbf{x}_2$ is

$$\frac{\pi(\mathbf{x}_1)/(1 - \pi(\mathbf{x}_1))}{\pi(\mathbf{x}_2)/(1 - \pi(\mathbf{x}_2))} = \underline{\qquad \exp((\mathbf{x}_1 - \mathbf{x}_2)^\top \boldsymbol{\beta}_0). \qquad}$$

If the j-th variable changes by 1 unit while all other variables remain constant, i.e.

$$x_{2j} = x_{1j} \pm 1 \;\text{ while }\; x_{1j'} = x_{2j'} \;\text{ for all }\; 1 \le j' \ne j \le p,$$

we have the odds ratio

$$\frac{\pi(\mathbf{x}_1)/(1 - \pi(\mathbf{x}_1))}{\pi(\mathbf{x}_2)/(1 - \pi(\mathbf{x}_2))} = \underline{\qquad \exp(\mp\beta_j), \qquad}$$

i.e. the odds is *multiplied* by ___exp(β_j)___ or ___exp($-\beta_j$)___, depending on the direction of the change. In the case of a categorical covariate taking values 0 or 1, a unit change reflects a change in the exposure status.

Example 13.1 (Pneumonia data)**.** Recall Example 9.1 where we model the number of deaths by pneumonia Y_i out of m_i mice which are given the dose level x_i (in logarithm), for $i = 1, \ldots, n = 5$. All subjects are assumed to be independent. In summary,

$$Y_i \sim \text{Binomial}(m_i, \pi_i) \;\text{ where }\; g(\mu_i) = \beta_1 + \beta_2 x_i. \tag{13.1}$$

With the choice of the logit link function $g(\mu_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right)$, from Example 10.1, the model

fitted for the pneumonia data is

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = -9.1894 - 1.8296\ \log(\texttt{dose}_i).$$

From this, we conclude that the odds of death is multiplied by $\exp(-1.8296) = 0.16$ as $\log(\texttt{dose})$ increases by 1, i.e. odds of death by pneumonia decreases by 84% as the logarithm of the dose level increases by 1.

13.2 Other link functions for binomial responses

Considering the problem from a less abstract and more applied perspective can motivate some other choices of link functions besides the logit link function for binary responses. For illustration, we continue with the model in (13.1).
Consider a random variable T_i called *tolerance*, which is the maximum exposure dose that the i-th subject can tolerate before it becomes toxic. Thus,

$$\mathsf{P}(\text{subject } i \text{ with exposure } x_i \text{ is affected}) = \mathsf{P}(T_i \leq x_i).$$

The distribution of T_i may be related to the covariates as

$$\mathsf{P}(T_i \leq x_i) = \Psi(\beta_1 + \beta_2 x_i),$$

where Ψ is some cumulative distribution function (CDF) and β_1 and β_2 are the parameters of interest. To summarise, $Y_i \sim \text{Binomial}(m_i, \Psi(\mathbf{x}_i^\top \boldsymbol{\beta}_0))$, where

$$\mathsf{P}(\text{subject } i \text{ with exposure } x_i \text{ is affected}) = \Psi(\beta_1 + \beta_2 x_i) = \Psi(\mathbf{x}_i^\top \boldsymbol{\beta}_0) \equiv \pi_i,$$

for all $1 \leq i \leq n$. Then,

$$\mu_i = \mathsf{E}(Y_i) = m_i \pi_i = m_i \Psi(\mathbf{x}_i^\top \boldsymbol{\beta}_0) \quad \text{and} \quad \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 = \Psi^{-1}(\mu_i/m_i),$$

which gives the link function $g(\mu_i) = \Psi^{-1}(\mu_i/m_i)$.
Recall that in logistic regression, we replace the inverse function Ψ^{-1} with the logit function whose inverse, the logistic function $f(x) = e^x/(1+e^x)$, is a CDF. Other choices are also possible, some of which are given below.

Probit link function We can set Ψ to be the CDF of the standard normal distribution,

$$\Psi(\eta) = \Phi(\eta) \equiv \int_{-\infty}^{\eta} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du.$$

We refer to the inverse function Φ^{-1} as the probit link function. In `R`, the command `binomial(link = 'probit')` in `glm` calls for the probit link function.

Extreme value distribution Extreme value distributions characterise the distribution of the extreme order statistics $X_{(n)} := \max_{1 \le i \le n} X_i$, where $X_{(i)}$ is the i-th largest value of $X_1, \ldots, X_n$. The following distribution is one such example,

$$\Psi(\eta) = 1 - \exp(-e^{\eta}),$$

whose inverse gives the complementary log-log link function

$$g(\pi) = \Psi^{-1}(\pi) = \underline{\quad \log(-\log(1-\pi)). \quad}$$

In `R`, the command `binomial(link = 'cloglog')` in `glm` calls for the complementary log-log link function.

13.3 Empirical logit

Let us continue with a GLM fitted to the pneumonia data in (13.1). Recall that we set as x_i the dose level *in logarithm* given to the i-th group, rather than the raw dose level.
Without taking account of the covariate information, we can obtain a crude estimator of π_i as $\underline{\quad Y_i/m_i. \quad}$ Replacing π_i in the logit function by Y_i/m_i, we obtain a crude estimator of $\text{logit}(\pi_i)$ as

$$\log\left(\frac{Y_i}{m_i - Y_i}\right)$$

This quantity is $-\infty$ and ∞, respectively, when Y_i is $\underline{\quad 0 \quad}$ and $\underline{\quad m_i. \quad}$ To avoid this, we adopt the following:

$$\log\left(\frac{Y_i + 0.5}{m_i - Y_i + 0.5}\right) = z_i(0.5) \ ,$$

which is referred to as the *empirical logit*.
The empirical logit values for our pneumonia data are given below, calculated as

$$z_1(0.5) = \log\left(\frac{35 + 0.5}{40 - 35 + 0.5}\right) = 1.865,$$

and so forth.

dose_i	m_i	y_i	$z_i(0.5)$
0.0028	40	35	1.865
0.0056	40	21	0.098
0.0112	40	9	-1.199
0.0225	40	6	-1.669
0.0450	40	1	-3.271

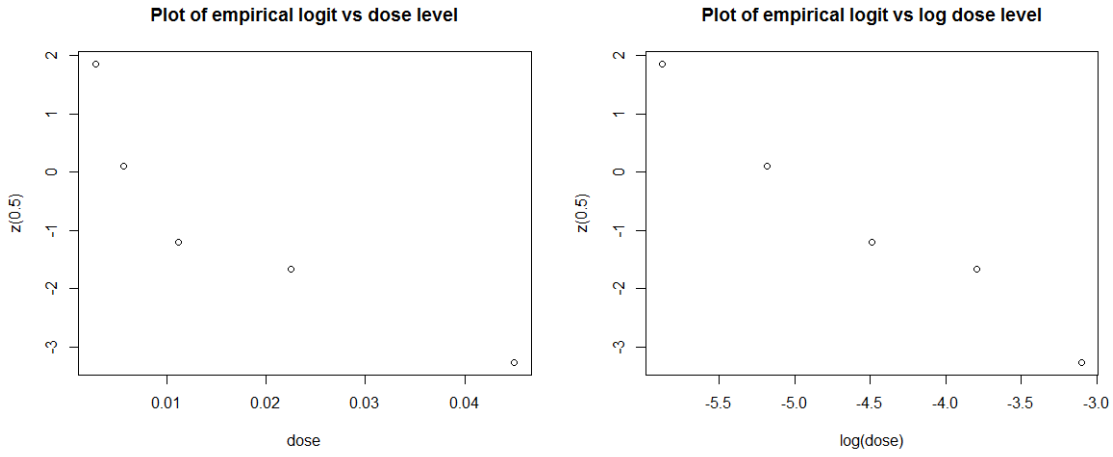


Figure 13.1: Empirical logit against the raw dose level (left) and the logarithm of the dose level (right).

The left panel of Figure 13.1 suggests that the empirical logit is not a linear function of raw dose level. On the other hand, the right panel of Figure 13.1 indicates that it is a linear function of the logarithm of the dose level, i.e. $x_i = \log(\text{dose}_i)$ is an appropriate choice with the logit link function

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \eta_i = \beta_1 + \beta_2 x_i.$$

13.4 Use of deviance for binary responses

For the GLM in (13.1) with $\mu_i = \mathsf{E}(Y_i) = m_i\pi_i$, the log-likelihood is calculated as

$$L(\boldsymbol{\mu}|\mathbf{Y}) = \prod_{i=1}^{n} \binom{m_i}{Y_i} \pi_i^{Y_i}(1-\pi_i)^{m_i-Y_i} = \prod_{i=1}^{n} \binom{m_i}{Y_i} \underline{\qquad \left(\frac{\mu_i}{m_i}\right)^{Y_i} \left(\frac{m_i-\mu_i}{m_i}\right)^{m_i-Y_i}},$$

$$\ell(\boldsymbol{\mu}|\mathbf{Y}) = \sum_{i=1}^{n} \left(\log \binom{m_i}{Y_i} + Y_i \log\left(\frac{\mu_i}{m_i}\right) + (m_i - Y_i)\log\left(\frac{m_i-\mu_i}{m_i}\right)\right).$$

Hence the scaled deviance $D^*(\mathbf{Y};\widehat{\boldsymbol{\mu}}) = 2(\ell(\mathbf{Y}|\mathbf{Y}) - \ell(\widehat{\boldsymbol{\mu}}|\mathbf{Y}))$, is given by

$$D^*(\mathbf{Y};\widehat{\boldsymbol{\mu}}) = \underline{\qquad 2\sum_{i=1}^{n}\left(Y_i \log\left(\frac{Y_i}{\widehat{\mu}_i}\right) + (m_i - Y_i)\log\left(\frac{m_i - Y_i}{m_i - \widehat{\mu}_i}\right)\right),}$$

and since $\phi = 1$, the scaled deviance is the same as the unscaled deviance. The use of deviance for the pneumonia data is illustrated in Examples 11.5–11.6.

Exercise 13.1 (Ungrouped binary data)**.** Let Y_i, $1 \le i \le n$, denote *ungrouped* binary data, i.e. independent random variables following the Bernoulli distribution with $\pi_i = \mathsf{P}(Y_i = 1)$. Given p-variate vectors of covariates $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top$, the logistic regression models the log odds of π_i to be linear in $\boldsymbol{\beta}_0 = (\beta_1, \ldots, \beta_p)^\top$, as

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \mathbf{x}_i^\top \boldsymbol{\beta}_0 = \beta_1 x_{i1} + \ldots + \beta_p x_{ip}. \tag{13.2}$$

Using the notation $\widehat{\pi}_1, \ldots, \widehat{\pi}_n$ for the fitted values of the observed $y_1, \ldots, y_n$, we obtain the scaled deviance of the fitted model (with the convention that $0 \cdot \log(0) = 0$):

$$\begin{aligned} D^*(\mathbf{y};\widehat{\boldsymbol{\mu}}) &= 2\sum_{i=1}^{n}(y_i\log(y_i) - y_i\log(\widehat{\pi}_i) + (1-y_i)\log(1-y_i) - (1-y_i)\log(1-\widehat{\pi}_i)) \\ &= 2\sum_{i=1}^{n}(y_i\log(\widehat{\pi}_i) + (1-y_i)\log(1-\widehat{\pi}_i)), \end{aligned}$$

since $y_i\log(y_i) = (1-y_i)\log(1-y_i) = 0$ when either $y_i = 0$ or $y_i = 1$. We have shown in

Example 11.1 that the likelihood equations are given by

$$\frac{\partial \ell}{\partial \beta_j}(\widehat{\boldsymbol{\beta}}) = \sum_{i=1}^{n}(y_i - \widehat{\pi}_i)x_{ij} = 0, \quad 1 \leq j \leq p.$$

Since $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 = \text{logit}(\pi_i)$, we can further show that

$$0 = \sum_{j=1}^{p} \widehat{\beta}_j \frac{\partial \ell}{\partial \beta_j}(\widehat{\boldsymbol{\beta}}) = \sum_{j=1}^{p} \widehat{\beta}_j \sum_{i=1}^{n}(y_i - \widehat{\pi}_i)x_{ij} = \sum_{i=1}^{n}(y_i - \widehat{\pi}_i)\sum_{j=1}^{p} \widehat{\beta}_j x_{ij}$$
$$= \sum_{i=1}^{n}(y_i - \widehat{\pi}_i)\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right),$$

i.e. $\sum_{i=1}^{n}(y_i - \widehat{\pi}_i)\text{logit}(\widehat{\pi}_i) = 0$ using (13.2). Then,

$$\sum_{i=1}^{n}(y_i \log(\widehat{\pi}_i) - y_i \log(1-\widehat{\pi}_i)) = \sum_{i=1}^{n}(\widehat{\pi}_i \log(\widehat{\pi}_i) - \widehat{\pi}_i \log(1-\widehat{\pi}_i)), \quad \text{so that}$$
$$\sum_{i=1}^{n}(y_i \log(\widehat{\pi}_i) + (1-y_i)\log(1-\widehat{\pi}_i)) = \sum_{i=1}^{n}(\widehat{\pi}_i \log(\widehat{\pi}_i) + (1-\widehat{\pi}_i)\log(1-\widehat{\pi}_i))$$

and therefore

$$D^*(\mathbf{y}; \widehat{\boldsymbol{\mu}}) = -2\sum_{i=1}^{n}(\widehat{\pi}_i \log(\widehat{\pi}_i) + (1-\widehat{\pi}_i)\log(1-\widehat{\pi}_i))$$
$$= -2\widehat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \widehat{\boldsymbol{\pi}} - 2\sum_{i=1}^{n}\log(1-\widehat{\pi}_i).$$

It shows that given $\widehat{\boldsymbol{\beta}}$, the conditional distribution of $D^*(\mathbf{y}; \widehat{\boldsymbol{\mu}})$ is a degenerate one and it cannot be used as a measure of the goodness-of-fit. This exact degeneracy occurs for the logistic regression model with $m_i = 1$, but near degeneracy occurs for any link function provided that m_i are small. The *difference* between scaled deviance values can still be used to compare two nested models via analysis of deviance as described in Section 11.4. For more information, see Section 4.4.5 of McCullagh and Nelder (1989).

13.5 R examples

13.5.1 Pneumonia data analysis

Read the pneumonia dataset in R as

```
mice <- read.table('file location\\mice.data', header=TRUE)
mice
```

```
    dose numberMice numberDeaths
1 0.0028         40           35
2 0.0056         40           21
3 0.0112         40            9
4 0.0225         40            6
5 0.0450         40            1
```

For binary responses, we need to provide information of both Y_i and $m_i - Y_i$ to the function glm:

```
mice$ymat <- cbind(mice$numberDeaths, mice$numberMice - mice$numberDeaths)
mice
```

```
    dose numberMice numberDeaths ymat.1 ymat.2
1 0.0028         40           35     35      5
2 0.0056         40           21     21     19
3 0.0112         40            9      9     31
4 0.0225         40            6      6     34
5 0.0450         40            1      1     39
```

The first column of mice$ymat denotes the number of deaths and the second column the number of survivors.
Now we use the function glm as

```
pneumoniaLogit <- glm(ymat ~ log(dose),
    family = binomial(link = logit), data = mice)
summary(pneumoniaLogit)
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -9.1894     1.2551  -7.322 2.45e-13 ***
log(dose)    -1.8296     0.2545  -7.188 6.59e-13 ***

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 90.0516  on 4  degrees of freedom
Residual deviance:  2.8089  on 3  degrees of freedom
AIC: 23.577

Number of Fisher Scoring iterations: 4
```

We can interpret $\widehat{\beta}_2 = -1.8296$ as that with $\exp(-1.8296) = 0.16$, the increase of $\log(\text{dose}_i)$ by unit 1 (which amounts to the dose level increasing by the multiplicative factor of e), the odds of death decreases by 84%.
The fitted values of π_i can be extracted as

```
pihat <- fitted.values(pneumoniaLogit)
pihat
```

```
         1          2          3          4          5
0.82712267 0.57375151 0.27467666 0.09557661 0.02887255
```

and the Fisher information matrix $I(\widehat{\boldsymbol{\beta}})$ can be computed as

```
I <- matrix(nrow=2, ncol=2)
I[1, 1] <- sum(serum$numberMice*pihat*(1-pihat))
I[2, 2] <- sum(serum$numberMice*pihat*(1-pihat)*log(serum$dose)^2)
I[1, 2] <- sum(serum$numberMice*pihat*(1-pihat)*log(serum$dose))
I[2, 1] <- I[1, 2]
I
```

```
           [,1]      [,2]
[1,]   28.05046 -136.7361
[2,] -136.73610  681.9737
```

The covariance matrix of $\widehat{\beta}_1$ and $\widehat{\beta}_2$ is the inverse of the above information matrix,

```
solve(I)
```

```
           [,1]       [,2]
[1,] 1.5753003 0.31584856
[2,] 0.3158486 0.06479414
```

Compare the diagonal elements with the standard error reported in the summary table.

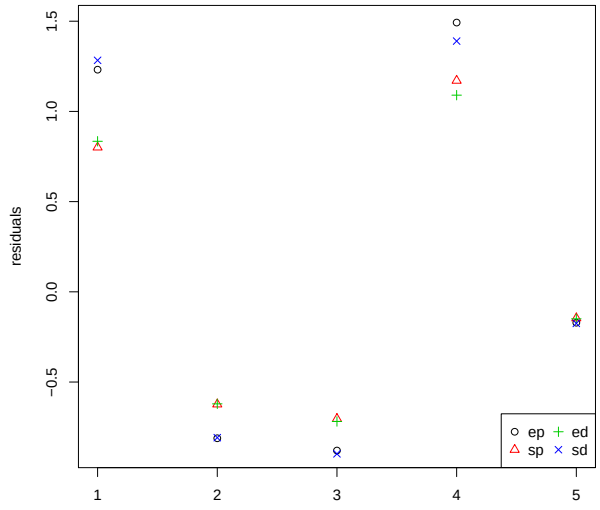


Figure 13.2: Pearson and deviance residuals, both standardised and unstandardised, for `pneumoniaLogit`.

The Pearson residuals can be calculated both using the function `resid` and manually:

```
ep <- resid(pneumoniaLogit, 'pearson')
ep <- (mice$numberDeaths - mice$numberMice*pihat)/
                    sqrt(mice$numberMice*pihat*(1-pihat))
```

Deviance residuals can be calculated similarly:

```
ed <- resid(pneumoniaLogit, 'deviance')
```

To further standardise the residuals, we access the leverages using the function `hatvalues`. Figure 13.2 plots both the standardised and unstandardised Pearson and deviance residuals.

```
hatvalues(pneumoniaLogit)

        1         2         3         4         5
0.5770939 0.4097906 0.3597682 0.3847795 0.2685679

sp <- ep/sqrt(1 - hatvalues(pneumoniaLogit))
sd <- ed/sqrt(1 - hatvalues(pneumoniaLogit))
```

We perform the prediction using the method `predict`. Setting the argument `type = 'link'` gives a predicted value of the linear predictor η, while setting `type = 'response'` gives a fitted value of π. The standard error of the predicted value is reported by setting `se.fit = TRUE`, from which we can compute the prediction intervals around the predicted values of η and π. When $\text{dose}_* = 0.01$ (such that $x_* = \log(\text{dose}_*) = \log(0.01)$), the predicted values for η_{x_*} and μ_{x_*} and their standard errors are

```
preds <- predict(pneumoniaLogit, data.frame(dose = c(0.01)),
                     type = 'link', se.fit = TRUE)
preds

$`fit`
         1
-0.7636743
$se.fit
[1] 0.2008861
$residual.scale
[1] 1

predict(pneumoniaLogit, data.frame(dose = c(0.01)),
          type = 'response', se.fit = TRUE)

$`fit`
```

```
       1
0.3178491
$se.fit
        1
0.04355633
$residual.scale
[1] 1
```

and the associated 95% confidence intervals are

```
fit <- preds$fit; fit # eta
upr <- preds$fit + (1.96 * preds$se.fit); upr # upper bound
lwr <- preds$fit - (1.96 * preds$se.fit); lwr # lower bound

-0.7636743
-0.3699376
-1.157411

pneumoniaLogit$family$linkinv(fit) # pi
pneumoniaLogit$family$linkinv(upr) # upper bound
pneumoniaLogit$family$linkinv(lwr) # lower bound

0.3178491
0.4085561
0.239138
```

Figure 13.3 plots the predicted values over a range of $\log(\text{dose}_i)$.

Finally, an analysis of deviance table is obtained as below.

```
anova(pneumoniaLogit, test = 'Chisq')

          Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                         4     90.052
log(dose)  1   87.243         3      2.809 < 2.2e-16
```

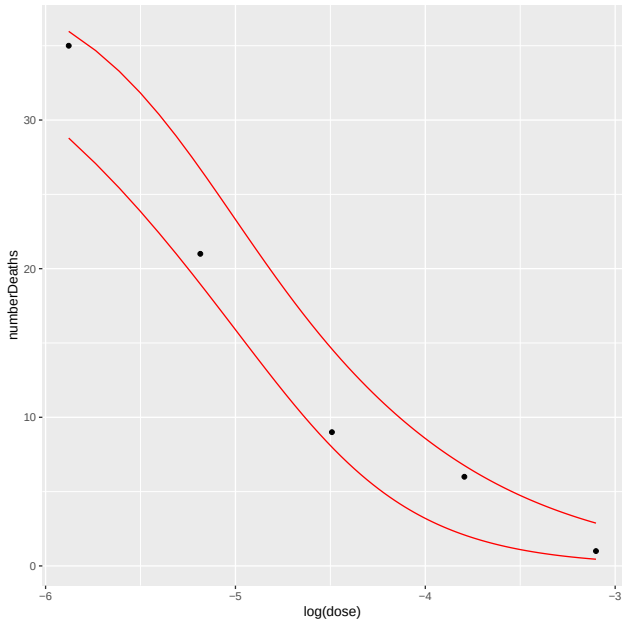


Figure 13.3: 95% confidence interval about $\mathsf{E}(Y_i|x_i) = \mu(x_i) = m_i\,\pi(x_i)$ over the range of $x_i = \log(\text{dose}_i)$, $i = 1, \ldots, 5$ for `pneumoniaLogit`.

13.5.2 Contraceptive use data

Table 13.1, adapted from Little (1978), shows the distribution of 1607 currently married and fertile women interviewed in the Fiji Fertility Survey of 1975, classified by current age, level of education, desire for more children and the use of contraceptive. We view current use of contraception as the binary response with two categories (use and non-use) and age, education and desire for more children as covariates. In this example all variables are categorical, i.e. factors. We aim describe the way in which contraceptive use varies by age, education and desire. Further, we ask e.g. whether the association between education and contraceptive use, is affected by the fact that women with high education tend to be younger and prefer smaller families, than women with lower primary education or less.
As all covariates are discrete factors, the 1607 subjects are partitioned into $n = 16 = \underline{\quad 4 \times 2 \times 2 \quad}$ groups, and the response variable Y_i represents the number of subjects using contraception for the i-th group, out of the m_i subjects belonging to the group. Due to that all regressors are categorical factors, it is easier to adopt the notation $Y_i \equiv Y_{jkl}$ when the i-th group is defined by the j-th level of age ($j = 1, 2, 3, 4$), the k-th level of desire

Table 13.1: Current use of contraception among married women by age, education and desire for more children, Fiji Fertility Survey, 1975.

Age	Education	Desires more children?	Contraceptive use no	yes	Total
< 25	low	yes	53	6	59
		no	10	4	14
	high	yes	212	52	264
		no	50	10	60
$25 - 29$	low	yes	60	14	74
		no	19	10	29
	high	yes	155	54	209
		no	65	27	92
$30 - 39$	low	yes	112	33	145
		no	77	80	157
	high	yes	118	46	164
		no	68	78	146
$40 - 49$	low	yes	35	6	41
		no	46	48	94
	high	yes	8	8	16
		no	12	31	43

for more children ($k = 1, 2$) and the l-th level of education ($l = 1, 2$), such that

$$Y_{jkl} \sim \text{Binomial}(m_{jkl}, \pi_{jkl}),$$

where π_{jkl} denotes the probability of a randomly selected woman belonging to the category (j, k, l) using the contraceptive.
First, we consider the model with all the *additive* terms

$$\log\left(\frac{\pi_{jkl}}{1 - \pi_{jkl}}\right) = \mu + \alpha_j + \beta_k + \gamma_l, \tag{13.3}$$

where α_j, β_k and γ_l denote the effects of the age, desire and education, respectively. As noted in Section 3.3, for model identifiability, we assume that all parameters with index 1 are set to be zero, i.e. $\alpha_1 = \beta_1 = \gamma_1 = 0$.
The dataset is contained in `cuse.dat` and in `R`, we fit the model in (13.3) as below.

```
cuse <- read.table('data/cuse.dat', header = TRUE)
```

```
cuse$ymat <- cbind(cuse$using, cuse$notUsing)
fit1 <- glm(ymat ~ age + education + wantsMore,
        family = binomial(link = 'logit'), data = cuse)
summary(fit1)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.8082     0.1590  -5.083 3.71e-07 ***
age25-29       0.3894     0.1759   2.214  0.02681 *
age30-39       0.9086     0.1646   5.519 3.40e-08 ***
age40-49       1.1892     0.2144   5.546 2.92e-08 ***
educationlow  -0.3250     0.1240  -2.620  0.00879 **
wantsMoreyes  -0.8330     0.1175  -7.091 1.33e-12 ***

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 165.772  on 15  degrees of freedom
Residual deviance:  29.917  on 10  degrees of freedom
AIC: 113.43

Number of Fisher Scoring iterations: 4
```

It appears that all three variables are significant. At given levels of education and desire, more women tend to use contraceptive as they grow older; compared to women with age < 25, the odds of using contraceptive for women of age 25–29, 30–39 and 40–49 are increased by $\exp(0.3894) = 1.48$, $\exp(0.9086) = 2.48$ and $\exp(1.1892) = 3.28$ times. Similarly, among those of the same age and desire for more children, the odds of using contraceptive is higher for women of higher education by $\exp(0.3250) = 1.38$ times. Also, women who want more children have odds of using contraceptive 57% lower ($\exp(-0.8330) = 0.43$) than those who do not want. To assess whether the model in (13.3) provide a good fit to the data, we assess the p-value

$$\mathsf{P}(\chi^2_{10} \geq 29.917) = \texttt{1 - pchisq(29.917, 10)} = 0.0008$$

($n - p = 16 - (1 + 3 + 1 + 1) = 10$), which indicates that it does not. Thus we consider including *interactions* between the variables to explain the data better.

The following model includes all interaction terms:

$$\log\left(\frac{\pi_{jkl}}{1-\pi_{jkl}}\right) = \mu + \alpha_j + \beta_k + \gamma_l + \delta_{jk} + \epsilon_{jl} + \phi_{kl} + \psi_{jkl}. \tag{13.4}$$

Here, $\delta_{jk}, \epsilon_{jl}$ and ϕ_{kl} denote the two-way interaction terms and ψ_{jkl} the three-way interaction term. As before, for identifiability, we set the parameters with any of their indices equal to one, at zero, e.g. $\delta_{1k} = \delta_{j1} = 0$ for all j, k and $\psi_{1kl} = 0$ for all k, l.

```
fit2 <- glm(ymat ~ age * wantsMore * education,
+             family = binomial(link = 'logit'), data = cuse)
summary(fit2)

Deviance Residuals:
 [1]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0

Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                        -1.6094     0.3464  -4.646 3.38e-06 ***
age25-29                            0.7309     0.4152   1.760  0.07838 .
age30-39                            1.7466     0.3841   4.547 5.43e-06 ***
age40-49                            2.5585     0.4854   5.271 1.36e-07 ***
wantsMoreyes                        0.2041     0.3794   0.538  0.59062
educationlow                        0.6931     0.6856   1.011  0.31199
age25-29:wantsMoreyes              -0.3800     0.4705  -0.808  0.41928
age30-39:wantsMoreyes              -1.2833     0.4491  -2.858  0.00427 **
age40-49:wantsMoreyes              -1.1532     0.7138  -1.615  0.10620
age25-29:educationlow              -0.4565     0.8216  -0.556  0.57852
age30-39:educationlow              -0.7921     0.7232  -1.095  0.27338
age40-49:educationlow              -1.5997     0.7926  -2.018  0.04356 *
wantsMoreyes:educationlow          -1.4663     0.8243  -1.779  0.07526 .
age25-29:wantsMoreyes:educationlow  0.8288     0.9988   0.830  0.40666
age30-39:wantsMoreyes:educationlow  1.2854     0.8955   1.435  0.15119
age40-49:wantsMoreyes:educationlow  0.6093     1.1326   0.538  0.59063

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1.6577e+02  on 15  degrees of freedom
Residual deviance: 6.2395e-14  on  0  degrees of freedom
```

```
AIC: 103.51
```

Is the inclusion of all the interaction terms well-justified? Why are the deviance residuals exactly zero with the deviance at `6.2395e-14` $\simeq 0$? To see how much contribution is made by the inclusion of each model term, we perform the analysis of deviance.

```
anova(fit2, test = 'Chisq')
```

```
                        Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                                       15    165.772
age                      3   79.192        12     86.581 < 2.2e-16 ***
wantsMore                1   49.693        11     36.888 1.798e-12 ***
education                1    6.971        10     29.917 0.0082860 **
age:wantsMore            3   17.288         7     12.630 0.0006167 ***
age:education            3    6.831         4      5.798 0.0774741 .
wantsMore:education      1    3.357         3      2.441 0.0669289 .
age:wantsMore:education  3    2.441         0      0.000 0.4859584
```

In sequentially adding model terms, we iteratively perform hypothesis testing between two nested models. The line starting with `age:wantsMore` corresponds to the hypothesis test

$$\mathcal{H}_0 : \log\left(\frac{\pi_{jkl}}{1-\pi_{jkl}}\right) = \mu + \alpha_j + \beta_k + \gamma_l, \quad \text{against}$$
$$\mathcal{H}_1 : \log\left(\frac{\pi_{jkl}}{1-\pi_{jkl}}\right) = \underline{\quad \mu + \alpha_j + \beta_k + \gamma_l + \delta_{jk}, \quad}$$

and the test statistic $D_0^* - D_1^* = 17.288 \sim_{\text{approx}} \underline{\quad \chi^2_3. \quad}$ Since $\mathsf{P}(\chi^2_3 \geq D_0^* - D_1^*) = 0.0006 \ll 0.05$, we conclude that the interaction between age and desire makes a significant contribution to explaining the response variable. Under $\mathcal{H}_1$, the goodness-of-fit test with the test statistic $D_1^* = 12.63$ yields $\mathsf{P}(\chi^2_7 \geq D_1^*) = 0.081$, and thus we may conclude that this model provides a reasonably good fit to the data. All the subsequent addition of model terms does not bring in significant reduction in the deviance at the significance level 0.05. Based on the above analysis, our final model is fitted as

```
fit3 <- glm(ymat ~ age * wantsMore + education,
        family = binomial(link = 'logit'), data = cuse)
summary(fit3)
```

```
Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)           -1.39630    0.29782  -4.688 2.75e-06 ***
age25-29               0.65378    0.35700   1.831  0.06705 .
age30-39               1.65933    0.32207   5.152 2.58e-07 ***
age40-49               1.94120    0.35076   5.534 3.13e-08 ***
wantsMoreyes          -0.06622    0.33071  -0.200  0.84130
educationlow          -0.34065    0.12577  -2.709  0.00676 **
age25-29:wantsMoreyes -0.25918    0.40975  -0.633  0.52704
age30-39:wantsMoreyes -1.11266    0.37404  -2.975  0.00293 **
age40-49:wantsMoreyes -1.36167    0.48433  -2.811  0.00493 **

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 165.77  on 15  degrees of freedom
Residual deviance:  12.63  on  7  degrees of freedom
AIC: 102.14
```

The significance of the interaction between age and desire implies that there is an association between women's age and whether they want more children or not, and that it has a joint effect on their use of contraceptive. In other words, the amount of reduction in odds of using contraceptive, due to their wish to have more children, differs based on women's age: Women wanting more children have odds of using contraceptive reduced by 6%, 28%, 69% and 76% ($\exp(-0.06622) = 0.94, \exp(-0.06622 - 0.25918) = 0.72, \exp(-0.06622 - 1.11266) = 0.31, \exp(-0.06622 - 1.36167) = 0.24$)) in age groups < 25, 25–29, 30–39 and 40–49, respectively. This was not seen when we considered the additive effects only, under the model (13.3), where we observed increasing odds of using contraceptive with age and decreasing odds with the desire for more children, separately.

14 Count responses and contingency tables

Count data are commonly found in medical applications, e.g. when recording the number of people falling ill with some disease within one year. Other classical examples include radiation counts as measured in particles per second by a Geiger counter. In behavioural studies, counts of incidents are often recorded in a time interval of specified length.

14.1 Log-linear model

The standard model for count data is the *log-linear* model, which treats the response Y_i as a random variable following a Poisson distribution with the parameter μ_i, and adopts the canonical log link function, i.e.

$$Y_i \sim_{\text{i.i.d}} \text{Poisson}(\mu_i) \quad \text{where} \quad \log(\mu_i) = \eta_i = \mathbf{x}_i{}^\top \boldsymbol{\beta}_0, \quad \text{for} \quad 1 \leq i \leq n.$$

Writing $\mu(\mathbf{x}) = \mathsf{E}(Y|\mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\beta}_0)$, the parameters in the log-linear model can be interpreted as below. At the vector of covariates $\mathbf{x}_1$,

$$\log(\mu(\mathbf{x}_1)) = \mathbf{x}_1^\top \boldsymbol{\beta}_0$$

and at $\mathbf{x}_2$,

$$\log(\mu(\mathbf{x}_2)) = \mathbf{x}_2^\top \boldsymbol{\beta}_0.$$

Then, we obtain the rate ratio (RR) as

$$\text{RR} = \frac{\mathsf{E}(Y|\mathbf{x}_1)}{\mathsf{E}(Y|\mathbf{x}_2)} = \frac{\mu(\mathbf{x}_1)}{\mu(\mathbf{x}_2)} = \underline{\qquad \exp((\mathbf{x}_1 - \mathbf{x}_2)^\top \boldsymbol{\beta}_0). \qquad}$$

If $\mathbf{x}_1$ and $\mathbf{x}_2$ are identical except for their j-th covariates, where $x_{2j} = x_{1j} \pm 1$, it yields $\text{RR} = \underline{\qquad \exp(\mp\beta_j) \qquad}$, i.e. an additive effect on the j-th covariate leads to a multiplicative effect on RR.

14.2 Contingency tables

Count data frequently arise in the form of a *contingency table* that cross-classifies individuals according to two or more categorical variables, one of which plays the role of the response

variable. For example, suppose that subjects have been cross-classified according to two factors with I and J levels each, which gives rise to a *two-way* contingency table:

	Level 1	Level 2	...	Level J	
Level 1	Y_{11}	Y_{12}	...	Y_{1J}	$Y_{1\cdot}$
Level 2	Y_{11}	Y_{12}	...	Y_{2J}	$Y_{2\cdot}$
$\vdots$			$\ddots$		$\vdots$
Level I	Y_{I1}	Y_{I2}	...	Y_{IJ}	$Y_{I\cdot}$
	$Y_{\cdot 1}$	$Y_{\cdot 2}$	...	$Y_{\cdot J}$	$Y_{\cdot\cdot}$

where we adopt the 'dot' notations

$$Y_{\cdot j} = \sum_{i=1}^{I} Y_{ij},\ Y_{i\cdot} = \sum_{j=1}^{J} Y_{ij} \text{ and } Y_{\cdot\cdot} = \sum_{i=1}^{I}\sum_{j=1}^{J} Y_{ij}.$$

Unlike the data analysed by ANOVA models in Section 4, the contingency tables do not report repeated measurements but only counts, which are non-negative integers.

14.2.1 Sampling scenarios

Conceivably, a contingency table could have been generated by one of several *sampling scenarios*. Often, it may be impossible to tell which one is most appropriate just by looking at the table without further information on how the data was collected.
We explore three different sampling scenarios using an example of a researcher studying the association between the favourite music genre (pop, rock, hip hop, classical; 4 levels) and the mode of listening to the music (streaming, CD, vinyl; 3 levels). Different sample scenarios give rise to different models yet in Section 14.3, we demonstrate that ultimately the results of our analyses are independent of the sampling scheme.

14.2.2 Poisson sampling

The researcher may decide to stand in front of the Fry Building for an hour and interview anyone who is willing to talk to them, recording which music they listen to and how. There is no constraint on how many people preferring specific genres are included in the study (= the row totals) or the total number of interviewees. This gives rise to a Poisson sampling scenario, which is summarised as follows.

(i) Subjects are chosen at random.

(ii) The study continues until fixed time elapses.

(iii) Each subject is categorised according to the factors under investigation, and the count for each cell defined by the categorical variables is recorded.

Let Y_{ij} be a random variable representing the count in cell (i, j). Under Poisson sampling scenario, we can model $Y_{ij},\ 1 \leq j \leq I,\ 1 \leq j \leq J$, as

$$Y_{ij} \sim \text{Poisson}(\mu_{ij})$$

for some $\mu_{ij} = \mathsf{E}(Y_{ij})$, which will be the focus of estimation and inference. Since Y_{ij} are independent, the joint distribution of $\mathbf{Y} = (Y_{11}, \ldots, Y_{IJ})^\top$ at $\mathbf{y} = (y_{11}, \ldots, y_{IJ})^\top$ is just the product of $\mathsf{P}(Y_{ij} = y_{ij})$ for all i and j, i.e.

$$\Pr(\mathbf{Y} = \mathbf{y}) = \prod_{i=1}^{I} \prod_{j=1}^{J} \mathsf{P}(Y_{ij} = y_{ij}) = \prod_{i=1}^{I} \prod_{j=1}^{J} \frac{\mu_{ij}^{y_{ij}}}{y_{ij}!} e^{-\mu_{ij}}.$$

Here, the number of observations is $n = IJ$.

14.2.3 Single multinomial sampling

The researcher may set out with the intention of interviewing a fixed number of individuals, say N. The multinomial sampling scenario is summarised as:

(i) The total number of subjects is fixed at N, i.e. $\sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij} = N$.

(ii) Each of the N subjects is classified into one of the IJ cells according to their music preference and how they listen to music, such that the joint distribution of $\mathbf{Y}$ is multinomial.

Re-labelling the cells in the contingency table as

$$\mathbf{Y} = (Y_{11}, \ldots, Y_{IJ})^\top = (Y_1, \ldots, Y_n)^\top \text{ with } n = IJ,$$

we have

$$(Y_1, \ldots, Y_n) \sim \text{Multinomial}(N, \boldsymbol{\pi})$$

where $\boldsymbol{\pi} = (\pi_{11}, \ldots, \pi_{IJ})^\top = (\pi_1, \ldots, \pi_n)^\top$ is the vector of probabilities summing up to one. The joint distribution is written as below.

$$\mathsf{P}(\mathbf{Y} = \mathbf{y}) = \frac{N!}{\prod_{k=1}^{n} y_k!} \prod_{k=1}^{n} \pi_k^{y_k} = \underline{\qquad \frac{N!}{\prod_{i=1}^{I} \prod_{j=1}^{J} y_{ij}!} \prod_{i=1}^{I} \prod_{j=1}^{J} \pi_{ij}^{y_{ij}}, \qquad}$$

$$\text{where } \sum_{k=1}^{n} \pi_k = \sum_{i=1}^{I} \sum_{j=1}^{J} \pi_{ij} = \underline{\quad 1. \quad}$$

14.2.4 Product multinomial sampling

The researcher may set out to interview a fixed number of people for each music genre, which fixes the ___row totals___ $Y_{i\cdot} = \sum_{j=1}^{J} Y_{ij}$ in advance. Essentially, this treats the row categories as sub-populations, and the column categories as the response. The properties of the product multinomial sampling scenario are:

(i) Row totals are fixed at $Y_{i\cdot} = y_{i\cdot}$.

(ii) ___For each row,___ we perform an independent multinomial sampling so that the joint distribution of the elements of $\mathbf{Y}$ is the ___product___ of the I multinomial distributions of $\mathbf{Y}_i = (Y_{i1}, \ldots, Y_{iJ})^\top$, the J-dimensional vector of counts in the i-th row.

The joint distribution is written as

$$\mathsf{P}(\mathbf{Y} = \mathbf{y}) = \prod_{i=1}^{I} \mathsf{P}(\mathbf{Y}_i = \mathbf{y}_i) = \underline{\qquad \prod_{i=1}^{I} \left(\frac{(y_{i\cdot})!}{\prod_{j=1}^{J} y_{ij}!} \prod_{j=1}^{J} \pi_{ij}^{y_{ij}} \right), \qquad} \text{ where}$$

$$\underline{\qquad \sum_{j=1}^{J} \pi_{1j} = \ldots = \sum_{j=1}^{J} \pi_{Ij} = 1. \qquad}$$

14.3 Relationship between sampling scenarios

In what follows, we demonstrate that regardless of the sampling scenarios, we may adopt the Poisson log-linear modelling by identifying the model equivalent for that arising under the single multinomial sampling scheme. We continue our discussion with the two-way contingency table, but the same arguments are applicable to contingency tables of any dimensions.

Random component

Let $\mathbf{Y} = (Y_1, \ldots, Y_n)^\top$ denote the vector of re-labelled random variables representing the counts from $n = IJ$ cells. Under the Poisson sampling scheme, each Y_i is drawn independently from the Poisson distribution with parameter μ_i, such that the joint probability mass function is

$$f(y_1, \ldots, y_n; \mu_1, \ldots, \mu_n) = \prod_{i=1}^{n} \mu_i^{y_i} \frac{e^{-\mu_i}}{y_i!}.$$

It is also conceivable that $\mathbf{Y}$ is drawn from a multinomial distribution with parameters $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_n)^\top$ such that the joint probability mass function is given as

$$f(y_1, \ldots, y_n; \pi_1, \ldots, \pi_n) = \frac{N!}{\prod_{i=1}^{n} y_i!} \prod_{i=1}^{n} \pi_i^{y_i},$$

where $\sum_{i=1}^{n} y_i = N$ and $\sum_{i=1}^{n} \pi_i = 1$.
The difference of the two approaches is that the total count, say $Y = \sum_{i=1}^{n} Y_i$, is random under Poisson sampling scheme while it is fixed at $Y = N$ under the multinomial sampling scheme. The distribution of Y is Poisson with parameter $\mu = \sum_{i=1}^{n} \mu_i$.
Then, the probability of observing $(Y_1, \ldots, Y_n)^\top = (y_1, \ldots, y_n)^\top$ under the Poisson sampling scheme conditional on $Y = N$, is

$$\begin{aligned}
\mathsf{P}(Y_1 = y_1, \ldots, Y_n = y_n | Y = N) &= \frac{\mathsf{P}(Y_1 = y_1, \ldots, Y_n = y_n)}{\mathsf{P}(Y = N)}. \\
&= \left(\frac{N!}{\mu^N e^{-\mu}}\right) \prod_{i=1}^{n} \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!} = \frac{\frac{N!}{\prod_{i=1}^{n} y_i!} \frac{e^{-\sum_{i=1}^{n} \mu_i}}{e^{-\mu}} \frac{\mu_1^{y_1} \cdots \mu_n^{y_n}}{\mu^N}}{} \\
&= \frac{\frac{N!}{\prod_{i=1}^{n} y_i!} \prod_{i=1}^{n} \left(\frac{\mu_i}{\mu}\right)^{y_i}}{} = \frac{N!}{\prod_{i=1}^{n} y_i!} \prod_{i=1}^{n} \pi_i^{y_i},
\end{aligned}$$

with $\pi_i = \mu_i / \mu$, where $\pi_i \in (0, 1)$ and $\sum_{i=1}^{n} \pi_i = 1$.
In other words, conditioned on the total $Y = N$, the joint distribution of the Poisson random variables becomes that of a multinomial random variable,

$$(Y_1, \ldots, Y_n) | Y = N \sim \text{Multinomial}(N, \pi_1, \ldots, \pi_n).$$

Similarly, it can be shown that conditional on ___the row totals,___ the single multinomial sampling scheme is identical to the product multinomial sampling.

Systematic component

We illustrate that in Poisson log-linear models, the systematic component, i.e. the linear predictor, models the dependence between the row and the column factors by the inclusion or exclusion of the interaction between the two factors.

No interaction. Under the multinomial sampling scheme, the independence between the row and the column factors is equivalent to the independence between U_1 and U_2, which represent the *membership* of a randomly selected subject according to the row and the column factors, respectively, with $U_1 \in \{___1, \ldots, I___\}$ and $U_2 \in \{___1, \ldots, J___\}$. Then,

$$\mathsf{P}(U_1 = i, U_2 = j) = \mathsf{P}(U_1 = i) \cdot \mathsf{P}(U_2 = j) \quad \Longleftrightarrow \quad \pi_{ij} = \pi_{i\cdot} \cdot \pi_{\cdot j},$$

where $\pi_{i\cdot} = \sum_{j=1}^{J} \pi_{ij}$ and $\pi_{\cdot j} = \sum_{i=1}^{I} \pi_{ij}$. Equivalently, we write

$$\mu_{ij} = \frac{\mu_{i\cdot} \cdot \mu_{\cdot j}}{N}, \quad \text{i.e.} \quad \log(\mu_{ij}) = -\log(N) + \log(\mu_{i\cdot}) + \log(\mu_{\cdot j}) \tag{14.1}$$

since $\mu_{ij} = \mathsf{E}(Y_{ij}) = N\pi_{ij}$, where $\mu_{i\cdot} = \sum_{j=1}^{J} \mu_{ij}$ and $\mu_{\cdot j} = \sum_{i=1}^{I} \mu_{ij}$.
Under the Poisson sampling scheme, we may consider the following log-linear model

$$\log(\mu_{ij}) = \theta + \alpha_i + \beta_j, \tag{14.2}$$

which indicates that the interaction between the row and the column factors does not have any influence on the mean responses μ_{ij} i.e. they are independent of one another. To see this, let us write

$$\theta = \frac{1}{IJ} \sum_{i=1}^{I} \sum_{j=1}^{J} \log(\mu_{ij}), \;\; \alpha_i = \frac{1}{J} \sum_{j=1}^{J} \log(\mu_{ij}) - \theta \;\; \text{and} \;\; \beta_j = \frac{1}{I} \sum_{i=1}^{I} \log(\mu_{ij}) - \theta. \tag{14.3}$$

Then, under (14.1),

$$\theta = -\log(N) + \frac{1}{I} \sum_{i=1}^{I} \log(\mu_{i\cdot}) + \frac{1}{J} \sum_{j=1}^{J} \log(\mu_{\cdot j}),$$

$$\alpha_i = -\log(N) + \log(\mu_{i\cdot}) + \frac{1}{J}\sum_{j=1}^{J}\log(\mu_{\cdot j}) - \theta = \log(\mu_{i\cdot}) - \frac{1}{I}\sum_{i=1}^{I}\log(\mu_{i\cdot}),$$
$$\beta_j = -\log(N) + \frac{1}{I}\sum_{i=1}^{I}\log(\mu_{i\cdot}) + \log(\mu_{\cdot j}) - \theta = \log(\mu_{\cdot j}) - \frac{1}{J}\sum_{j=1}^{J}\log(\mu_{\cdot j}),$$

so that (14.2) holds. The multinomial model restricts the joint and marginal probabilities to add up to one, i.e. $\sum_{i=1}^{I}\sum_{j=1}^{J}\pi_{ij} = \sum_{i=1}^{I}\pi_{i\cdot} = \sum_{j=1}^{J}\pi_{\cdot j} = 1$. The log-linear model in (14.2) also imposes some constraints on the parameters corresponding to factor variables for model identifiability: While one approach is to set $\alpha_1 = \beta_1 = 0$, it is also valid to impose that $\sum_{i=1}^{I}\alpha_i = \sum_{j=1}^{J}\beta_j = 0$, which is the case for the parameterisation in (14.3).

Interaction. Allowing for the dependence between the row and the column factors under the multinomial sampling scheme, we have

$$\pi_{ij} \neq \pi_{i\cdot}\pi_{\cdot j}, \quad \text{equivalently} \quad \mu_{ij} \neq \frac{\mu_{i\cdot}\cdot\mu_{\cdot j}}{N}.$$

This is modelled under the Poisson sampling scheme, by considering the log-linear model containing the interaction between row and column factors, i.e.

$$\log(\mu_{ij}) = \theta + \alpha_i + \beta_j + (\alpha\beta)_{ij}.$$

We may parameterise μ, α_i and β_j as in (14.3) and additionally, set

$$(\alpha\beta)_{ij} = \log(\mu_{ij}) - \alpha_i - \beta_j - \theta,$$

such that

$$\sum_{i=1}^{I}\alpha_i = \sum_{j=1}^{J}\beta_j = 0 \quad \text{and} \quad \sum_{i=1}^{I}(\alpha\beta)_{ij} = \sum_{j=1}^{J}(\alpha\beta)_{ij} = 0.$$

From the above, we conclude that modelling under Poisson and multinomial sampling schemes results in the same likelihood and thus the same estimation and inference outcomes. The morale of this story is that we can always resort to the equivalent Poisson log-linear model even when different sampling schemes have been used to produce the contingency table. While the gain is trivial in the case of a two-way contingency table, it can be

significant as we move to higher order contingency tables, since the log-linear model makes it easier to model complex dependence structure between the factor variables by means of interaction terms.

14.3.1 Example: Serum cholesterol and heart disease

The table below was taken from the Framingham longitudinal study of coronary heart disease. It shows 1329 patients cross-classified according to the level of their serum cholesterol (below or above 260) and the presence or absence of heart disease.

Serum cholesterol	Heart disease		Total
	Present	Absent	
< 260	51	992	1043
> 260	41	245	286
Total	92	1237	1329

Single multinomial sampling scenario. Let us assume that the data were collected by sampling $N = 1329$ (fixed) patients who were then classified according to their cholesterol level and heart disease status. The likelihood or joint distribution of the counts is given as

$$L(\boldsymbol{\pi}|\mathbf{y}) = \mathsf{P}(\mathbf{Y} = \mathbf{y}; \boldsymbol{\pi}) = \frac{N!}{y_{11}!y_{12}!y_{21}!y_{22}!}\pi_{11}^{y_{11}}\pi_{12}^{y_{12}}\pi_{21}^{y_{21}}\pi_{22}^{y_{22}}$$

subject to $\sum_{i=1}^{2}\sum_{j=1}^{2} y_{ij} = N$ and $\sum_{i=1}^{2}\sum_{j=1}^{2} \pi_{ij} = 1$. The corresponding log-likelihood is given by

$$\ell(\boldsymbol{\pi}|\mathbf{y}) = \sum_{i=1}^{I}\sum_{j=1}^{J} y_{ij}\log(\pi_{ij}) + \text{constant}.$$

The main hypothesis of interest is whether the heart disease and the serum cholesterol level are *independent*. For this, we perform the hypothesis testing where

$$\mathcal{H}_0 : \underline{\quad \pi_{ij} = \pi_{i\cdot} \cdot \pi_{\cdot j} \quad} \text{ or, equivalently } \underline{\quad \mu_{ij} = \frac{\mu_{i\cdot}\mu_{\cdot j}}{N}, \quad}$$

against the alternative model $\mathcal{H}_1$: the row and the column factors are dependent. We use the deviance to test the validity of $\mathcal{H}_0$ under the single multinomial sampling scheme.

To obtain the MLE of $\boldsymbol{\pi}$ **under** $\mathcal{H}_1$**,** say $\widetilde{\boldsymbol{\pi}}$, we derive the likelihood equations $\frac{\partial \ell}{\partial \pi_{ij}}$ subject to the constraint that the probabilities add up to one, i.e. $\sum_{i=1}^{I}\sum_{j=1}^{J}\pi_{ij}=1$. Here, we are able to obtain the MLE analytically: Noting that $\pi_{IJ}=1-\sum_{(i,j)\neq(I,J)}\pi_{ij}$,

$$\frac{\partial \ell(\widetilde{\boldsymbol{\pi}}|\mathbf{y})}{\partial \pi_{ij}} = \underline{\qquad \frac{y_{ij}}{\widetilde{\pi}_{ij}}-\frac{y_{IJ}}{\widetilde{\pi}_{IJ}}=0, \qquad}$$

$$\therefore \widetilde{\pi}_{ij}=\frac{y_{ij}\widetilde{\pi}_{IJ}}{y_{IJ}} \quad \text{subject to} \quad \sum_{i=1}^{I}\sum_{j=1}^{J}\widetilde{\pi}_{ij}=1,$$

which leads to $\widetilde{\pi}_{ij} = \underline{\quad y_{ij}/N \quad}$ and $\widetilde{\mu}_{ij} = \underline{\quad y_{ij\cdot} \quad}$ This is as expected since the model has $\underline{\quad IJ-1 \quad}$ parameters to estimate from $\underline{\quad IJ-1 \quad}$ free observations.

Under $\mathcal{H}_0$**,** we similarly derive the MLE of $\pi_{ij}=\pi_{i\cdot}\cdot\pi_{\cdot j}$, say $\widehat{\pi}=\widehat{\pi}_{i\cdot}\cdot\widehat{\pi}_{\cdot j}$: From the log-likelihood

$$\ell(\boldsymbol{\pi}|\mathbf{y})=\sum_{i=1}^{I}\sum_{j=1}^{J}\underline{\quad y_{ij}(\log(\pi_{i\cdot})+\log(\pi_{\cdot j})) \quad}=\sum_{i=1}^{I}y_{i\cdot}\log(\pi_{i\cdot})+\sum_{j=1}^{J}y_{\cdot j}\log(\pi_{\cdot j}),$$

we derive

$$\frac{\partial \ell(\widehat{\boldsymbol{\pi}}|\mathbf{y})}{\partial \pi_{i\cdot}}=\frac{y_{i\cdot}}{\widehat{\pi}_{i\cdot}}-\frac{y_{I\cdot}}{\widehat{\pi}_{I\cdot}}=0, \quad \therefore \widehat{\pi}_{i\cdot}=\frac{y_{i\cdot}\widehat{\pi}_{I\cdot}}{y_{I\cdot}} \text{ subject to } \sum_{i=1}^{I}\widehat{\pi}_{i\cdot}=1,$$

$$\frac{\partial \ell(\widehat{\boldsymbol{\pi}}|\mathbf{y})}{\partial \pi_{\cdot j}}=\frac{y_{\cdot j}}{\widehat{\pi}_{\cdot j}}-\frac{y_{\cdot J}}{\widehat{\pi}_{\cdot J}}=0, \quad \therefore \widehat{\pi}_{\cdot j}=\frac{y_{\cdot j}\widehat{\pi}_{\cdot J}}{y_{\cdot J}} \text{ subject to } \sum_{j=1}^{J}\widehat{\pi}_{\cdot j}=1,$$

from which we yield $\widehat{\pi}_{i\cdot}=y_{i\cdot}/N$, $\widehat{\pi}_{\cdot j}=y_{\cdot j}/N$ and $\widehat{\pi}_{ij}=N^{-2}y_{i\cdot}y_{\cdot j}$. Therefore the fitted values for μ_{ij} under $\mathcal{H}_0$ are

$$\widehat{\mu}_{ij} = \underline{\quad \frac{y_{i\cdot}y_{\cdot j}}{N}. \quad}$$

The fitted values $\widehat{\mu}_{ij}$ are reported in Table 14.1.

Consequently, the scaled deviance is

$$D^*(\mathbf{y};\widehat{\boldsymbol{\mu}})=2(\ell(\mathbf{y}|\mathbf{y})-\ell(\widehat{\boldsymbol{\mu}}|\mathbf{y}))=\underline{\qquad 2\sum_{i=1}^{I}\sum_{j=1}^{J}y_{ij}\log\left(\frac{y_{ij}}{\widehat{\mu}_{ij}}\right) \qquad}=26.43.$$

Table 14.1: Fitted values under $\mathcal{H}_0$.

Serum	Heart disease		
cholesterol	Present	Absent	Total
< 260	72.20	970.80	1043
> 260	19.80	266.20	286
Total	92	1237	1329

Under $\mathcal{H}_0$, we have $D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}}) \sim \chi^2_{n-p}$ with

$$n - p = \underline{\quad IJ - 1 - (I-1) - (J-1) = (I-1)(J-1) = 1 \quad}$$

under the constraint $\sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij} = N$. Clearly, we reject the hypothesis of independence with p-value $\mathsf{P}(\chi^2_1 > 26.43) = 2.73 \times 10^{-7}$, concluding that heart disease and serum cholesterol level are associated.

Poisson sampling scenario. An alternative approach to modelling the contingency table is to treat the four counts as realisations of independent Poisson random variables: There are four groups of people, one for each cell in the table, and the members from each group arrive randomly over a period of time e.g. for a health check, and their status is recorded. The joint distribution of the counts is given as

$$\mathsf{P}(\mathbf{Y} = \mathbf{y}; \boldsymbol{\mu}) = \prod_{i=1}^{I} \prod_{j=1}^{J} e^{-\mu_{ij}} \frac{\mu_{ij}^{y_{ij}}}{y_{ij}!}.$$

For a two-way contingency table, the log-linear model without the interaction term as in (14.2), corresponds to the assumption of independence between the row and column factors. On the contrary, the log-linear model with the most number of parameters we can fit to the two-way contingency table, is

$$\log(\mu_{ij}) = \theta + \alpha_i + \beta_j + (\alpha\beta)_{ij} \tag{14.4}$$

which includes the interaction between the row and the column factors. The presence of an interaction term implies the existence of association (dependence) between the row and the column factors, which is a saturated model that leads to $\widehat{\mu}_{ij} = y_{ij}$. We can verify this by showing that the number of parameters involved in the model equals the number of

observations (total number of cells), as

$$1 + (I-1) + (J-1) + (I-1)(J-1) = IJ.$$

Therefore, the (scaled) deviance for this full model is simply zero.
Let us verify this. We first fit the log-linear model with two factors *without* the interaction term. In the `R` code below, `rf` and `cf` contain the row and column factors, respectively.

```
y <- c(51, 992, 41, 245)
rf <- gl(2, 2, 4)
cf <- gl(2, 1, 4)

loglinear.fit <- glm(y ~ rf + cf, family = poisson(link = 'log'))
fitted(loglinear.fit)

        1         2         3         4
 72.20166 970.79834  19.79834 266.20166

sum(fitted(loglinear.fit))

[1] 1329

summary(loglinear.fit)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.27946    0.10524   40.66   <2e-16 ***
rf2         -1.29386    0.06675  -19.38   <2e-16 ***
cf2          2.59866    0.10806   24.05   <2e-16 ***
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1658.18  on 3  degrees of freedom
Residual deviance:   26.43  on 1  degrees of freedom
AIC: 59.836
```

```
Number of Fisher Scoring iterations: 4
```

Note that `loglinear.fit` returns identical fitted values as those obtained under the independence between the row and column factors in Table 14.1, and the fitted values sum up to the total number of patients $N = 1329$, i.e. $\sum_{i=1}^{I}\sum_{j=1}^{J}\widehat{\mu}_{ij} = N$. Also, the residual deviance, $D^* = 26.43$, is identical to that obtained under the multinomial sampling. Recalling that $\phi = 1$ for Poisson distributions, we perform the analysis of deviance as below.

```
anova(loglinear.fit, test = 'Chisq')

     Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                     3    1658.18
rf    1   458.19         2    1199.98 < 2.2e-16 ***
cf    1  1173.55         1      26.43 < 2.2e-16 ***
```

It appears that both row and column factors are significant, but a good deal of the data is not explained well by the model as $D^* = 26.43$ lies in the extreme end of the χ^2_{n-p}-distribution with $n =$ $IJ = 4$ and $p =$ $1 + (I-1) + (J-1) = 3$. Therefore, as before, the null hypothesis

$$\mathcal{H}_0 : \quad \log(\mu_{ij}) = \theta + \alpha_i + \beta_j,$$

is rejected and we choose the full (saturated) model

$$\mathcal{H}_1 : \quad \log(\mu_{ij}) = \theta + \alpha_i + \beta_j + (\alpha\beta)_{ij},$$

i.e. there is a statistically significant association between the level of serum cholesterol and the presence or absence of heart disease.

14.4 Models for higher order contingency tables

For higher order contingency tables, interactions between the factors can be fitted without yielding a saturated model. For example, for a three-way table, the saturated log-linear

model is of the form

$$\log(\mu_{ijk}) = \theta + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk},$$

with all the usual constraints, e.g. for the three-way interactions, we have either $(\alpha\beta\gamma)_{ijk} = 0$ if $\{i = 1\} \cup \{j = 1\} \cup \{k = 1\}$ holds. This is a saturated model, since the number of parameters equals the number of observations $n = IJK$. Fitting a reduced model, e.g.

$$\log(\mu_{ijk}) = \theta + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} \tag{14.5}$$

and testing the validity of such a model for the data allows us to make a statement about the dependence or independence among the row, column and layer factors. Consider the multinomial sampling scenario where $N = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} Y_{ijk}$ is fixed. Let (U_1, U_2, U_3) denote the vector of random variables representing the membership of a randomly chosen subject to the three categorical variables. Then, under the model (14.5), we have

$$\begin{aligned}\mathsf{P}(U_1 = i, U_2 = j, U_3 = k) = \pi_{ijk} &= \frac{\mu_{ijk}}{N} \\ &= \frac{1}{N}\exp(\theta + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik}).\end{aligned}$$

The conditional probability of U_2 and U_3 given U_1 is

$$\begin{aligned}\mathsf{P}(U_2 = j, U_3 = k | U_1 = i) &= \frac{\mathsf{P}(U_1 = i, U_2 = j, U_3 = k)}{\sum_{j'=1}^{J}\sum_{k'=1}^{K}\mathsf{P}(U_1 = i, U_2 = j', U_3 = k')} \\ &= \frac{\exp(\theta + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \gamma_k + (\alpha\gamma)_{ik})}{\sum_{j'=1}^{J}\sum_{k'=1}^{K}\exp(\theta + \alpha_i + \beta_{j'} + (\alpha\beta)_{ij'} + \gamma_{k'} + (\alpha\gamma)_{ik'})} \\ &= \frac{\exp(\beta_j + (\alpha\beta)_{ij})}{\sum_{j'=1}^{J}\exp(\beta_{j'} + (\alpha\beta)_{ij'})} \cdot \frac{\exp(\gamma_k + (\alpha\gamma)_{ik})}{\sum_{k'=1}^{K}\exp(\gamma_{k'} + (\alpha\gamma)_{ik'})} \\ &= \mathsf{P}(U_2 = j | U_1 = i) \cdot \mathsf{P}(U_3 = k | U_1 = i).\end{aligned}$$

That is, under the model (14.5), U_2 and U_3 are *conditionally independent* given U_1. This follows from that $\pi_{ijk} = \mathsf{P}(U_1 = i, U_2 = j, U_3 = k)$ is factorised as

$$\mathsf{P}(U_1 = i, U_2 = j, U_3 = k) = \frac{1}{N}\exp(\theta + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \gamma_k + (\alpha\gamma)_{ik})$$

$$= \frac{1}{N} \exp(\theta) \cdot \exp(\beta_j + (\alpha\beta)_{ij}) \cdot \exp(\gamma_k + (\alpha\gamma)_{ik}) \cdot \exp(\alpha_i).$$

Writing the model syntax as in `R`, if you choose the models

(i) `count ~ a + b + c`, then `a`, `b` and `c` are mutually independent;

(ii) `count ~ a + b*c`, then `a` is independent of `b` and `c`;

(iii) `count ~ a*b + b*c`, then `a` is dependent on `b`, and `c` is dependent on `b`, but there is no direct dependence between `a` and `c` given `b`.

Note that in `R`, the model syntax is regarded to be *hierarchical*, meaning that if a higher order interaction term is included in the model, then all the related lower-order terms are also included. For example, including the interaction `b*c` in the model, the terms `b` and `c` are all entered into the model automatically, along with their interaction term `b:c`. The example in (iii) can be alternatively expressed as '`a` and `c` are conditionally independent given `b`'. This implies that if we wish to predict `a`, we need only to know `b` and `c` is irrelevant for the purpose. However, if we consider the two-way contingency table between `a` and `c`, we may get a statistically significant association due to the dependence of `a` and `c` on `b`.

14.5 `R` examples

14.5.1 Educational aspirations in Wisconsin

The table below classifies $N = 4991$ Wisconsin male high school seniors according to three factors.

(i) College attendance; whether they plan to attend college: No and yes.

(ii) Socio-economic status: Low, lower middle, upper middle and high.

(iii) The degree of parental encouragement they receive: Low and high.

Social stratum	Parental encouragement	College plans		Total
		No	Yes	
Low	Low	749	35	784
	High	233	133	366
Lower Middle	Low	627	38	665
	High	330	303	633
Upper Middle	Low	420	37	457
	High	374	467	841
High	Low	153	26	179
	High	266	800	1066
Total		3152	1938	4991

We denote the number of levels for each of the three factors by $I = 2$ (college attendance, 1: no, 2: yes), $J = 4$ (socio-economic status, 1: low, 2: lower middle, 3: upper middle, 4: high) and $K = 2$ (parental encouragement, 1: low, 2:high), respectively. There are in total $n = IJK = 16$ cells. The saturated log-linear model is given by

$$\log(\mu_{ijk}) = \theta + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk}.$$

We study the dependence structure between the factors by removing some of the interaction terms in the above model.
Let's read the data into `R` and assign the factor vectors `c` (college plan), `s` (social stratum) and `p` (parental encouragement).

```
y <- c(749, 35, 233, 133, 627, 38, 330, 303, 420, 37, 374,
467, 153, 26, 266, 800)
c <- gl(2, 1, 16) # college attendance
s <- gl(4, 4, 16) # socio-economic status
p <- gl(2, 2, 16) # parental encouragement
```

We perform the analysis of deviance by sequentially adding terms to the null model until it is saturated.

```
fit0 <- glm(y ~ c*s*p, family = poisson(link = 'log'))
anova(fit0, test = 'Chisq')
```

```
      Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                    15     3211.0
c      1   349.51        14     2861.5 < 2.2e-16 ***
s      3    11.87        11     2849.6  0.007856 **
p      1   135.67        10     2714.0 < 2.2e-16 ***
c:s    3   793.55         7     1920.4 < 2.2e-16 ***
c:p    1  1621.91         6      298.5 < 2.2e-16 ***
s:p    3   296.91         3      1.576 < 2.2e-16 ***
c:s:p  3     1.58         0        0.0  0.664965
```

Repeatedly fitting the models with different combinations of the interactions, we obtain the following table summarising the goodness-of-fit test results.

	Model	df	D^*	$\mathsf{P}(\chi^2_{n-p} > D^*)$
1	c+s+p	10	2714.0	0
2	c∗s+p	7	1920.4	0
3	c∗p+s	9	1092.0	0
4	s∗p+c	7	1877.4	0
5	c∗s+c∗p	6	298.5	0
6	s∗c+s∗p	4	1083.8	0
7	p∗s+p∗c	6	255.5	0
8	c∗s+c∗p+s∗p	3	1.576	0.665
9	c∗s∗p	0	0	–

Based on the analysis of deviance table, we choose Model 8, which involves all three two-factor interactions ($D^* = 1.576$, df $= 1$). Thus, planning for a college is associated with parental encouragement as well as the socio-economic status and, in addition, parental encouragement and socio-economic status are dependent on each other. The lack of a three-factor interaction implies that the association between any two factors is the same at all levels of the third. For example, the interaction between the social class and college plan remains the same, whether the parental encouragement is low or high.

Multinomial sampling scheme. Let us denote the membership of a randomly selected subject according to $\mathtt{c} \times \mathtt{s} \times \mathtt{p}$, by (U_1, U_2, U_3). Then,

$$\mathsf{P}(U_1 = i, U_2 = j, U_3 = k) = \pi_{ijk} = \mu_{ijk}/N.$$

Performing the goodness-of-fit test of Model 1 is equivalent to testing the null hypothesis of complete independence:

$$\mathcal{H}_0 : \pi_{ijk} = \pi_{i\cdot\cdot} \cdot \pi_{\cdot j\cdot} \cdot \pi_{\cdot\cdot k},$$

against the saturated one. The log-likelihood is given by

$$\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} Y_{ijk} \log(\pi_{ijk}) = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} Y_{ijk}(\log(\pi_{i\cdot\cdot}) + \log(\pi_{\cdot j\cdot}) + \log(\pi_{\cdot\cdot k})),$$

ignoring the terms not involving π_{ijk}. Solving the likelihood equations under the constraints $\sum_{i=1}^{I} \pi_{i\cdot\cdot} = 1$, $\sum_{j=1}^{J} \pi_{\cdot j\cdot} = 1$ and $\sum_{k=1}^{k} \pi_{\cdot\cdot k} = 1$, we obtain the MLE

$$\widehat{\pi}_{i\cdot\cdot} = \frac{Y_{i\cdot\cdot}}{N}, \quad \widehat{\pi}_{\cdot j\cdot} = \frac{Y_{\cdot j\cdot}}{N} \text{ and } \widehat{\pi}_{\cdot\cdot k} = \frac{Y_{\cdot\cdot k}}{N}.$$

Hence, the deviance is

$$D^*(\mathbf{Y}; \widehat{\boldsymbol{\mu}}) = 2\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} Y_{ijk} \log\left(\frac{Y_{ijk}}{\widehat{\mu}_{ijk}}\right) \quad \text{with} \quad \widehat{\mu}_{ijk} = \frac{Y_{i\cdot\cdot}Y_{\cdot j\cdot}Y_{\cdot\cdot k}}{N^2},$$

which is calculated as $D^* = 2714.0$ and we reject $\mathcal{H}_0$ with p-value $\simeq 0$.
Next, consider Model 5 which includes the interactions between U_1 and U_2, and U_1 and U_3, but no interaction between U_2 and U_3. Then, we can show that U_2 and U_3 are conditionally independent given U_1, i.e.

$$\begin{aligned}
&\mathsf{P}(U_2 = j, U_3 = k | U_1 = i) = \mathsf{P}(U_2 = j | U_1 = i)\mathsf{P}(U_3 = k | U_1 = i), \qquad \text{hence} \\
&\mathsf{P}(U_1 = i, U_2 = j, U_3 = k) = \mathsf{P}(U_2 = j | U_1 = i)\mathsf{P}(U_3 = k | U_1 = i)\mathsf{P}(U_1 = i).
\end{aligned}$$

The null hypothesis associated with the goodness-of-fit test of Model 5 is

$$\mathcal{H}_0 : \pi_{ijk} = \frac{\pi_{ij\cdot} \cdot \pi_{i\cdot k}}{\pi_{i\cdot\cdot}}$$

and, as before, by identifying the MLE of π_{ijk}, we are able to calculate the deviance which coincides with $D^* = 298.5$ reported in the table above.

Model interpretation. The fitted model containing all the two-way interactions is:

```
fit <- glm(y ~ c*s + c*p + s*p, family = poisson(link = "log"))
summary(fit)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  6.62426    0.03602 183.918  < 2e-16 ***
c2          -3.19497    0.11850 -26.962  < 2e-16 ***
s2          -0.18496    0.05304  -3.487 0.000489 ***
s3          -0.58183    0.05931  -9.810  < 2e-16 ***
s4          -1.62046    0.08450 -19.178  < 2e-16 ***
p2          -1.19117    0.07166 -16.622  < 2e-16 ***
c2:s2        0.42013    0.11768   3.570 0.000357 ***
c2:s3        0.73851    0.11382   6.488 8.69e-11 ***
c2:s4        1.59311    0.11527  13.820  < 2e-16 ***
c2:p2        2.68292    0.09867  27.191  < 2e-16 ***
s2:p2        0.55410    0.09469   5.852 4.87e-09 ***
s3:p2        1.07056    0.09649  11.095  < 2e-16 ***
s4:p2        1.78588    0.11444  15.606  < 2e-16 ***
```

All the interaction terms are highly significant. Also, it is clear that higher socio-economic status and stronger parental encouragement positively influence the college attendance. Besides, there is a strong association between the socio-economic status and parental encouragement, as evidenced by positive coefficients estimated for the interactions.
We interpret the fitted model using the rate ratio. For example, the ratio between the mean counts corresponding to the cells $(U_1, U_2, U_3) = (2, 3, 2)$ and $(U_1, U_2, U_3) = (2, 3, 1)$ is

$$\frac{\mu_{232}}{\mu_{231}} = \frac{\exp(\theta + \alpha_2 + \beta_3 + \gamma_2 + (\alpha\beta)_{23} + (\alpha\gamma)_{22} + (\beta\gamma)_{32})}{\exp(\theta + \alpha_2 + \beta_3 + \gamma_1 + (\alpha\beta)_{23} + (\alpha\gamma)_{21} + (\beta\gamma)_{31})}$$

$$= \exp(\gamma_2 - \gamma_1 + (\alpha\gamma)_{22} - (\alpha\gamma)_{21} + (\beta\gamma)_{32} - (\beta\gamma)_{31})$$
$$= \exp(\gamma_2 + (\alpha\gamma)_{22} + (\beta\gamma)_{32}),$$

and replacing the parameters with their estimates, the rate ratio between the two cells is $\exp(-1.19117 + 2.68292 + 1.07056) = 12.966$. This indicates that for students belonging to the upper middle social stratum, the college attendance is almost 13 times higher for the students with high parental encouragement.

Product multinomial sampling scheme. An alternative scenario that could have given rise to the identical contingency table as follows: The researchers interviewed the students from different combinations of socio-economic status and parental encouragement, and classified them according to their college aspiration. This corresponds to the product multinomial sampling scheme where the totals of (socio-economic status $\times$ parental encouragement) are fixed, and the joint probability mass function is simply the product of those of 8 binomial distributions. We explore the model fitting under this scenario in problem sheet Q2.

14.5.2 Contraceptive use data

We re-visit the data given in Table 13.1: $N = 1607$ women in Fiji in 1975 were cross-classified according to their use of contraceptive, age, level of education and desire to have more children. In Section 13.5.2, we modelled the numbers of women using contraceptive per group, as independent random variables following binomial distributions with parameters varying according to the factor variables. In doing so, we regarded the data as being collected under the product multinomial (binomial) sampling scheme, with (age $\times$ (desire more children) $\times$ education) margins as fixed.

Viewing Table 13.1 as a contingency table and regarding the use of contraceptive as an extra factor, we may fit a log-linear model and yield the same inferential results. For this, let Y_{ijkl} denote the count of the cell (i, j, k, l) cell in the contingency table in Table 13.1, where i corresponds to the index of the level for the use of contraceptive (1: `not using`, 2: `using`), j the age group ($1 \leq j \leq 4$), k the desire for more children ($1 \leq k \leq 2$) and l the level of education ($1 \leq l \leq 2$). In (13.3), we modelled

$$\pi_{jkl} = \mathsf{P}(\text{A women belonging to the cell } (j, k, l) \text{ uses contraceptive})$$

using a logistic regression model including the additive terms only.

$$\log\left(\frac{\pi_{jkl}}{1-\pi_{jkl}}\right) = \widetilde{\mu} + \widetilde{\alpha}_j + \widetilde{\beta}_k + \widetilde{\gamma}_l. \tag{14.6}$$

In doing so, it is implicitly assumed that the probability of using contraceptive differs from one cell to another, i.e. according to the *joint* effect of the three factors. This justifies the inclusion of their three-way interaction term and all its lower-order terms, in the corresponding log-linear model for $\mu_{ijkl} = \mathsf{E}(Y_{ijkl})$, as

$$\begin{aligned} &Y_{ijkl} \sim \text{Poisson}(\mu_{ijkl}), \quad \text{where} \\ &\log(\mu_{ijkl}) = \mu + \alpha_j + \beta_k + \gamma_l + (\alpha\beta)_{jk} + (\alpha\gamma)_{jl} + (\beta\gamma)_{kl} + (\alpha\beta\gamma)_{jkl} + \\ &\qquad \underline{\delta_i + (\delta\alpha)_{ij} + (\delta\beta)_{ik} + (\delta\gamma)_{il},} \end{aligned} \tag{14.7}$$

with the usual assumption for model identifiability. To verify the equivalence between the model (14.6) and (14.7), let U, X, Y, Z denote the random variable indicating the membership of a randomly chosen subject in the study with respect to the four factor variables contraceptive use, age, desire and education, respectively. Then, in (14.6),

$$\pi_{jkl} = \pi_{2|jkl} = \underline{\mathsf{P}(U = 2|X = j, Y = k, Z = l)}$$

which, under (14.7),

$$\begin{aligned} &\mathsf{P}(U = i, X = j, Y = k, Z = l) = \pi_{ijkl} = \frac{\mu_{ijkl}}{N}, \quad \text{hence} \\ &\mathsf{P}(U = 2|X = j, Y = k, Z = l) = \pi_{2|jkl} = \underline{\qquad \frac{\mu_{2jkl}}{\sum_{i=1}^{2} \mu_{ijkl}} \qquad} \\ &\qquad\qquad = \frac{\exp(\delta_2 + (\delta\alpha)_{2j} + (\delta\beta)_{2k} + (\delta\gamma)_{2l})}{1 + \exp(\delta_2 + (\delta\alpha)_{2j} + (\delta\beta)_{2k} + (\delta\gamma)_{2l})}. \end{aligned}$$

Here, we adopt the identifiability condition imposed by `R`, that $\delta_1 = (\delta\alpha)_{1j} = (\delta\beta)_{1k} = (\delta\gamma)_{1l} = 0$. Therefore, the logistic regression model in (14.6) is given by

$$\log\left(\frac{\pi_{2|jkl}}{1-\pi_{2|jkl}}\right) = \delta_2 + (\delta\alpha)_{2j} + (\delta\beta)_{2k} + (\delta\gamma)_{2l} \equiv \widetilde{\mu} + \widetilde{\alpha}_j + \widetilde{\beta}_k + \widetilde{\gamma}_l. \tag{14.8}$$

In other words, the effect of individual main effect in (14.6) correspond to the interaction between age/desire/education and the contraceptive use under the log-linear model (14.7).

We verify this by comparing the following R output to `summary(fit1)` in page 143:

```
cuse <- read.table('data/cuse.dat', header = TRUE)
count <- c(cuse$notUsing, cuse$using)
using <- gl(2, 16)
wants <- rep(cuse$wantsMore, 2) # gl(2, 1, 32)
edu <- rep(cuse$education, 2) # gl(2, 2, 32)
age <- rep(cuse$age, 2) # gl(4, 4, 32)

ll_fit1 <- glm(count ~ using * age + using * edu +
        using * wants + age * edu * wants,
        family = poisson(link = 'log'))
summary(ll_fit1)

Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)              2.91757    0.16960  17.203  < 2e-16 ***
using2                  -0.80822    0.15900  -5.083 3.71e-07 ***
...
using2:age25-29          0.38938    0.17585   2.214  0.02681 *
using2:age30-39          0.90861    0.16462   5.519 3.40e-08 ***
using2:age40-49          1.18924    0.21443   5.546 2.92e-08 ***
using2:edulow           -0.32499    0.12404  -2.620  0.00879 **
using2:wantsyes         -0.83295    0.11747  -7.091 1.33e-12 ***
...

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1192.403  on 31  degrees of freedom
Residual deviance:   29.917  on 10  degrees of freedom
```

Similarly, we can show that the log-linear including all the terms up to the four-way interaction term, $(\delta\alpha\beta\gamma)_{ijkl}$ corresponds to the saturated model of (13.4).
The final model we chose previously, fitted by `fit3` in page 146, includes the two-way interaction between age and desire, and it corresponds to:

$$\log(\mu_{ijkl}) = \mu + \alpha_j + \beta_k + \gamma_l + (\alpha\beta)_{jk} + (\alpha\gamma)_{jl} + (\beta\gamma)_{kl} + (\alpha\beta\gamma)_{jkl} + \\ \delta_i + (\delta\alpha)_{ij} + (\delta\beta)_{ik} + (\delta\gamma)_{il} + (\delta\alpha\beta)_{ijk}.$$

```
ll_fit2 <- glm(count ~ using * age * wants + using * edu +
         age * wants * edu, family = poisson(link = 'log'))
summary(ll_fit2)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)                3.87319    0.14198  27.280  < 2e-16 ***
using2                    -1.39630    0.29782  -4.688 2.75e-06 ***
...
using2:age25-29            0.65378    0.35700   1.831  0.06705 .
using2:age30-39            1.65933    0.32207   5.152 2.58e-07 ***
using2:age40-49            1.94120    0.35076   5.534 3.13e-08 ***
using2:wantsyes           -0.06622    0.33071  -0.200  0.84130
...
using2:edulow             -0.34065    0.12577  -2.709  0.00676 **
...
using2:age25-29:wantsyes -0.25918    0.40975  -0.633  0.52704
using2:age30-39:wantsyes -1.11266    0.37404  -2.975  0.00293 **
using2:age40-49:wantsyes -1.36167    0.48432  -2.811  0.00493 **
...

    Null deviance: 1192.40  on 31  degrees of freedom
Residual deviance:   12.63  on  7  degrees of freedom
AIC: 232.43
```

References

Box, G. E. and Cox, D. R. (1964). An analysis of transformations. *Journal of the Royal Statistical Society: Series B*, pages 211–252.

Breiman, L. (1992). The little bootstrap and other methods for dimensionality selection in regression: X-fixed prediction error. *Journal of the American Statistical Association*, 87:738–754.

Friedman, J., Hastie, T., and Tibshirani, R. (2001). *The Elements of Statistical Learning*. Springer, Berlin.

Hoerl, A. E. and Kennard, R. W. (1970). Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12:55–67.

McCullagh, P. and Nelder, J. A. (1989). *Generalized Linear Models*. Routledge, New York.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, pages 267–288.

Wood, S. (2006). *Generalized Additive Models: An Introduction with R*. CRC press.

Wood, S. (2015). *Core Statistics*. Cambridge University Press.